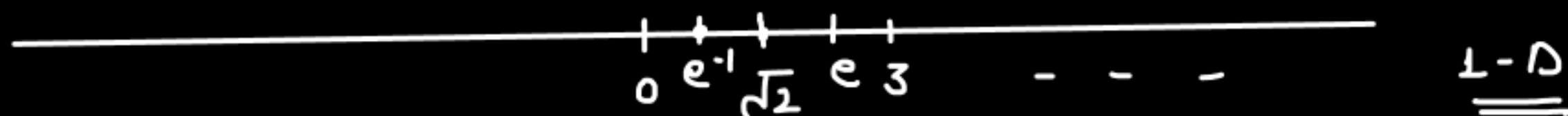


Complex Analysis

↳ Real Number:-

$$\sqrt{2}, 3, 3+\sqrt{2}, -2+\sqrt{3}, \pi/2, e, e^{-1}$$



↳ Complex Number:-

$$-7-3i$$

$$z_1 =$$

$$2+3i$$

$$z_2 =$$

$$z_3 =$$

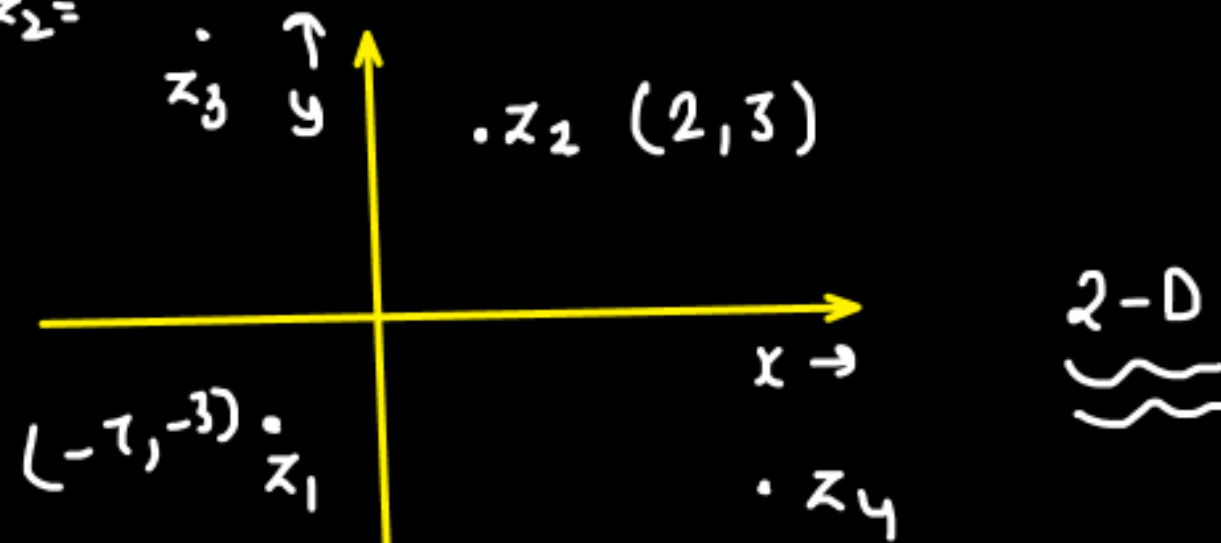
$$-2+7i$$

$$z_4 =$$

$$9-\pi i$$

- - -

$$z = x + iy$$



↳ Calculus of Real Variables

- Basics ✓
- Limit
- Continuity
- Differentiability }
→ Differential calculus ✓
- Integral calculus ✓

↳ Calculus of Complex Variables

- Basics ✓
- Limit
- Continuity
- Differentiability }
→ Analytic f^n }
→ Entire f^n }
- Integral calculus →

Q. Represent the following complex numbers on the complex plane and write their polar form:

$$Z_1 = 1 + 2j, \quad Z_2 = -1 + 2j, \quad Z_3 = -1 - 2j, \quad Z_4 = 1 - 2j.$$

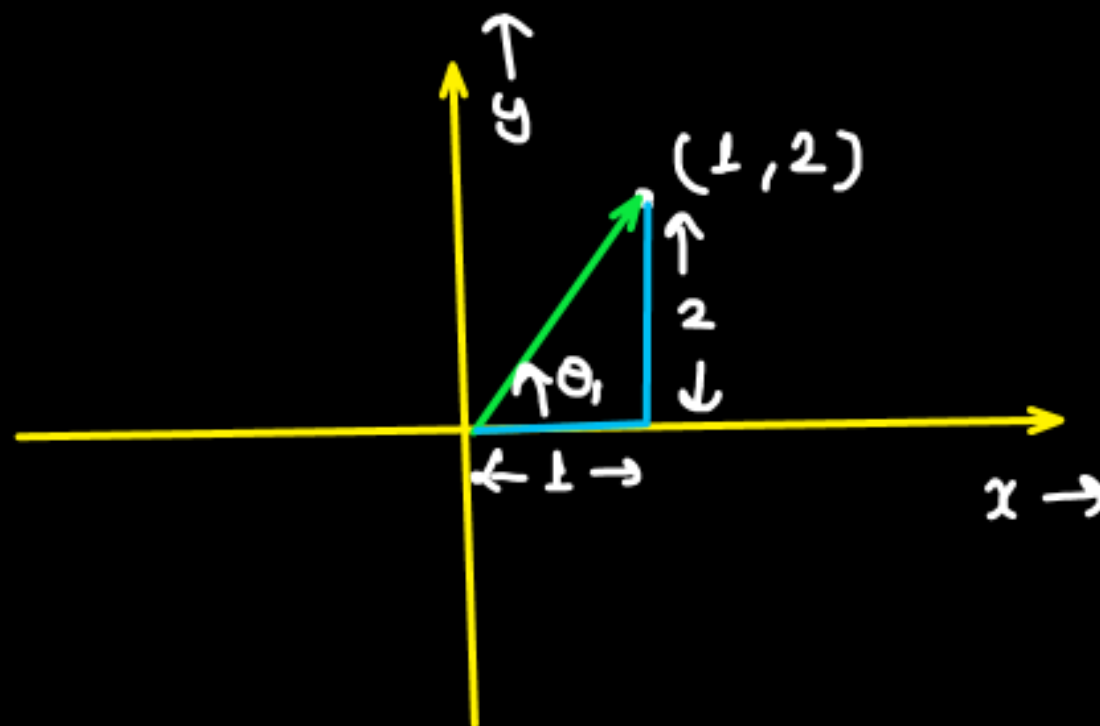
$z = x + jy \rightarrow$ Rectangular

$z = r e^{j\theta}$; $r = |z| = \text{Magnitude of } z = \sqrt{x^2 + y^2}$; $r > 0$
 $\theta = \text{Angle made by } z \text{ with +ve } x\text{-axis in acw dir}^n$

$$z_1 = 1 + 2j = r_1 e^{j\theta_1}$$

$$|z_1| = \sqrt{1 + 4} = \sqrt{5} = r_1$$

$$\theta_1 = \tan^{-1}(2) \\ = 63.43^\circ$$



$$\tan \theta_1 = \frac{2}{1}$$

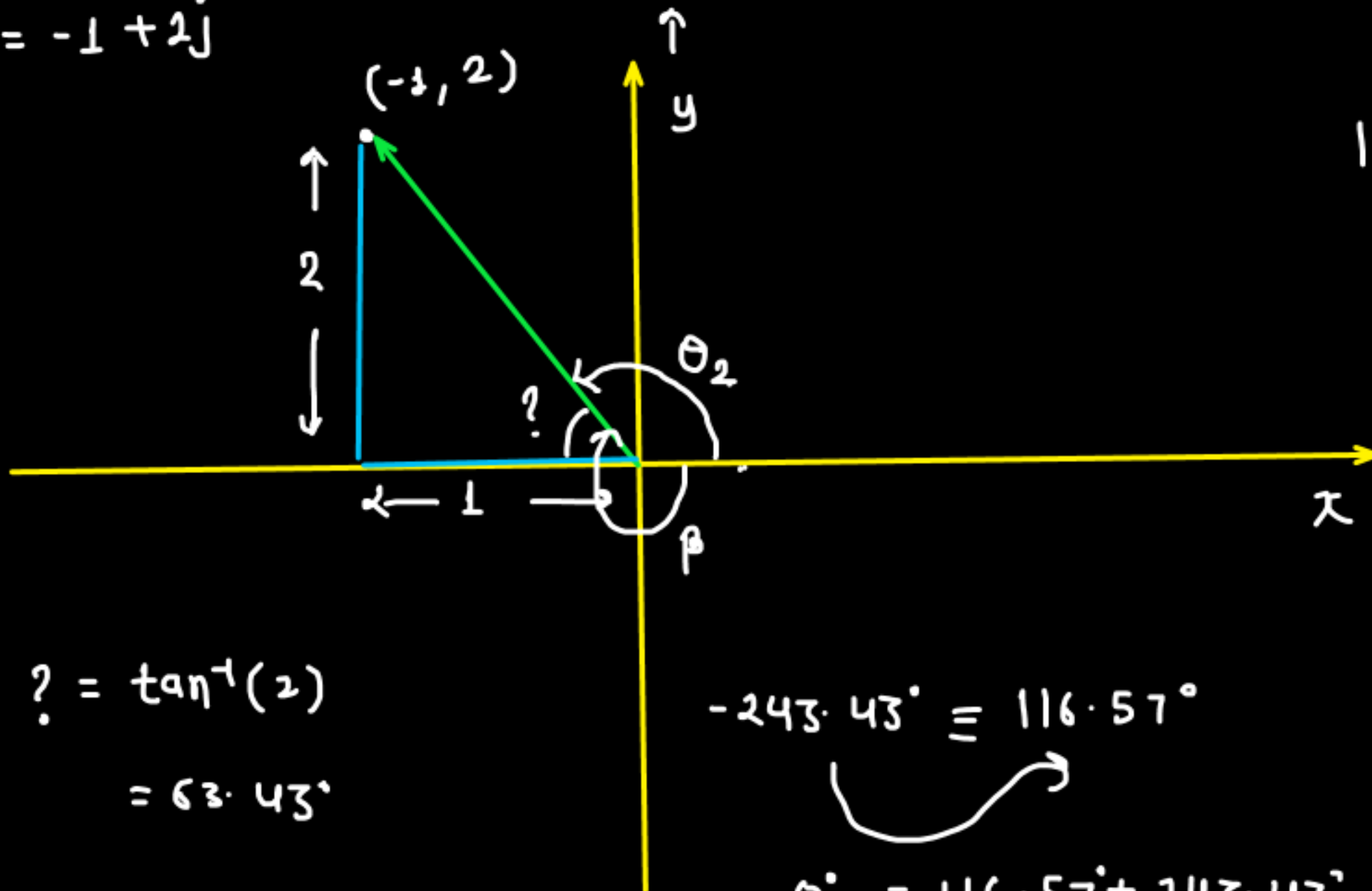
complex plane
 x - y plane / z -plane

$$z_1 = 1 + 2j = \sqrt{5} e^{i 63.43^\circ}$$

$$i \rightarrow \text{jota}$$

$$j \rightarrow \text{jota}$$

$$\hookrightarrow z_2 = -1 + 2j$$



$$|z_2| = \sqrt{5} = r_2$$

$$z_2 = \sqrt{5} e^{j 116.57^\circ}$$

$$\theta_2 = -\beta = -243.43^\circ$$

$$? = \tan^{-1}(2)$$

$$= 63.43^\circ$$

$$\theta_2 = 180 - 63.43^\circ$$

$$= 116.57^\circ$$

$$-243.43^\circ \equiv 116.57^\circ$$

$$0^\circ \equiv 116.57^\circ + 243.43^\circ$$

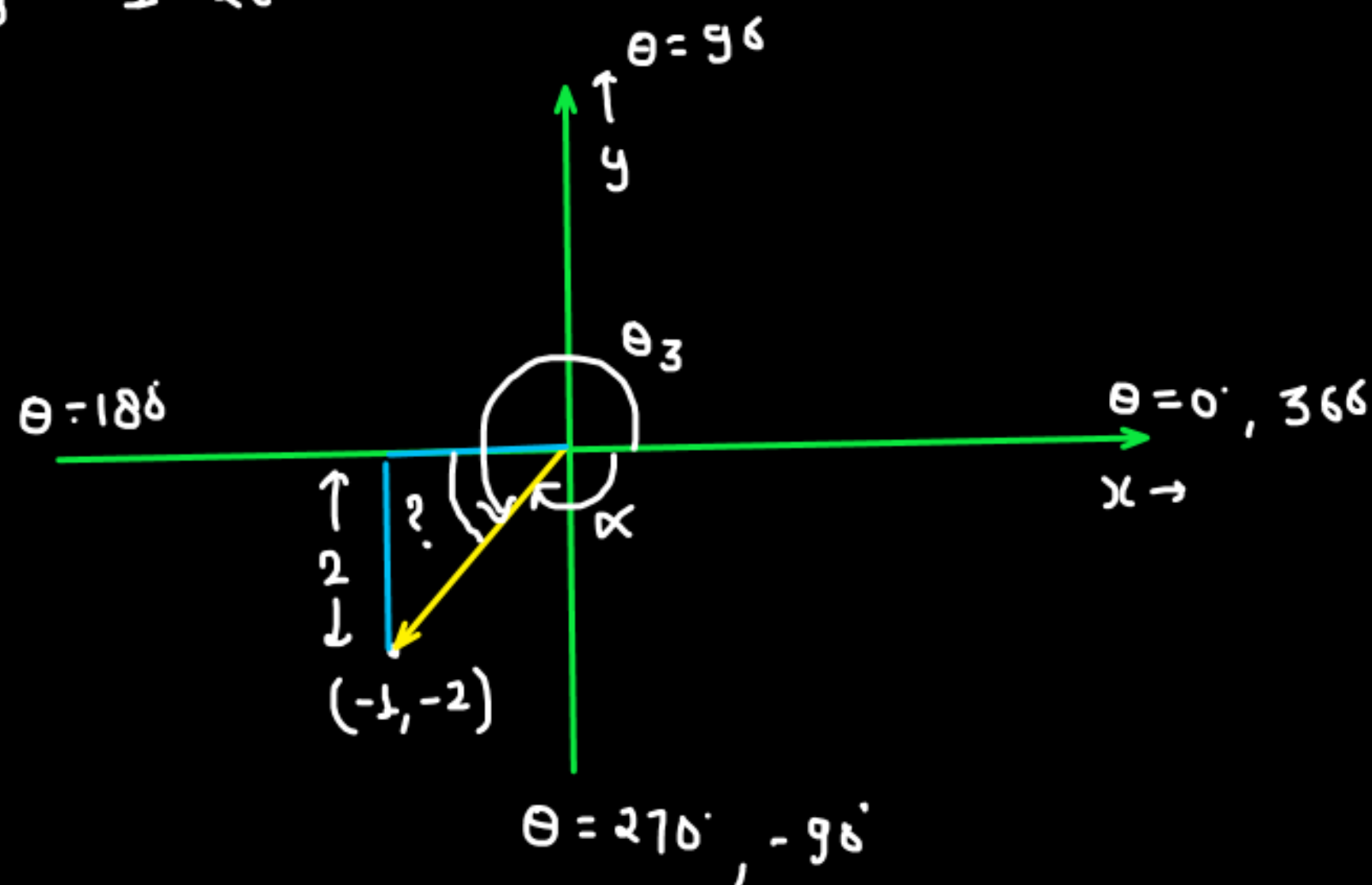
$$0^\circ \equiv 360^\circ$$

$$\theta_2 = 116.57^\circ$$

$$\beta = 180 + 63.43^\circ$$

$$= 243.43^\circ$$

$$z_3 = -1 - 2j$$



$$\alpha + \tan^{-1}(2) = 180$$

$$\alpha = 116.57^\circ$$

$$z_3 = \sqrt{5} e^{-j 116.57^\circ} = \sqrt{5} e^{j 243.43^\circ}$$

$$\begin{aligned} ? &= \tan^{-1}(2) \\ &= 63.43^\circ \end{aligned}$$

$$\begin{aligned} \theta_3 &= 180^\circ + 63.43^\circ \\ &= 243.43^\circ \end{aligned}$$

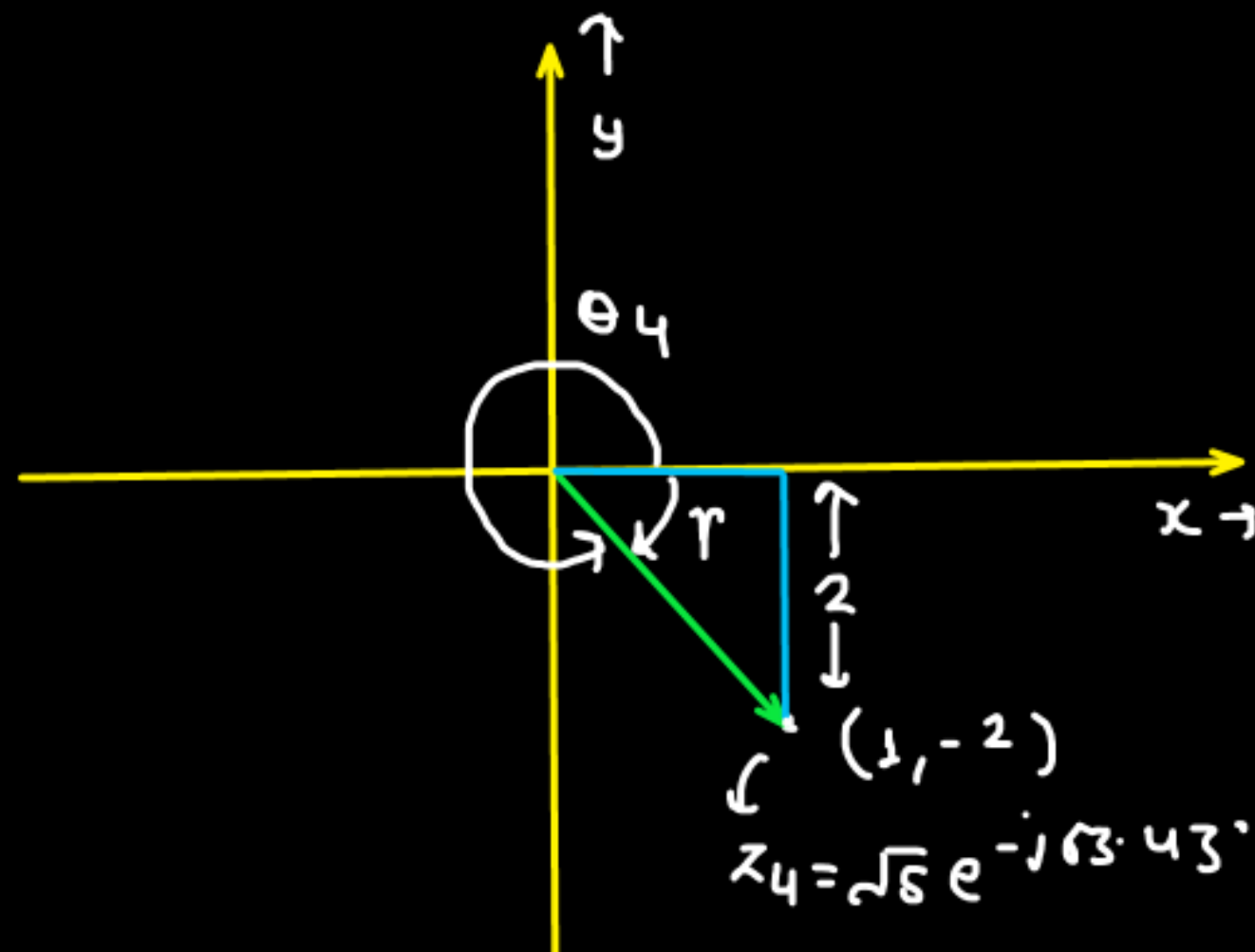
$$\theta_3 = -\alpha$$

$$\theta_3 = -116.57^\circ$$

$$243.43^\circ \equiv -116.57^\circ$$

$$360 \equiv 0^\circ$$

$$z_4 = 1 - 2j = r_4 e^{j\theta_4} \quad r_4 = \sqrt{5}$$



$$\gamma = \tan^{-1}(2) = 63.43^\circ$$

$$\theta_4 = -\gamma = -63.43^\circ$$

$$z_4 = \sqrt{5} e^{-j63.43^\circ} = \sqrt{5} e^{j296.57^\circ}$$

$$\gamma = \tan^{-1}(2)$$

$$\gamma = 63.43^\circ$$

$$\theta_4 = 360^\circ - 63.43^\circ$$

$$\theta_4 = 296.57^\circ$$

$$360 \equiv 0^\circ$$

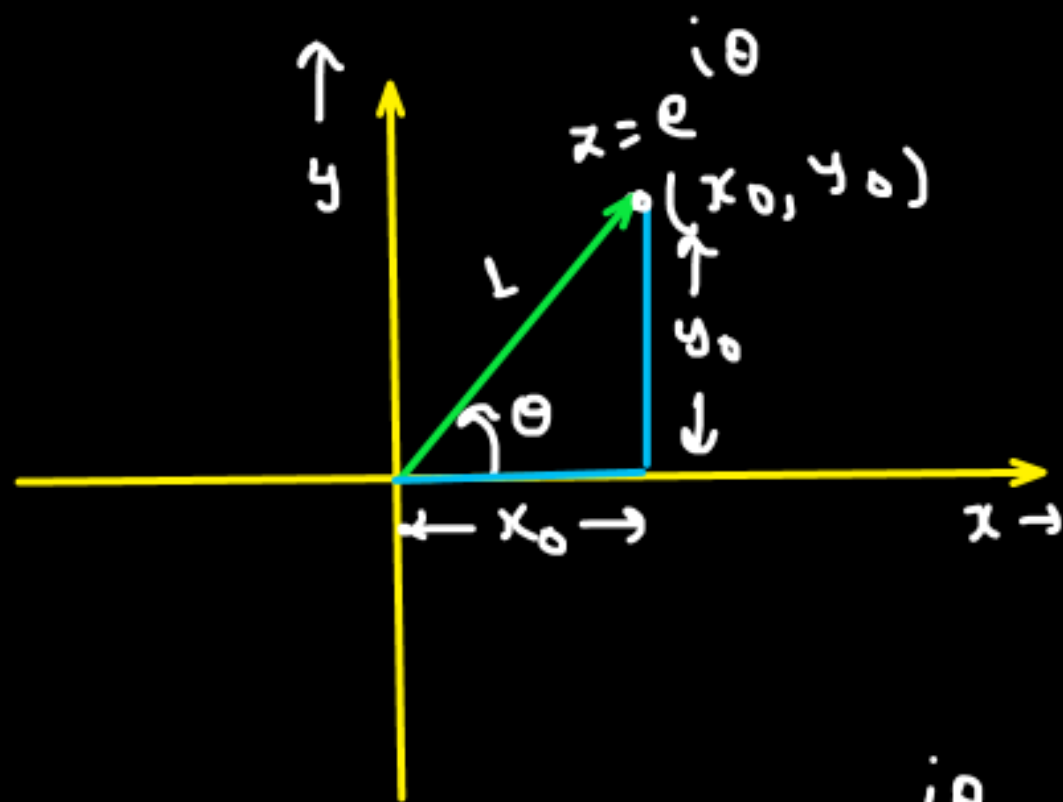
$$296.57^\circ + 63.43^\circ \equiv 0^\circ$$

$$296.57^\circ \equiv -63.43^\circ$$

$$\theta_4 = -63.43^\circ$$

↳ Understanding $e^{i\theta}$:-

$z = e^{i\theta}$: Polar form



$$x_0 = \cos \theta$$

$$y_0 = \sin \theta$$

$$z = e^{i\theta} = \cos \theta + i \sin \theta$$

$$\hookrightarrow \boxed{e^{i\theta} = \cos \theta + i \sin \theta}$$

$$z = r e^{i\theta}$$

$$r = 1$$

$$e^{i\theta} = \cos \theta + i \sin \theta$$

$$e^{i(\theta + 2n\pi)} = \cos(\theta + 2n\pi) + i \sin(\theta + 2n\pi)$$

$$e^{i(\theta + 2n\pi)} = \cos \theta + i \sin \theta = e^{i\theta}$$

$$\boxed{e^{i(\theta + 2n\pi)} = e^{i\theta}}$$

↳ 360° Angle

$$z = r e^{i\theta}$$

$$= r [\cos \theta + i \sin \theta]$$

$$\boxed{z = r \cos \theta + i r \sin \theta = r e^{i\theta}}$$

$$4. \quad e^{i\theta} = \cos\theta + i\sin\theta$$

$$4. \quad e^{i(\theta + 2n\pi)} = e^{i\theta} \cdot \underbrace{e^{i2n\pi}} = e^{i\theta}$$

Q. Convert polar form into rectangular form.

$$1. \quad e^{i0} = e^0 = 1 = \cos 0 + i\sin 0 = 1$$

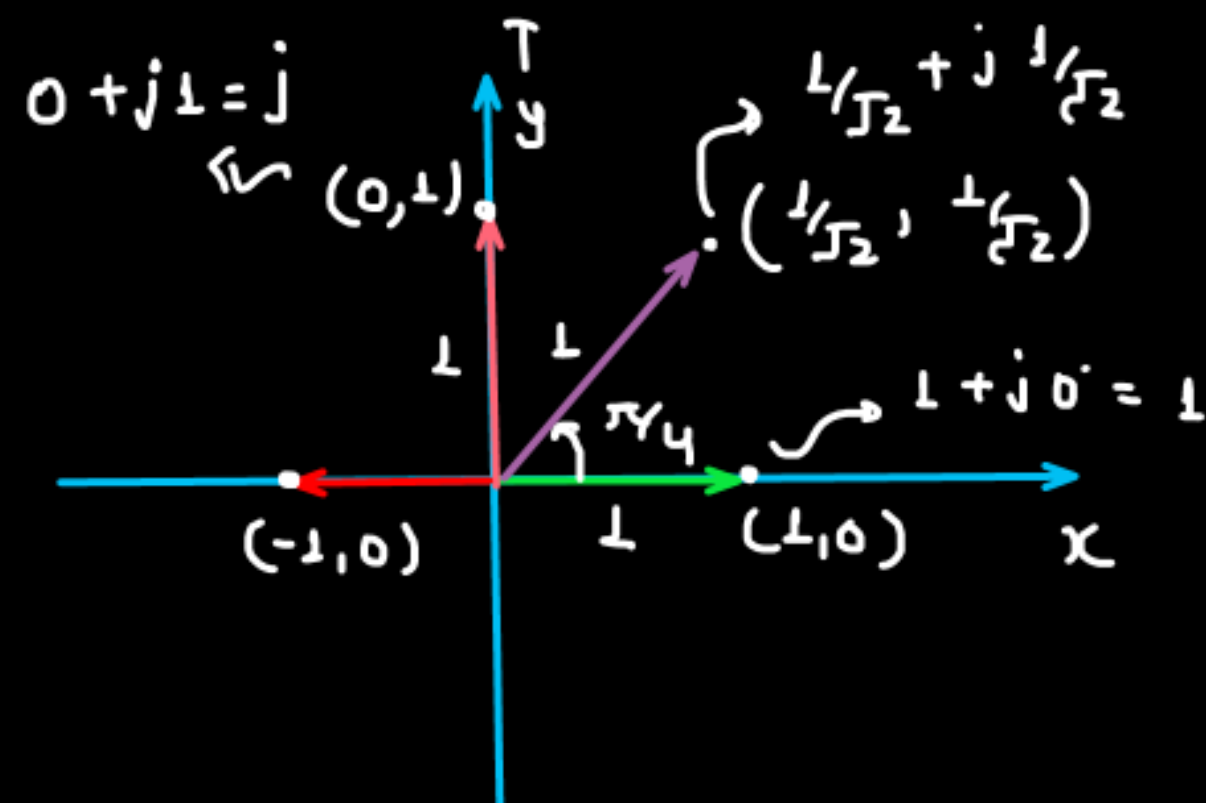
$$2. \quad e^{i\pi/4} = \frac{1}{\sqrt{2}} + j \frac{1}{\sqrt{2}}$$

$$3. \quad e^{i\pi/2} = i = \cos \pi/2 + i\sin \pi/2 = i$$

$$4. \quad e^{i\pi} = -1$$

$$e^{a+b} = e^a \cdot e^b$$

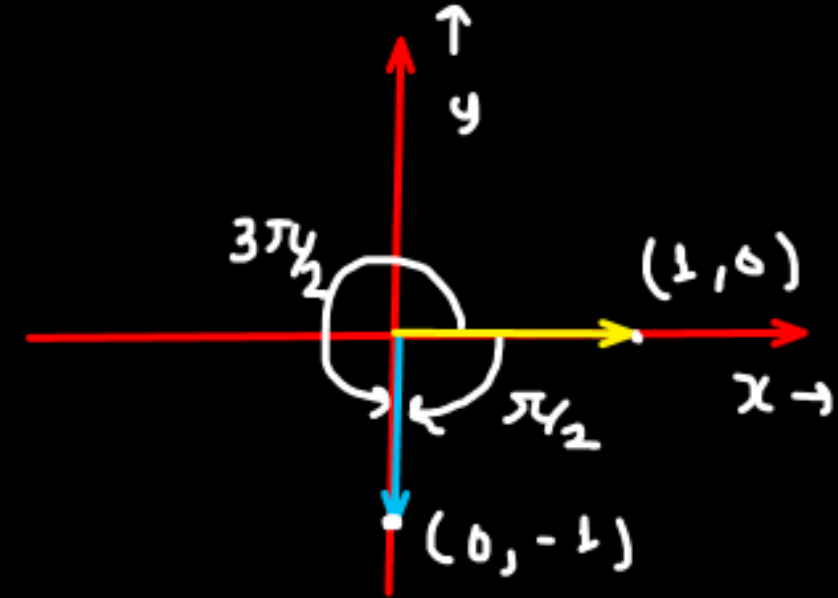
$$\begin{aligned} e^{i2\pi} &= \cos(2\pi) + i\sin(2\pi) \\ &= 1 + i(0) = 1 \end{aligned}$$



$$5. \underline{e^{j3\pi/2}} = -j = \underline{e^{-j\pi/2}} = e^{j(\pi+\pi/2)} = \underline{e^{j\pi}} \cdot e^{j\pi/2} = -e^{j\pi/2}$$

$$6. e^{j2\pi} = e^{j0} = 1$$

$$7. e^{j7\pi/3} = e^{j(2\pi+\pi/3)} = \underbrace{e^{j2\pi}}_1 \cdot e^{j\pi/3} = e^{j\pi/3}$$



Q. Find the following:-

$$1. i^2 = (e^{j\pi/2})^2 = e^{j\pi} = -1 \quad \checkmark$$

$$2. i^3 = i^2 \cdot i = -1 \cdot i = -i = -e^{j\pi/2} = e^{-j\pi/2} = e^{j3\pi/2}$$

$$4. i^4 = (i^2)^2 = (-1)^2 = 1 \quad \checkmark$$

$$5. i^7 = i^4 \cdot i^2 \cdot i = 1 \cdot (-1) \cdot i = -i$$

$$6. i^{4n} = (i^4)^n = (1)^n = 1 \quad n \in I$$

$$7. i^{2n} = (i^2)^n = (-1)^n$$

$$(e^a)^b = e^{ab}$$

* Represent $\cos \theta$ and $\sin \theta$ in terms of complex exponentials:-

$$e^{i\theta} = \cos \theta + i \sin \theta \text{ --- (1)} ; e^{-i\theta} = \cos \theta - i \sin \theta \text{ --- (2)}$$

$$(1) + (2)$$

$$2 \cos \theta = e^{i\theta} + e^{-i\theta} ;$$

$$\hookrightarrow \boxed{\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}} \quad \text{w}$$

$$(1) - (2)$$

$$e^{i\theta} - e^{-i\theta} = 2i \sin \theta$$

$$\hookrightarrow \boxed{\sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}}$$

$$\cosh x = \frac{e^x + e^{-x}}{2}$$

$$\cosh(i\theta) = \frac{e^{i\theta} + e^{-i\theta}}{2} = \cos \theta$$

$$\hookrightarrow \boxed{\cosh(i\theta) = \cos \theta}$$

$$\sinh x = \frac{e^x - e^{-x}}{2}$$

$$\sinh(i\theta) = \frac{e^{i\theta} - e^{-i\theta}}{2}$$

$$\sinh(i\theta) = \frac{e^{i\theta} - e^{-i\theta}}{2j} \times j = \sin\theta \times j$$

$$\hookrightarrow \boxed{\sinh(i\theta) = i\sin\theta}$$

Conclusion:-

$$\hookrightarrow e^{i\theta} = \cos\theta + i\sin\theta$$

$$\hookrightarrow \cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$$

$$\hookrightarrow \sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2j}$$

$$\hookrightarrow \sinh(i\theta) = i\sin\theta$$

$$\hookrightarrow \cosh(i\theta) = \cos\theta$$

$$\hookrightarrow \sinh x = \frac{e^x - e^{-x}}{2} ; \quad \cosh x = \frac{e^x + e^{-x}}{2}$$

Q. $z = \left(\frac{1+i}{1-i} \right)^i$. simplify

$$1+i = \sqrt{2} e^{j\pi/4}$$

$$1-i = \sqrt{2} e^{-j\pi/4}$$

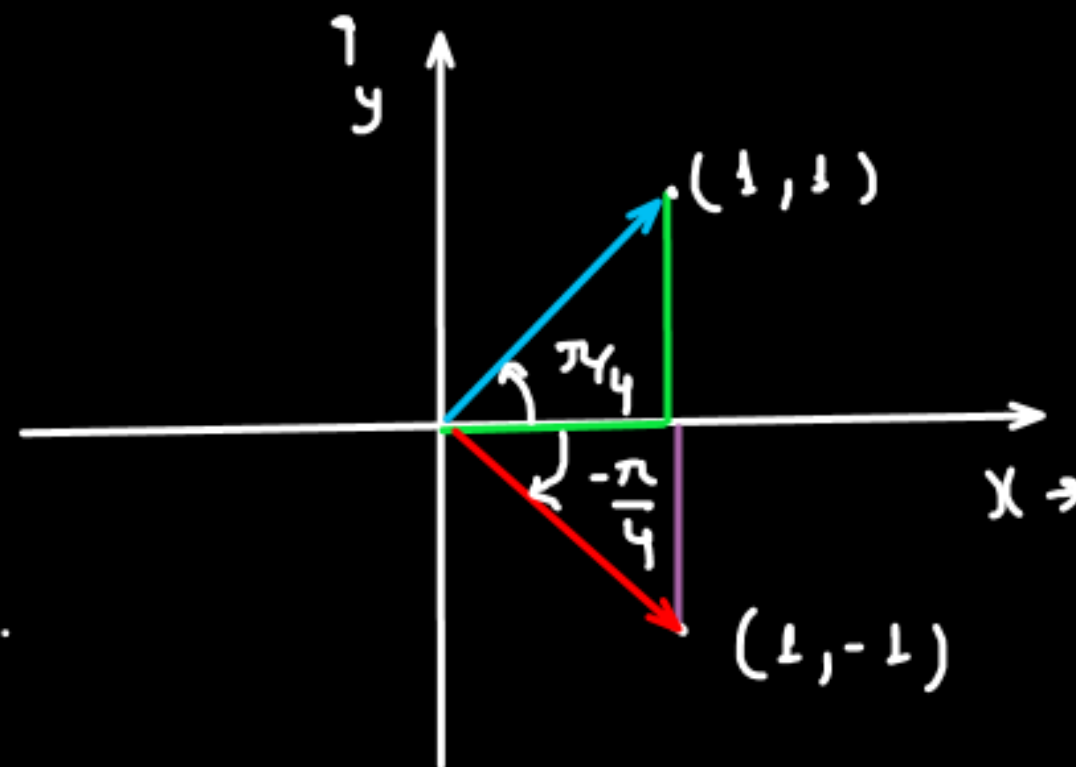
$$z = \left(\frac{\sqrt{2} e^{j\pi/4}}{\sqrt{2} e^{-j\pi/4}} \right)^i$$

$$z = \left(e^{j\pi/2} \right)^i = \left(e^{j5\pi/2} \right)^i = e^{-5\pi/2} \text{ Ans.}$$

$$z = e^{i^2 \pi/2}$$

$$z = e^{-\pi/2} : \text{Real}$$

J



$$\pi/2 + 2\pi = 5\pi/2$$

* Conjugate of a complex number:-

$$z = x + iy = re^{i\theta}$$

$$\bar{z} = z^* = x - iy = re^{-i\theta}$$

$$\operatorname{Re}(z) = x$$

$$\operatorname{Im}(z) = y$$

$$z = re^{i\theta}$$

$$z^* = (re^{i\theta})^* \\ = re^{-i\theta}$$

$$z = x + iy \quad ; \quad z^* = x - iy$$

—(1) —(2)

$$z + z^* = 2x$$

$$z - z^* = 2iy$$

$$x = \frac{z + z^*}{2} \Rightarrow$$

$$\operatorname{Re}(z) = \frac{z + z^*}{2}$$

$$y = \frac{z - z^*}{2i}$$

$$\operatorname{Im}(z) = \frac{z - z^*}{2i}$$

$$z = -1 - 3j$$

$$\operatorname{Re}(z) = -1$$

$$\operatorname{Im}(z) = -3$$

$$z \cdot z^* = re^{i\theta} \cdot re^{-i\theta} = r^2 \cdot e^{i\theta} \cdot e^{-i\theta} = r^2 = |z|^2$$

$$\hookrightarrow \boxed{|z|^2 = z \cdot z^*} \quad \downarrow$$

Imp
Q.

Let Z_1 and Z_2 be two complex numbers such that

$$|Z_1| = 3, \quad |Z_2| = 5.$$

Determine the following:

$$r_1 = 3, \quad r_2 = 5$$

(A) Find the maximum possible value of $|Z_1 + Z_2|$. **8**

(B) Find the minimum possible value of $|Z_1 + Z_2|$. **2**

(C) Find the maximum possible value of $|Z_1 - Z_2|$. **8**

(D) Find the minimum possible value of $|Z_1 - Z_2|$. **2**

(E) Find the value of

$$|Z_1 + Z_2|^2 + |Z_1 - Z_2|^2.$$

(F) A complex number Z is defined as

$$Z = \frac{Z_1 \overline{Z_2}}{|Z_2|^2}.$$

Find the magnitude of Z , i.e., $|Z|$.

$|z_1|$ = Magnitude of z_1

$$z_1 = r_1 e^{i\theta_1} \quad |z_1| = r_1$$

$$z_2 = r_2 e^{i\theta_2} \quad |z_2| = r_2$$

$$(z_1 + z_2)^* = z_1^* + z_2^*$$

$$\begin{aligned} (r_1 e^{i\theta_1} + r_2 e^{i\theta_2})^* &= r_1 e^{-i\theta_1} + r_2 e^{-i\theta_2} \\ &= z_1^* + z_2^* \end{aligned}$$

$$4 \quad z_1 + z_2 = r_1 e^{i\theta_1} + r_2 e^{i\theta_2}$$

$$|z|^2 = z \cdot z^*$$

$$|z_1 + z_2|^2 = (z_1 + z_2) \cdot (z_1 + z_2)^*$$

$$|z_1 + z_2|^2 = (r_1 e^{i\theta_1} + r_2 e^{i\theta_2}) (r_1 e^{-i\theta_1} + r_2 e^{-i\theta_2})$$

$$= r_1^2 + r_1 r_2 e^{i(\theta_1 - \theta_2)} + r_1 r_2 e^{-i(\theta_1 - \theta_2)} + r_2^2$$

$$= r_1^2 + r_2^2 + r_1 r_2 \left[\frac{e^{i(\theta_1 - \theta_2)} + e^{-i(\theta_1 - \theta_2)}}{2} \right] \times 2$$

$$|z_1 + z_2|^2 = r_1^2 + r_2^2 + 2r_1 r_2 \underbrace{\cos(\theta_1 - \theta_2)}$$

↓
max^m

↙ max
= 1

↘ min
= -1

$$\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$$

$$|z_1 + z_2|^2_{\max} = r_1^2 + r_2^2 + 2r_1 r_2 = (r_1 + r_2)^2$$

$$|z_1 + z_2|^2_{\max} = (r_1 + r_2)^2$$

$$\boxed{|z_1 + z_2|_{\max} = r_1 + r_2 = |z_1| + |z_2|}$$

$$|z_1 + z_2|^2_{\min} = r_1^2 + r_2^2 - 2r_1 r_2 = (r_1 - r_2)^2$$

$$|z_1 + z_2|^2_{\min} = (r_1 - r_2)^2$$

$$|z_1 + z_2|_{\min} = |r_1 - r_2| = ||z_1| - |z_2||$$

$$\boxed{||z_1| - |z_2|| \leq |z_1 + z_2| \leq |z_1| + |z_2|}$$

$$\left. \begin{array}{l} |z_1 + z_2| \geq 0 \\ |z_1 - z_2| \geq 0 \end{array} \right\} \rightarrow \text{Magnitude}$$

$$|3 - 5| \leq |z_1 + z_2| \leq 3 + 5$$

$$2 \leq |z_1 + z_2| \leq 8$$

$$z_1 - z_2 = r_1 e^{i\theta_1} - r_2 e^{i\theta_2}$$

$$|z|^2 = z \cdot z^*$$

$$|z_1 - z_2|^2 = (z_1 - z_2) \cdot (z_1 - z_2)^*$$

$$|z_1 - z_2|^2 = (r_1 e^{i\theta_1} - r_2 e^{i\theta_2}) (r_1 e^{-i\theta_1} - r_2 e^{-i\theta_2})$$

$$= r_1^2 - r_1 r_2 e^{i(\theta_1 - \theta_2)} - r_1 r_2 e^{-i(\theta_1 - \theta_2)} + r_2^2$$

$$= r_1^2 + r_2^2 - r_1 r_2 \left[\frac{e^{i(\theta_1 - \theta_2)} + e^{-i(\theta_1 - \theta_2)}}{2} \right] \times 2$$

$$|z_1 - z_2|^2 = r_1^2 + r_2^2 - 2r_1 r_2 \underbrace{\cos(\theta_1 - \theta_2)}$$

max

max
= 1

min
= -1

$$\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$$

$$|z_1 - z_2|_{\max} = r_1^2 + r_2^2 + 2r_1 r_2 = (r_1 + r_2)^2$$

$$|z_1 - z_2|_{\max} = r_1 + r_2 = |z_1| + |z_2|$$

$$|z_1 - z_2|_{\min} = r_1^2 + r_2^2 - 2r_1 r_2 = (r_1 - r_2)^2$$

$$|z_1 - z_2|_{\min} = |r_1 - r_2| = ||z_1| - |z_2||$$

$$||z_1| - |z_2|| \leq |z_1 - z_2| \leq |z_1| + |z_2|$$

★

$$||z_1| - |z_2|| \leq |z_1 \pm z_2| \leq |z_1| + |z_2|$$

; conclusion =

$$5. \quad |z_1 + z_2|^2 + |z_1 - z_2|^2 = r_1^2 + r_2^2 + 2\cancel{r_1}r_2 \cos(\theta_1 - \theta_2) \\ + r_1^2 + r_2^2 - 2\cancel{r_1}r_2 \cos(\theta_1 - \theta_2)$$

$$\boxed{|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2|z_1|^2 + 2|z_2|^2} = 2(9) + 2(25) = 68 \text{ ans.}$$

$$6. \quad z = \frac{z_1 \bar{z}_2}{|z_2|^2} = \frac{r_1 e^{i\theta_1} \cdot r_2 e^{-i\theta_2}}{r_2^2} \quad \bar{z}_2 = z_2^*$$

$$z = \frac{r_1}{r_2} e^{i(\theta_1 - \theta_2)}$$

$$\boxed{|z| = \frac{r_1}{r_2} = \frac{3}{5}}$$

Imp.
Q. Solving the following eqⁿ for z .

(a) $z^2 = 1$

(b) $z^2 = -1$

(c) $z^2 = j$

(d) $z^3 = 1$

(e) $z^3 = -1$

(f) $z^4 = 1$

(g) $z^4 = -j$

(a) $z^2 = 1$

$$z = \pm 1$$

Real

$$z^2 = 1$$

$$r^2 e^{2i\theta} = e^{i2n\pi}$$

$$r^2 = 1$$

$$2\theta = 2n\pi$$

$$r = 1, -1$$

$$\theta = n\pi$$

$$; z = r e^{i\theta} ;$$

$$1 = e^{i0}$$

$$1 = e^{i(0 + 2n\pi)}$$

$$1 = e^{j2\pi n}$$

$$z = 1 e^{in\pi}$$

$$n=0 \Rightarrow z = 1 ; n=1 \Rightarrow z = e^{i\pi} = -1$$

$$z = 1, -1$$

$$(b) \quad z^2 = -1 \quad ; \quad z = re^{i\theta}$$

$$r^2 e^{i2\theta} = e^{j(\pi + 2n\pi)}$$

$$r^2 = 1 \quad , \quad 2\theta = \pi + 2n\pi$$

$$r = 1 \quad , \quad \theta = \pi/2 + n\pi$$

$$\boxed{z^2 = -1 \Rightarrow z = i, -i}$$

$$-1 = e^{j(\pi + 2n\pi)}$$

$$z = e^{i(\pi/2 + n\pi)}$$

$$n=0 \Rightarrow z = e^{i\pi/2} = i$$

$$n=1 \Rightarrow z = e^{i(\pi/2 + \pi)} = e^{i(3\pi/2)} = -i$$

$$(c) \quad z^2 = i \quad ; \quad z = re^{i\theta}$$

$$r^2 e^{i2\theta} = e^{i(\pi/2 + 2n\pi)}$$

$$r^2 = 1 \quad ; \quad 2\theta = \pi/2 + 2n\pi$$

$$r = 1 \quad \theta = \pi/4 + n\pi$$

$$\boxed{z^2 = i \Rightarrow z = \pm \left(\frac{1+i}{\sqrt{2}} \right)}$$

$$i = e^{i(\pi/2 + 2n\pi)}$$

$$z = e^{i(\pi/4 + n\pi)}$$

$$n=0 \quad ; \quad z = e^{i\pi/4} = \frac{1}{\sqrt{2}} + i \frac{1}{\sqrt{2}} = \frac{1+i}{\sqrt{2}}$$

$$n=1 \quad ; \quad z = e^{i(\pi/4 + \pi)} = e^{i\pi} e^{i\pi/4} = -e^{i\pi/4}$$

$$= -\frac{1}{\sqrt{2}} - i \frac{1}{\sqrt{2}}$$

$$(d) \quad z^3 = 1 \quad z = re^{i\theta}$$

$$r^3 e^{j3\theta} = e^{i2n\pi}$$

$$r^3 = 1$$

$$3\theta = 2n\pi$$

$$r = 1$$

$$\theta = 2n\pi/3$$

$$z^3 = 1$$

$$\hookrightarrow z = 1, -e^{j\pi/3}, -e^{-j\pi/3}$$

$$z = 1, \frac{-1 - \sqrt{3}j}{2}, \frac{-1 + \sqrt{3}j}{2}$$

Ans

$$1 = e^{i(0+2n\pi)}$$

$$z = e^{i2n\pi/3}$$

$$n=0; \quad z=1$$

$$n=1; \quad z = e^{j2\pi/3} = e^{j(\pi - \pi/3)} = e^{j\pi} \cdot e^{-j\pi/3}$$

$$z = -e^{-j\pi/3}$$

$$n=2; \quad z = e^{j4\pi/3} = e^{j(\pi + \pi/3)} = e^{j\pi} e^{j\pi/3}$$

$$z = -e^{j\pi/3}$$

$$e^{j\pi/3} = \cos \pi/3 + i \sin \pi/3 = \frac{1}{2} + j\frac{\sqrt{3}}{2} = \frac{1 + \sqrt{3}j}{2}$$

$$e^{-j\pi/3} = \frac{1 - \sqrt{3}j}{2}$$

$$\hookrightarrow z^3 = 1$$

$$z^3 - 1 = 0 \quad \hookrightarrow \underbrace{(z-1)}_{z=1} \underbrace{(z^2+z+1)}_{z^2+z+1=0} = 0$$

$$z = \frac{-1 \pm \sqrt{1-4}}{2} = \frac{-1 \pm \sqrt{3}j}{2}$$

$$z = 1, \frac{-1 + \sqrt{3}j}{2}, \frac{-1 - \sqrt{3}j}{2}$$

$$(e) \quad z^3 = -1 \quad (z+1)(z^2-z+1) = 0$$

$$z^3 + 1^3 = 0$$

$$z = -1 \quad ; \quad z = \frac{1 \pm \sqrt{3}j}{2}$$

$$(z+1)(z^2-z+1)$$

$$z = -1, \frac{1 + \sqrt{3}j}{2}, \frac{1 - \sqrt{3}j}{2}$$

$$\hookrightarrow z^3 = -1$$

$$r^3 e^{j3\theta} = e^{j(\pi + 2n\pi)}$$

$$r = 1 ; \quad \theta = \frac{\pi}{3} + 2n\pi/3$$

$$\boxed{\begin{array}{l} z^3 = -1 \\ z = -1, e^{j\pi/3}, e^{-j\pi/3} \end{array}}$$

$$(f) z^4 = 1 ; \quad z^4 - 1 = 0$$

$$\Rightarrow \boxed{z = 1, -1, i, -i} \quad \begin{array}{l} (z^2)^2 - (1)^2 = 0 \\ (z^2 + 1)(z^2 - 1) = 0 \end{array}$$

$$-1 = e^{j(\pi + 2n\pi)}$$

$$z = e^{j(\pi/3 + 2n\pi/3)}$$

$$n=0 ; \quad z = e^{j\pi/3}$$

$$n=1 ; \quad z = e^{j\pi}$$

$$n=2 ; \quad z = e^{j5\pi/3} = e^{j2\pi - \pi/3} = e^{j2\pi} \cdot e^{-j\pi/3} = e^{-j\pi/3}$$

$$e^{j\pi/3} = \frac{1 + j\sqrt{3}}{2}$$

$$e^{-j\pi/3} = \frac{1 - j\sqrt{3}}{2}$$

$$z^2 - 1 = 0, \quad z = 1, -1$$

$$z^2 + 1 = 0$$

$$z = i, -i$$

$$\hookrightarrow z^4 = 1 \quad ; \quad z = re^{i\theta}$$

$$r^4 e^{i4\theta} = e^{i2\pi n}$$

$$r^4 = 1 \Rightarrow \underbrace{r = 1}$$

$$4\theta = 2n\pi$$

$$\underbrace{\theta = n\pi/2}$$

$$z = 1, j, -1, -j$$

$$z = e^{jn\pi/2}$$

$$n=0 \Rightarrow z = 1$$

$$n=1 \Rightarrow z = e^{j\pi/2} = j$$

$$n=2 \Rightarrow z = e^{j\pi} = -1$$

$$n=3 \Rightarrow z = e^{j3\pi/2} = -j$$

$$(g) \quad z^4 = -j$$

$$r^4 e^{i4\theta} = e^{j(-\pi/2 + 2n\pi)}$$

$$r = 1 \Rightarrow \theta = \frac{-\pi/2 + 2n\pi}{4} = -\pi/8 + \frac{n\pi}{2}$$

$$\hookrightarrow z = e^{j(-\pi/8)}, e^{j(3\pi/8)}, -e^{-j\pi/8}, -e^{j3\pi/8}$$

$$n=0 \Rightarrow z = e^{j(-\pi/8)}$$

$$n=1 \Rightarrow z = e^{j3\pi/8}$$

$$n=2 \Rightarrow z = -e^{-j\pi/8}$$

$$n=3 \Rightarrow z = -e^{j3\pi/8}$$

Q. e^z is periodic with ~~(a) 2π~~ (b) $j2\pi$ (c) π (d) None

$$\hookrightarrow e^{(z+?) } = e^z$$

$$e^{(z+j2\pi)} = e^z \cdot \underbrace{e^{j2\pi}}_{=1} = e^z \cdot 1$$

$$e^{(z+\underbrace{j2\pi}_{T=j2\pi})} = e^z$$

$f(x)$: periodic

$$f(x+T) = f(x)$$

$\downarrow$
Time period

$$\sin(\theta + \underline{2\pi}) = \sin \theta$$

V. Imp.

Q. Draw the following locus

$$(a) |z| = 1 \quad (b) |z - 3j| = 2 \quad (c) z = 2e^{j\theta} \quad (d) z = 2 + 4e^{j\theta}$$

$$(e) |z - 2 - 3j| = 2 \quad (f) |z| \leq 2 \quad (g) |z - 1 + 3j| \leq 2$$

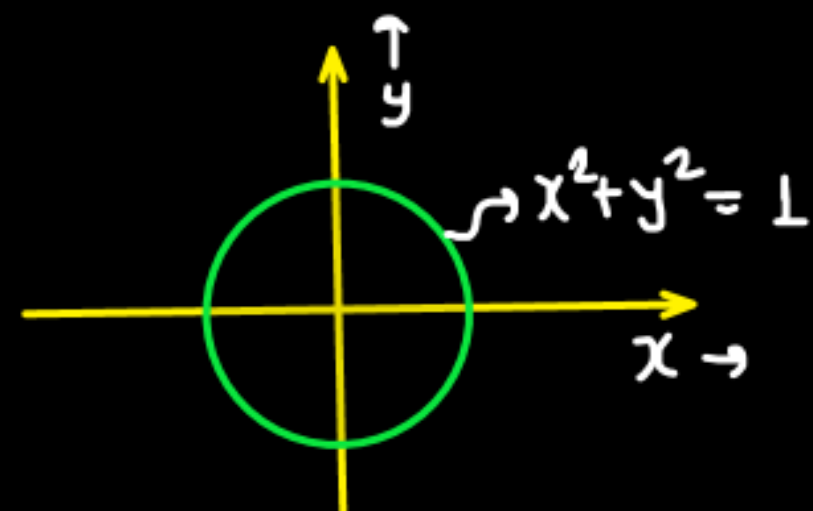
$$\hookrightarrow z = x + jy = re^{j\theta}$$

$$|z| = \sqrt{x^2 + y^2}$$

(a) $|z| = 1$ is closed line

$$\sqrt{x^2 + y^2} = 1$$

↳ $\boxed{x^2 + y^2 = 1}$ is circle (0,0) radius = 1



(c) $z = 2e^{i\theta}$ is closed line

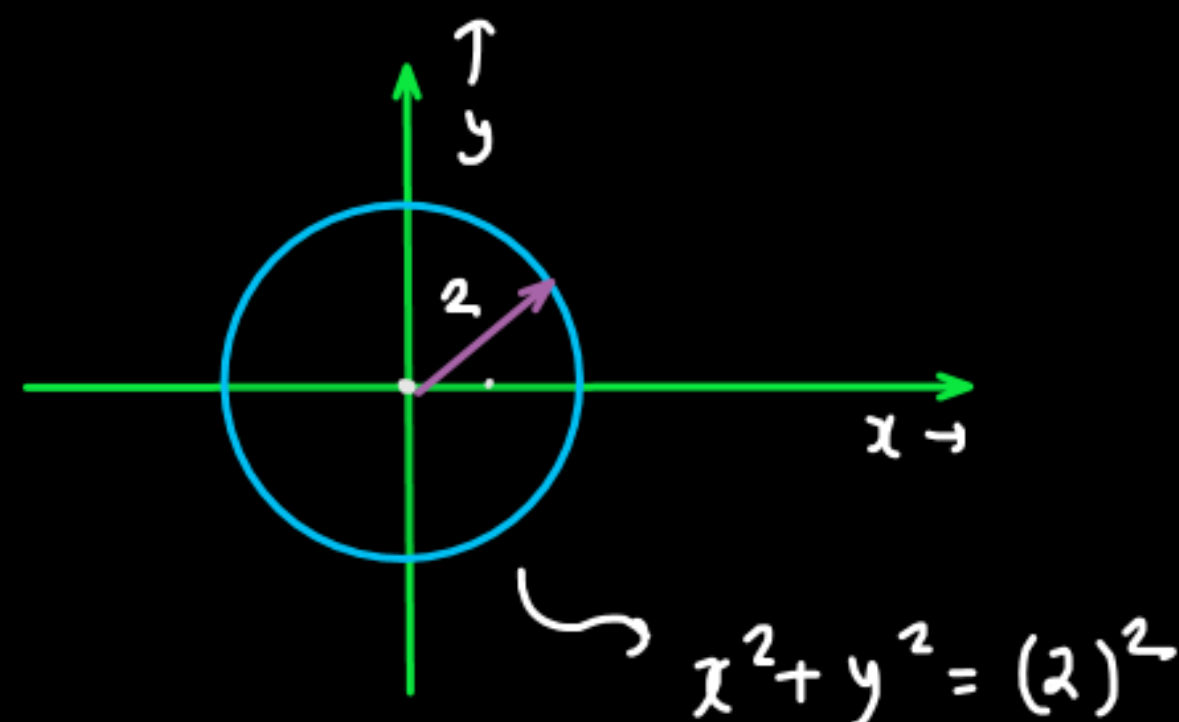
↳ $r = 2$

↓ $z = 2e^{i\theta}$

$$|z| = |2e^{i\theta}|$$

↳ $|z| = 2$

$$\sqrt{x^2 + y^2} = 2 \Rightarrow \boxed{x^2 + y^2 = (2)^2} : \text{circle (0,0) , radius} = 2$$

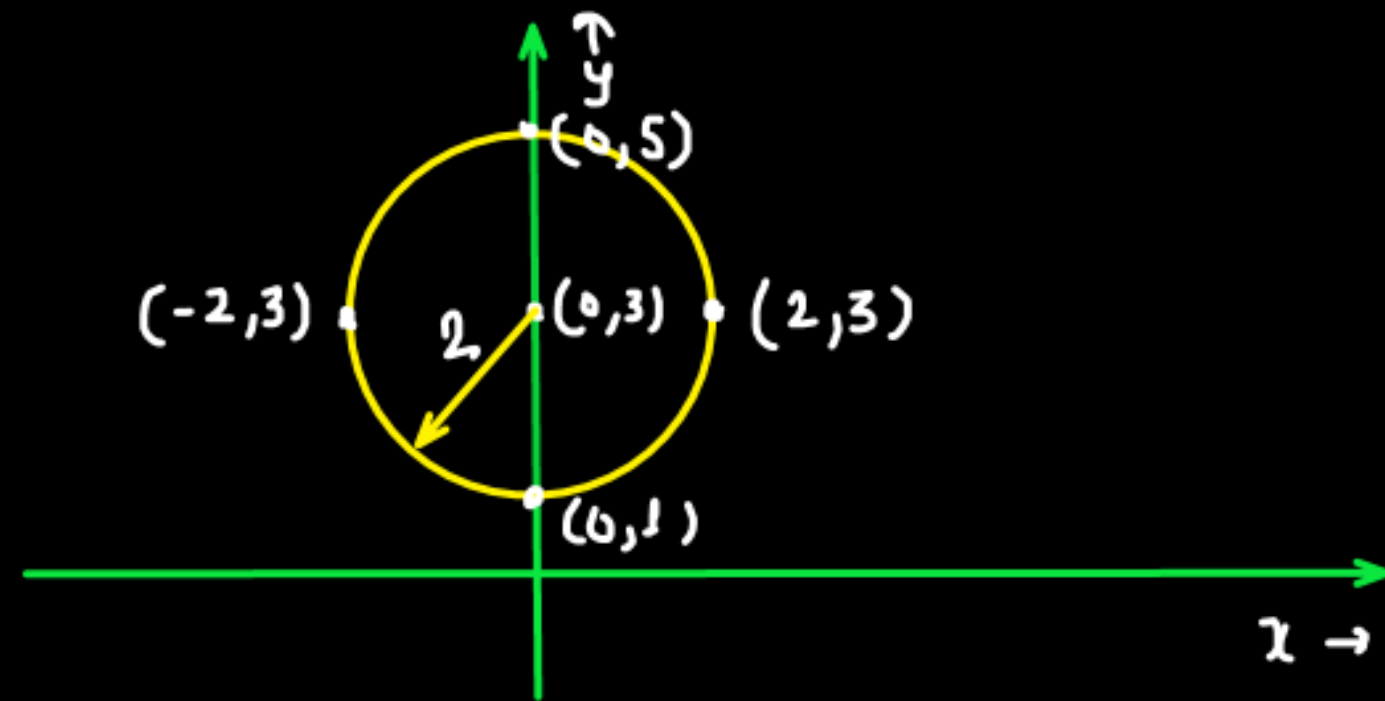


$$(b) |z - 3i| = 2 \quad \text{radius}$$

$$|x + iy - 3i| = 2$$

$$|x + i(y - 3)| = 2$$

$$\sqrt{x^2 + (y - 3)^2} = 2 \rightarrow x^2 + (y - 3)^2 = 4 : \text{circle } c(0, 3), \text{ radius} = 2$$



$$(d) z = 2 + 4e^{i\theta}$$

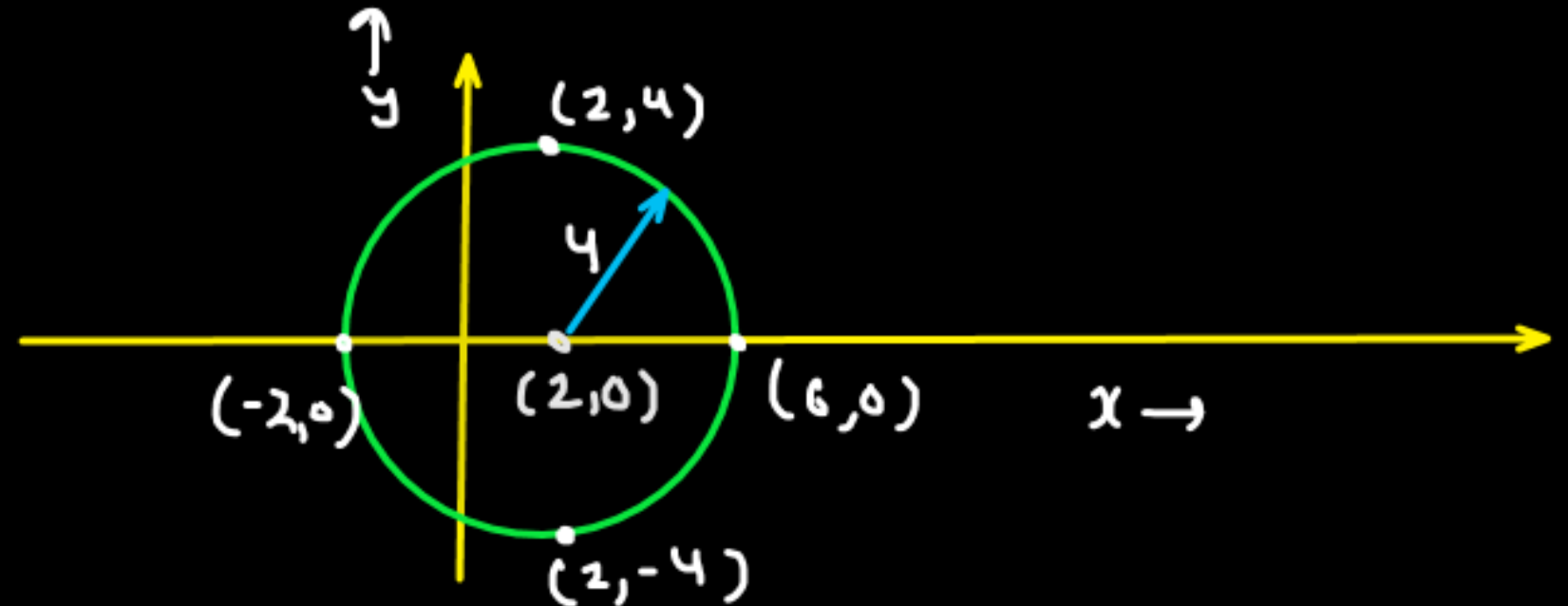
$$z - 2 = 4e^{i\theta}$$

$$|z - 2| = 4 \quad \leftarrow$$

$$|x + iy - 2| = 4$$

$$|x - 2 + iy| = 4$$

$$(x - 2)^2 + y^2 = (4)^2 ; c(2, 0), \text{ radius} = 4$$



$$|z - a - ib| = r$$

$$|x + iy - a - ib| = r$$

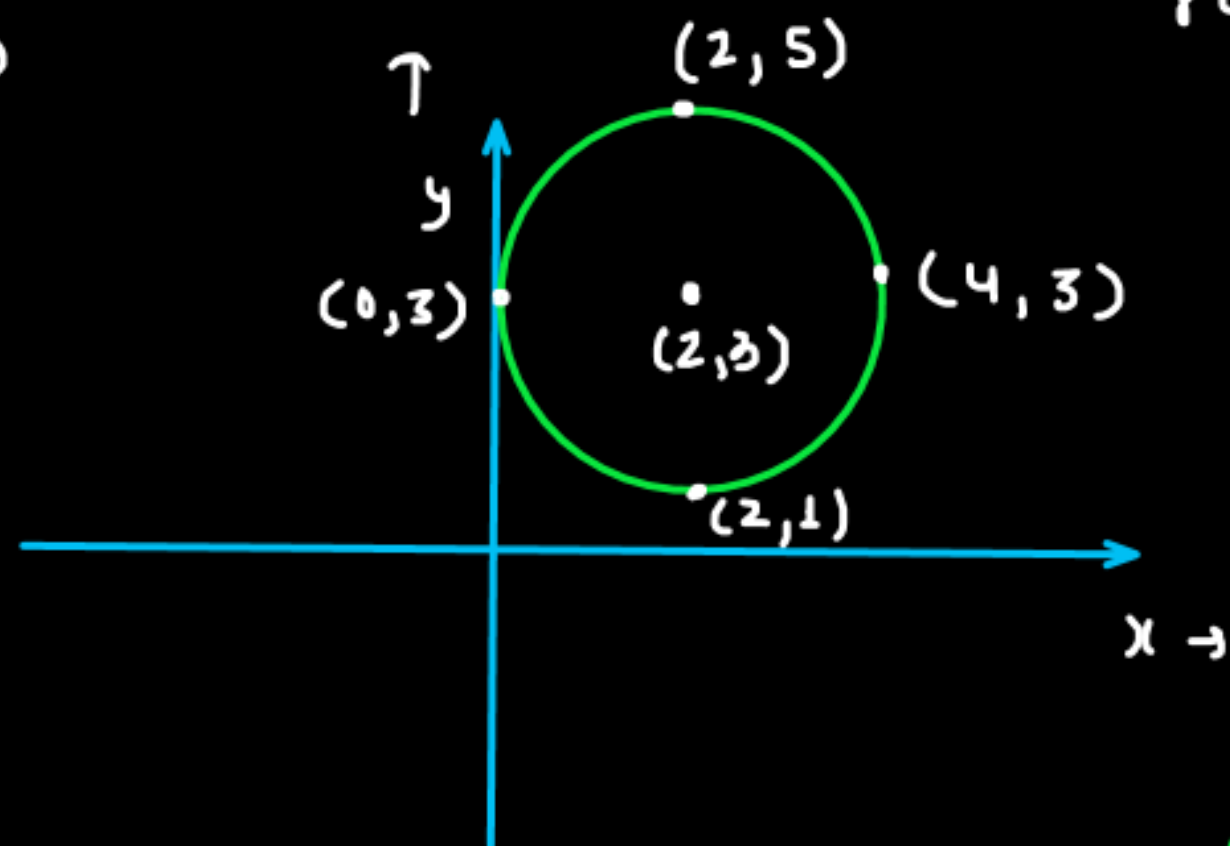
$$|x - a + i(y - b)| = r \quad \hookrightarrow$$

$$(x - a)^2 + (y - b)^2 = r^2 \quad \hookrightarrow \quad c(a, b) \\ \text{radius} = r$$

$$(e) \quad |z - 2 - 3j| = 2$$

$$c(2, 3) ; \text{radius} = 2$$

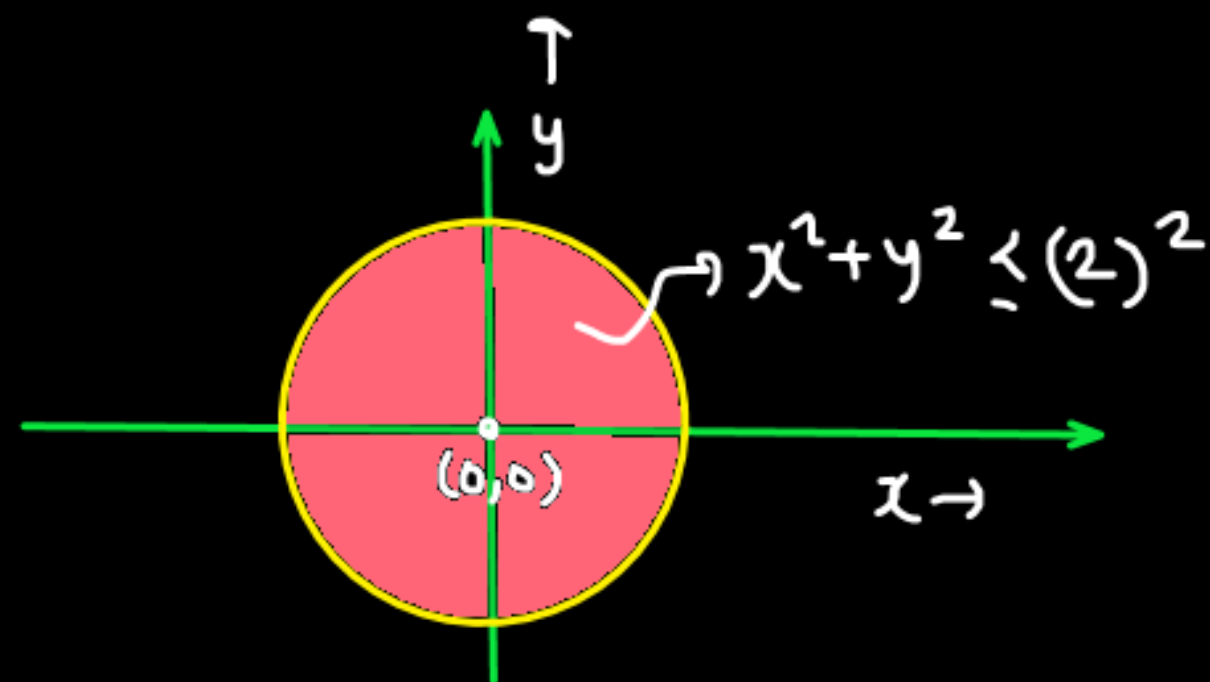
$$(x - 2)^2 + (y - 3)^2 = (2)^2$$



$$(f) \quad |z| \leq 2$$

$$x^2 + y^2 \leq (2)^2 \quad \hookrightarrow \text{disc}$$

$$0 \leq 4$$



$$(g) |z - 1 + 3j| \leq 2$$

$$C(1, -3) ; \text{radius} = 2$$

$$\hookrightarrow (x-1)^2 + (y+3)^2 \leq (2)^2$$

$$\text{at } (1, -3)$$

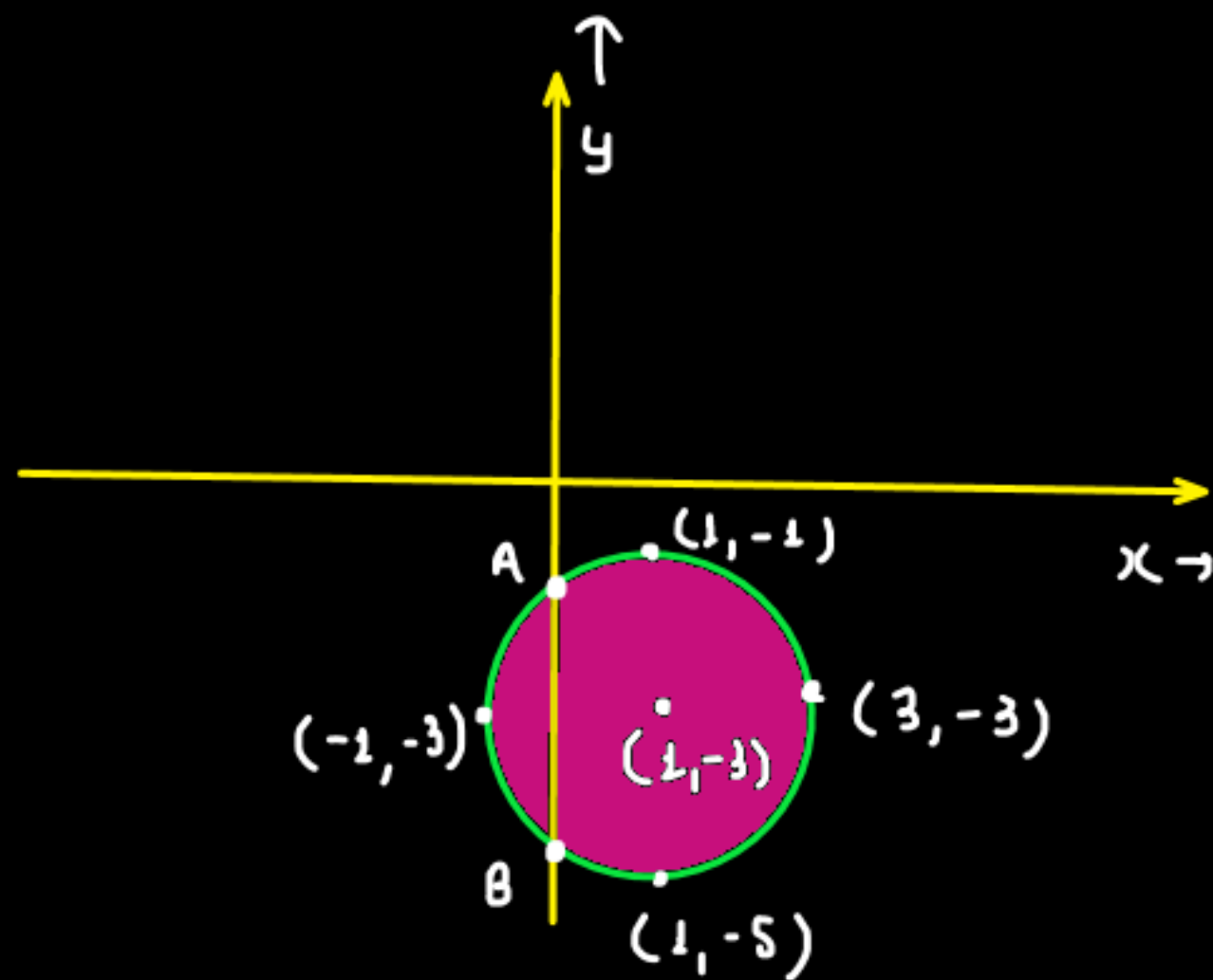
$$\hookrightarrow 0 \leq 4$$

$$(x-1)^2 + (y+3)^2 = (2)^2$$

$$\left(\begin{array}{l} A(0, ?), B(0, ?) \\ \hookrightarrow x=0 \end{array} \right)$$

$$1 + (y+3)^2 = 4$$

$$(y+3)^2 = 3 \quad \hookrightarrow y = -3 \pm \sqrt{3}$$



$$A(0, -3 + \sqrt{3}), B(0, -3 - \sqrt{3})$$



The curve C is given by

$$|z - (2 + 3j)| = 1.$$

Determine whether the following points lie inside the region bounded by the curve C or outside the region:

$\hookrightarrow$ inside

$$Z_1 = 1.5 + 2.5j,$$

$\hookrightarrow$ Inside

$$Z_2 = 2.6 + 3.5j,$$

$\hookrightarrow$ Inside

$$Z_3 = 2 + 2.5j.$$

$\hookrightarrow$ outside

$$Z_4 = 2.9 + 3.5j$$

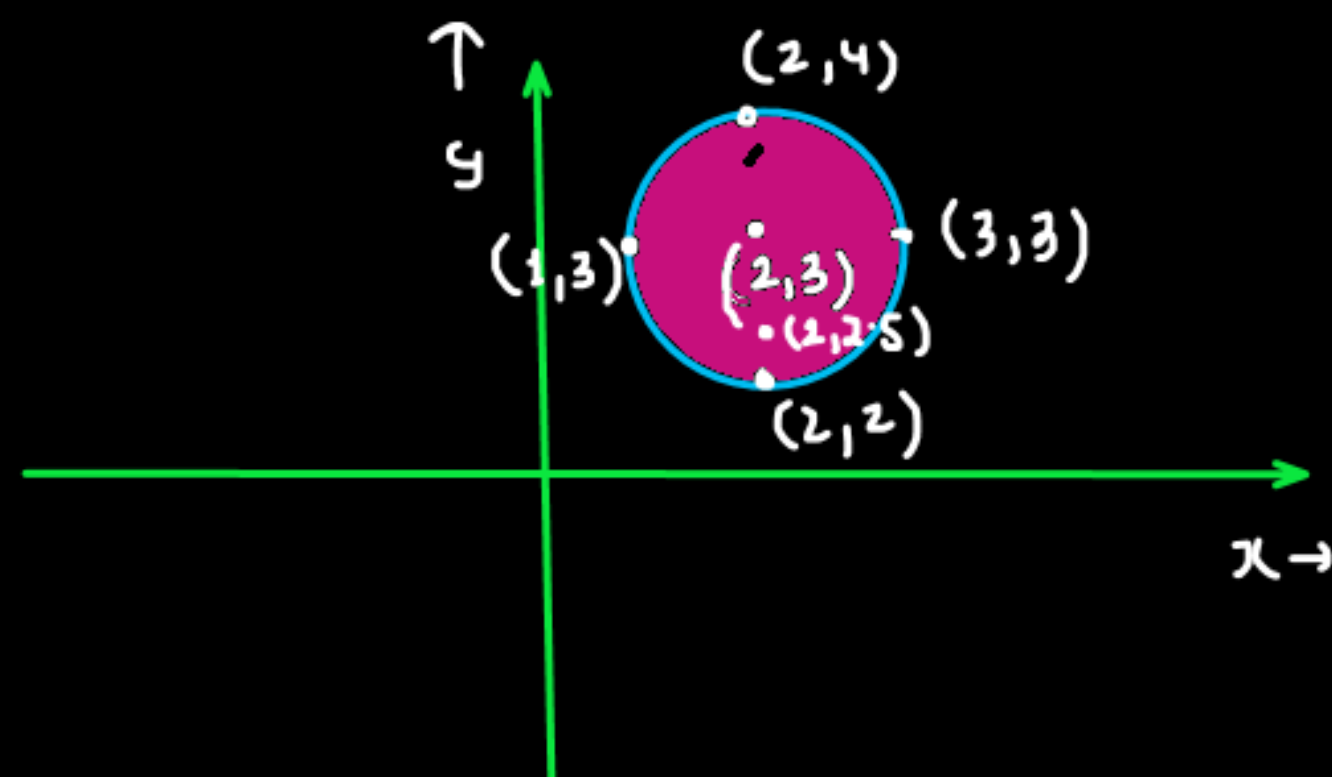
$$(x-2)^2 + (y-3)^2 = (1)^2 \hookrightarrow \text{circle } (2, 3), \text{ radius} = 1$$

$$Z_1 = 1.5 + 2.5j \quad Z_1 (1.5, 2.5)$$

(x_0, y_0) lies inside the circle

$$\text{then}; (x_0 - 2)^2 + (y_0 - 3)^2 \leq 1$$

$$Z_1: (-0.5)^2 + (-0.5)^2 = 0.5 \leq 1$$



$$z_2: (0.6)^2 + (0.5)^2 = 0.36 + 0.25 = 0.61 \leq 1$$

↙
lying inside

$$z_3: (0)^2 + (-0.5)^2 = 0.25 \leq 1$$

$$z_4: (0.9)^2 + (0.5)^2 = 0.81 + 0.25 = 1.06 > 1$$

MCQ
Q.

Question:

Let z be a non-zero complex number satisfying

$$z^3 = \bar{z}.$$

Then z is necessarily a solution of:

☒ (A) $z^2 = 1$ $z = i \Rightarrow i^2 = -1 \neq 1$

☒ (B) $z^3 = 1$ $z = i \Rightarrow i^3 = -i \neq 1$

☒ (C) $z^4 = 1$ $z = i \Rightarrow i^4 = 1 = 1$

☒ (D) $z^9 = 1$ $z = i \Rightarrow i^9 = i^8 \cdot i$
 $= i \neq 1$

$$z = 1, i, -1, -i$$

$$z^4 = 1$$

$$z = re^{i\theta}$$

$$r^3 e^{i3\theta} = r e^{-i\theta}$$

$$r^2 e^{i4\theta} = 1$$

$$r^2 e^{i4\theta} = e^{i(2n\pi)}$$

$$r = 1, \quad 4\theta = 2n\pi$$

$$\theta = n\pi/2$$

$$z = e^{in\pi/2}$$

$$\begin{aligned} n=0 &\Rightarrow z=1; \quad n=1, \Rightarrow z = e^{i\pi/2} = i, \quad n=2 \Rightarrow z=-1 \\ n=3 &\Rightarrow z=-i, \quad n=4 \Rightarrow z=1 \end{aligned}$$

Q. Question

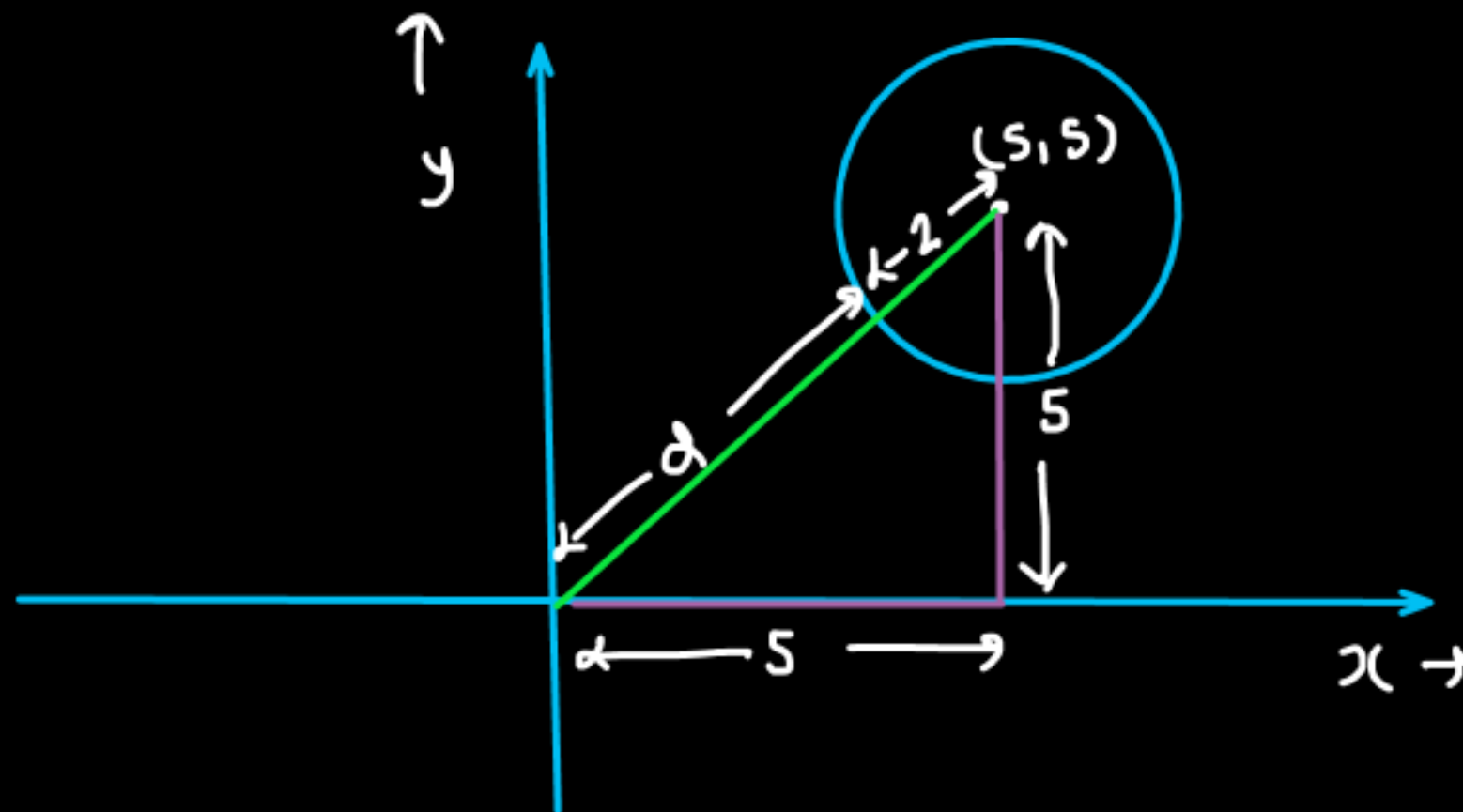
Consider the circle

$$|z - (5 + 5i)| = 2 \rightsquigarrow C(5, 5) \text{ radius} = 2$$

in the complex plane (x, y) , where

$$z = x + iy.$$

The minimum distance from the origin to the circle is:



$$d + 2 = 5\sqrt{2}$$

$$d = 5\sqrt{2} - 2$$

Q. $z = \frac{\sqrt{3} + i}{2}$; Find $z^4 = ?$

M-1 $z^2 = z \cdot z = \left(\frac{\sqrt{3}}{2} + \frac{i}{2}\right)^2$

$$= \frac{3}{4} - \frac{1}{4} + 2\left(\frac{\sqrt{3}}{2}\right)\left(\frac{i}{2}\right)$$

$$= \frac{1}{2} + i\frac{\sqrt{3}}{2}$$

$$z^4 = (z^2)^2 = \left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)^2$$

$$= \frac{1}{4} - \frac{3}{4} + 2\left(\frac{1}{2}\right)\left(i\frac{\sqrt{3}}{2}\right)$$

$$= -\frac{1}{2} + i\frac{\sqrt{3}}{2} \text{ Ans.}$$

M-2 $z = re^{i\theta}$ $z = \frac{\sqrt{3}}{2} + i\frac{1}{2}$

$$r = \sqrt{\frac{3}{4} + \frac{1}{4}} = 1$$

$$\theta = 30^\circ$$

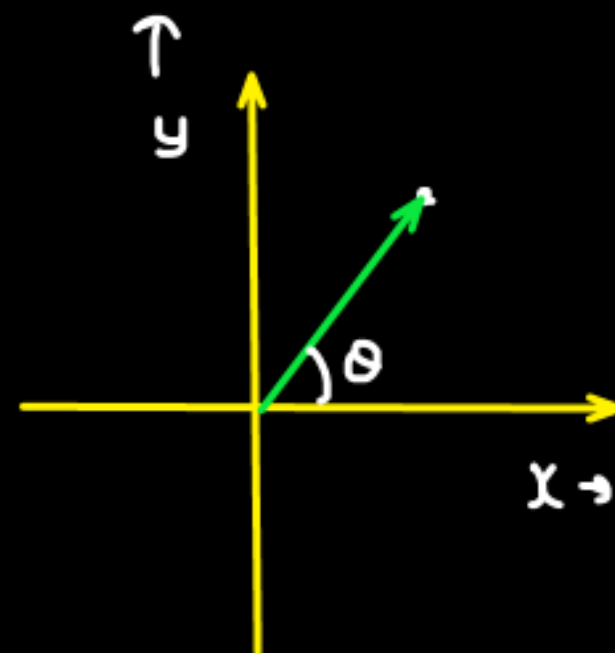
$$z = e^{i30^\circ}$$

$$z^4 = (e^{i30^\circ})^4 = e^{i120^\circ}$$

$$z^4 = \cos 120^\circ + i \sin 120^\circ$$

$$= -\frac{1}{2} + i\frac{\sqrt{3}}{2}$$

Ans. =



$$\tan \theta = \frac{1/2}{\sqrt{3}/2}$$

$$\theta = 30^\circ$$

Q. $Z = \frac{-5+10j}{3+4j}$

Simplify

↳ M-I

$$Z = \frac{-5+10j}{3+4j} \times \left(\frac{3-4j}{3-4j} \right)$$

$$= \frac{-15 + 20j + 30j + 40}{25} = \frac{25 + 50j}{25} = 1 + 2j \text{ Ans.}$$

M-II

$$Z = \frac{\sqrt{125}}{\sqrt{25}} \times \frac{e^{j116.56^\circ}}{e^{j53^\circ}} = \sqrt{5} e^{j63.43^\circ}$$

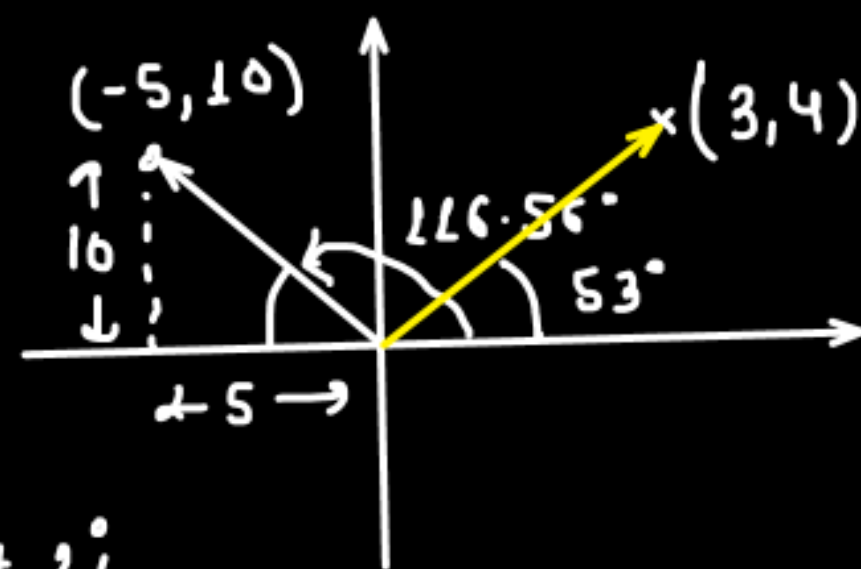
$$= \sqrt{5} [0.447 + j0.89] = 1 + 2j \text{ Ans.}$$

$$Z = \frac{a+ib}{c+id} \times \frac{c-id}{c-id}$$

$$(c+id)(c-id)$$

$$= (c)^2 - (id)^2$$

$$= c^2 + d^2$$



Q. Show; $(\overline{\sin z}) = \sin(\bar{z})$; $(\overline{\cos z}) = \cos(\bar{z})$ → = *
 z : complex No.

$$\hookrightarrow \sin z = \frac{e^{iz} - e^{-iz}}{2j}$$

$$z = x + iy, \quad z^* = x - iy$$

$$\hookrightarrow (\sin z)^* = \left[\frac{e^{iz} - e^{-iz}}{2j} \right]^* = \frac{e^{-iz^*} - e^{iz^*}}{-2j} = \frac{e^{iz^*} - e^{-iz^*}}{2i} = \sin(z^*)$$

$$\hookrightarrow (\cos z)^* = \cos(z^*) \quad \checkmark$$

$$\boxed{[f(z)]^* = f(z^*)}$$

$$\cos z = \cos(x + iy)$$

$$(\cos z)^* = [\cos(x + iy)]^* = \cos(x - iy) = \cos(z^*)$$

MSQ
Q. Let z be a complex number satisfying $\sin z = 10$. ↗ periodic

Which of the following statements is true?

~~(A)~~ No such complex number z exists.

~~(B)~~ $z = \frac{\pi}{2} + 2k\pi \pm i \ln(10 + \sqrt{99})$, $k \in \mathbb{Z}$.

~~(C)~~ $z = \frac{\pi}{2} + 2k\pi \pm i \cosh^{-1}(10)$, $k \in \mathbb{Z}$.

~~(D)~~ The equation has infinitely many complex solutions.

$$\hookrightarrow \sin z = 10$$

$$\sin(x + iy) = 10$$

$$\sin x \cdot \cos(iy) + \cos x \sin(iy) = 10$$

$$\hookrightarrow \sin x \cdot \cosh y + \cos x (i \sinh y) = 10$$

$$\boxed{-1 \leq \sin x \leq 1}$$

$\hookrightarrow x: \text{REAL}$

$$\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$$

$$\cos(iy) = \frac{e^{-y} + e^y}{2}$$

$$\hookrightarrow \cos(iy) = \cosh y$$

$$\hookrightarrow \sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$$

$$\sin(iy) = \frac{e^{-y} - e^y}{2i}$$

$$= i \left[\frac{e^y - e^{-y}}{2} \right]$$

$$= i \sinh y$$

$$\boxed{\frac{1}{i} = -i}$$

$$\frac{1}{i} \times \frac{i}{i} = \frac{i}{i^2} = -i$$

$$\underbrace{\sin x \cdot \cosh y}_{\text{REAL}} + i \underbrace{\cos x \sinh y}_{\text{Imag.}} = \underbrace{10}_{\hookrightarrow \text{REAL}} \Rightarrow \sin x \cdot \cosh y = 10$$

$$\cos x \cdot \sinh y = 0$$

$$\cos x = 0$$



$$x = \pi/2, 3\pi/2, 5\pi/2$$



$$x = \pi/2 + n\pi$$

$$\pi/2, 3\pi/2, \frac{5\pi}{2} \dots$$



$$\sinh y = 0 \rightarrow \text{xx}$$



$$y = 0$$



$$\cosh y = 1$$



$$\sin x = 10$$

↳ NOT possible

$$\cosh y = \frac{e^y + e^{-y}}{2}$$

$$\text{at } y=0$$

$$= \frac{e^0 + e^{-0}}{2}$$

$$\cosh(0) = 1$$

$$\sinh y = \frac{e^y - e^{-y}}{2}$$

$$\text{when } \sinh y = 0$$

$$\hookrightarrow y = 0$$

$$\frac{e^y - e^{-y}}{2} = 0$$

$$e^y = e^{-y}$$

$$y = -y$$

$$\underbrace{y = 0}$$

$$\sin x \cdot \cosh y = 10$$

if $\sin x = -1 \rightarrow$ NOT possible

then $\cosh y = -10 \rightarrow$ NOT possible

$$\frac{e^y + e^{-y}}{2} = -10 \quad \{e^{(\cdot)} > 0\}$$

$$\boxed{\cosh y > 0}$$

if $\sin x = 1$

$\hookrightarrow \cosh y = 10 \rightarrow$ possible =

$$\boxed{y = \pm \cosh^{-1}(10)}$$

$$x = \pi/2 \Rightarrow \sin x = 1 \checkmark$$

$$x \neq x = 3\pi/2 \Rightarrow \sin x = -1 \checkmark$$

$$x = 5\pi/2 \Rightarrow \sin x = 1$$

$$x \neq x = 7\pi/2 \Rightarrow \sin x = -1$$

$$\sin x = \underbrace{1 \text{ OR } -1}$$

$$x = \pi/2, 5\pi/2, \dots$$

$$\boxed{x = \pi/2 \pm 2n\pi} \checkmark$$

$$\downarrow \cosh(y) = \cosh(-y) \leftarrow \text{Even}$$

$$\cosh y = \frac{e^y + e^{-y}}{2}$$

$$\cosh(-y) = \frac{e^{-y} + e^y}{2}$$

$$z = x + iy = \pi/2 \pm 2n\pi \pm i \cosh^{-1}(10)$$

$$n \in \mathbb{Z}$$

$$n = \dots -2, -1, 0, 1, 2 \dots$$

$$\sinh z = 10$$

$$\frac{e^{iz} - e^{-iz}}{2i} = 10$$

$$z \rightarrow z + 2n\pi$$

$$\frac{e^{i(z+2n\pi)} - e^{-i(z+2n\pi)}}{2i}$$

$$= \sinh z$$

$$\sinh(z + 2n\pi) = \sinh z = 10$$

$$\text{at } z = \alpha$$

$$\begin{aligned} z = \alpha + 2\pi \checkmark & \quad \alpha + 4\pi \checkmark \\ = \alpha - 2\pi \checkmark & \quad \alpha - 4\pi \checkmark \dots \end{aligned}$$

$$Q. \sinh z = 10$$

$$z = x_0 + iy_0 ; y_0 = ?$$

$$\rightarrow \cosh^{-1}(10) = 2.99 \checkmark$$

(NAT)

4 NO NEED TO REMEMBER:-

$$4 \cosh^{-1}(a) = \ln(a + \sqrt{a^2 - 1})$$

$$4 \cosh^{-1}(10) = \ln(10 + \sqrt{99})$$

M-2 $\sin z = 10$; $e^{iz} = \alpha$

$$\frac{e^{iz} - e^{-iz}}{2i} = 10$$

$$\alpha - 1/\alpha = 20i$$

$$\alpha^2 - 1 = 20i\alpha$$

$$\alpha^2 - 20i\alpha - 1 = 0$$

$$\alpha = \frac{20i \pm \sqrt{-400 + 4}}{2} \Rightarrow \alpha = 10i \pm \sqrt{99}i$$

$$e^{iz} = 10i \pm \sqrt{99}i$$

$$e^{i(z+2n\pi)} = (10 \pm \sqrt{99})e^{i\pi/2}$$

$$i(z+2n\pi) = \ln[(10 \pm \sqrt{99})e^{i\pi/2}]$$

$$i(z + 2n\pi) = \ln[(10 \pm \sqrt{99})e^{i\pi/2}]$$

$$= \ln(10 \pm \sqrt{99}) + \ln(e^{i\pi/2})$$

$$z + 2n\pi = \frac{i\pi/2}{i} + \frac{\ln(10 \pm \sqrt{99})}{i}$$

$$z = \pi/2 - 2n\pi - i \ln(10 \pm \sqrt{99})$$

$$\ln(ab) = \ln a + \ln b$$

$$\hookrightarrow \ln(e^a) = a$$

$$n \in \mathbb{I}$$

$$= \pi/2 + 2k\pi + i \underbrace{[-\ln(10 \pm \sqrt{99})]}$$

$$\swarrow$$

$$-\ln(10 + \sqrt{99})$$

$$-2.99$$

$$\searrow$$

$$-\ln(10 - \sqrt{99})$$

$$2.99$$

$$\Rightarrow z = \pi/2 + 2k\pi + i(\pm 2.99)$$

$$= \pi/2 + 2k\pi \pm i \underbrace{(2.99)}_{\ln(10 + \sqrt{99})}$$

Q. $z = x + iy$; $|e^{iz}| = ?$

$$e^{iz} = e^{i(x+iy)}$$

$$= e^{(ix + i^2 y)}$$

$$= e^{(ix - y)}$$

$$e^{iz} = e^{-y} e^{ix} = r e^{i\theta}$$

$$|e^{iz}| = \underbrace{e^{-y}}_{\downarrow \text{positive}}$$

Always positive

$\text{Ans} = e^{-y}$

Q. $z = (3 - 2i)(3 + 4i) = ?$

$$= 9 + 12i - 6i + 8 = 17 + 6i$$

Q. $I = \int_0^{\pi/2} \frac{\cos x + i \sin x}{\cos x - i \sin x} \cdot dx$

$$I = \int_0^{\pi/2} \frac{e^{ix}}{e^{-ix}} \cdot dx = \int_0^{\pi/2} e^{2ix} \cdot dx$$

$$= \left[\frac{e^{2ix}}{2i} \right]_0^{\pi/2} = \frac{e^{i\pi} - e^{2i0}}{2i} = \frac{-1 - 1}{2i}$$

$$= i \text{ Ans.}$$

Q. Question:

Let ω be a complex number satisfying $\omega^4 = 16j$.

Which of the following cannot be the value of ω ?

✓ (A) $2e^{j\frac{2\pi}{8}}$

(B) $2e^{j\frac{\pi}{8}}$

(C) $2e^{j\frac{5\pi}{8}}$

(D) $2e^{j\frac{9\pi}{8}}$

$$\omega = re^{j\theta}$$
$$\omega^4 = r^4 e^{j4\theta}$$
$$16j = 16 e^{j(\pi/2 + 2n\pi)}$$
$$r^4 e^{j4\theta} = 16 e^{j(\pi/2 + 2n\pi)}$$

$$r^4 = 16$$

$$\boxed{r = 2}$$

$$4\theta = \pi/2 + 2n\pi$$

$$\boxed{\theta = \frac{\pi}{8} + \frac{n\pi}{2}}$$

✓

$$\omega^4 = 16j$$

$$(a) (2e^{j2\pi/8})^4 = 16 e^{j\pi}$$
$$= -16$$

$$(b) (2e^{j\pi/8})^4 = 16 e^{j\pi/2}$$

$$= 16j$$

$$(c) (2e^{j5\pi/8})^4 = 16 e^{j5\pi/2}$$

$$= 16 e^{j(2\pi + \pi/2)}$$

$$= 16 e^{j\pi/2} = 16j$$

$$(d) (2e^{j9\pi/8})^4 = 16 (e^{j9\pi/2})$$
$$= 16j$$

Q. $f(z) = \frac{2z+4}{5z+d}$; $f(z_1) = f(z_2)$; $z_1 \neq z_2$ find $d = ?$ (NAT)

↳ $f(z_1) = f(z_2)$

$$\frac{2z_1+4}{5z_1+d} = \frac{2z_2+4}{5z_2+d}$$

$$10z_1/z_2 + 2dz_1 + 20z_2 + 4/d = 10z_1/z_2 + 20z_1 + 2dz_2 + 4/d$$

$$2dz_1 + 20z_2 = 20z_1 + 2dz_2$$

$$2dz_1 - 2dz_2 = 20z_1 - 20z_2$$

$$2d(z_1/z_2) = 20(z_1/z_2)$$

$$d = 10$$

$$\underline{\underline{n-2}} \quad f(z_1) = f(z_2) \quad ; \quad f(z) = \frac{2z+4}{5z+d}$$

$$\text{let } z_1 = 0, \quad z_2 = 1$$

$$\frac{4}{d} = \frac{6}{5+d}$$

$$20 + 4d = 6d$$

$$\boxed{d=10}$$

$$f(z) = \frac{2z+4}{5z+10} = \frac{2(z+2)}{5(z+2)} = \frac{2}{5}$$

$$f(z) = \frac{2}{5}$$

$$z = x + iy$$

$$x=0, \quad y=0$$

$$z=0$$

$$x=1, \quad y=0$$

$$z=1$$

$$\hookrightarrow f(x) = \frac{2x}{2x}$$

$$f(x_1) = f(x_2)$$

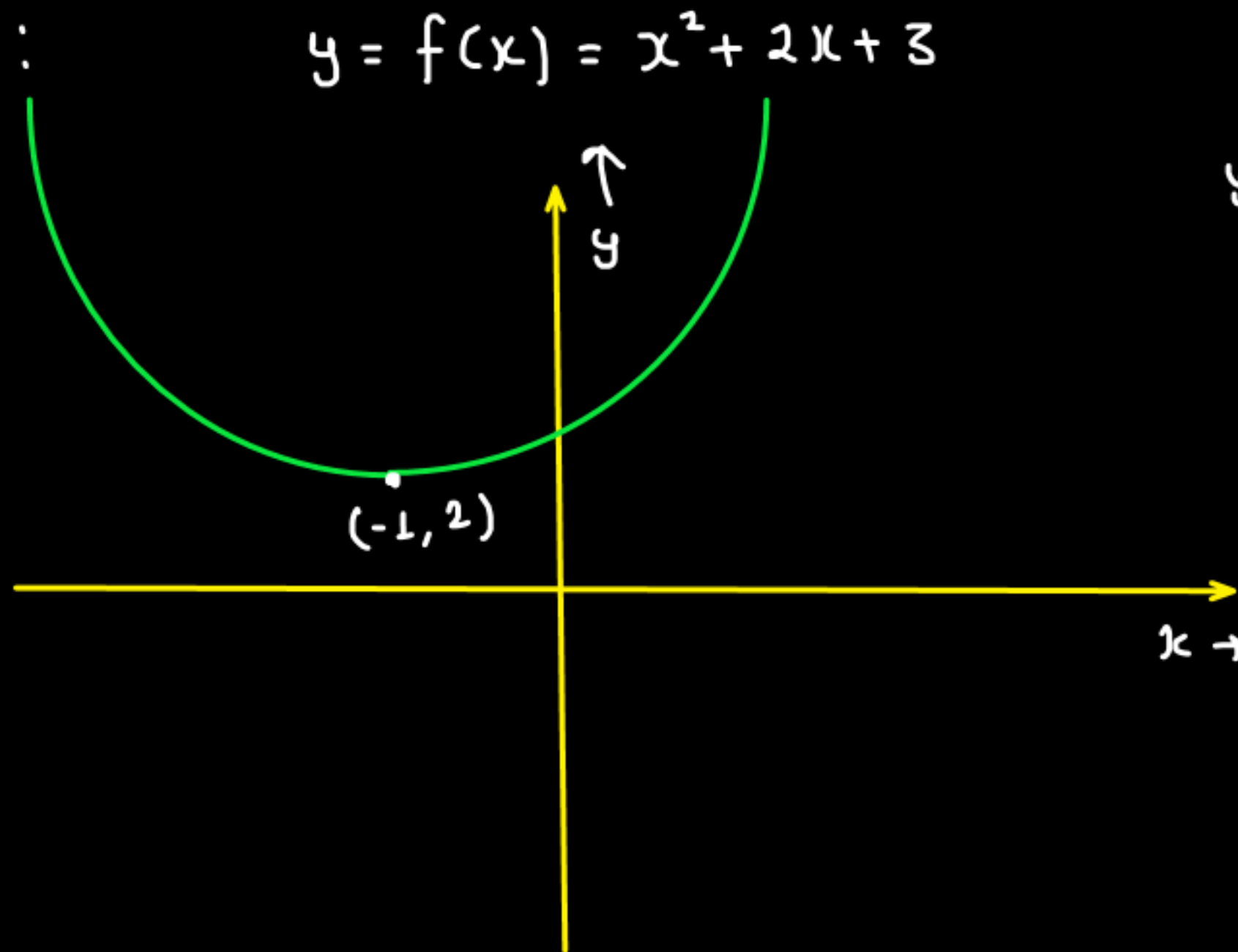
$f(x) =$ can't not be a f^n
of x

Q. $z = 2e^{i\pi/3}$; Real part of z ____ . 2X

$$\begin{aligned} z &= 2e^{i\pi/3} = 2 \left[\cos \pi/3 + i \sin \pi/3 \right] \\ &= 2 \cos \pi/3 + 2i \sin \pi/3 \end{aligned}$$

$$\operatorname{Re}(z) = 2 \cos \pi/3 = 2 \times 1/2 = \boxed{1} \text{ Ans.}$$

* Function of Real Variable:-



x : Independent
Real variable

y : dependent real variable

$\rightarrow$ 2-D plane

* Functions of complex variables:-

$$w = f(z) = z^2$$

$$w = (x+iy)^2 = x^2 - y^2 + 2ixy$$

$$w = \underbrace{(x^2 - y^2)} + i \underbrace{(2xy)}$$

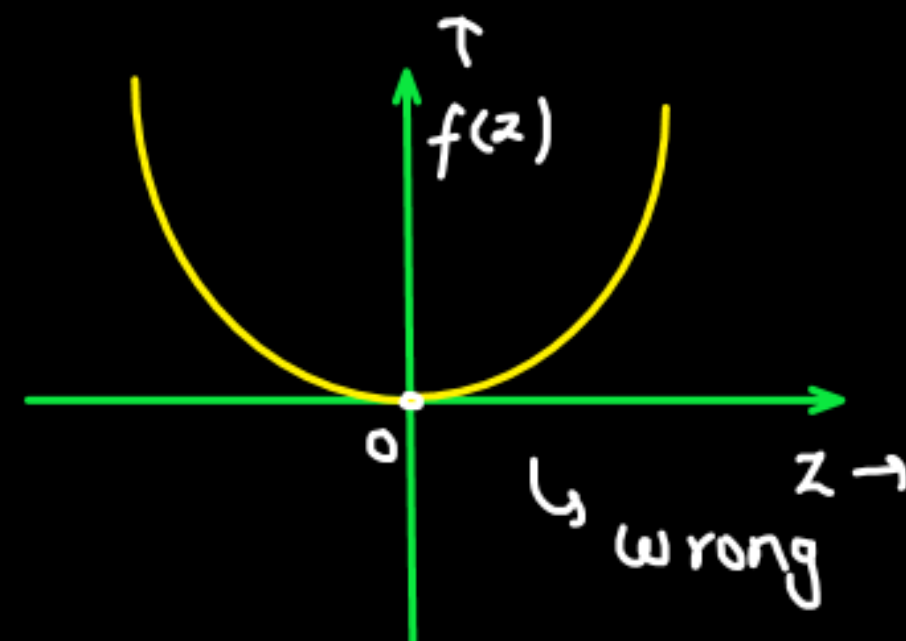
$$\boxed{w = u + iv} = (x^2 - y^2) + i(2xy)$$

$$u = x^2 - y^2 \quad ; \quad v = 2xy$$

$$z = 1+i \quad ; \quad f(z) = (1+i)^2 = 1 - 1 + 2i = 2i$$

$$z = -2+4i \quad ; \quad f(z) = (-2+4i)^2 = 4 - 16 - 16i = -12 - 16i$$

$$z = 2+i \quad ; \quad u = 3, \quad v = 4 \quad f(z) = 3 + i4$$



$$z = 1+i$$

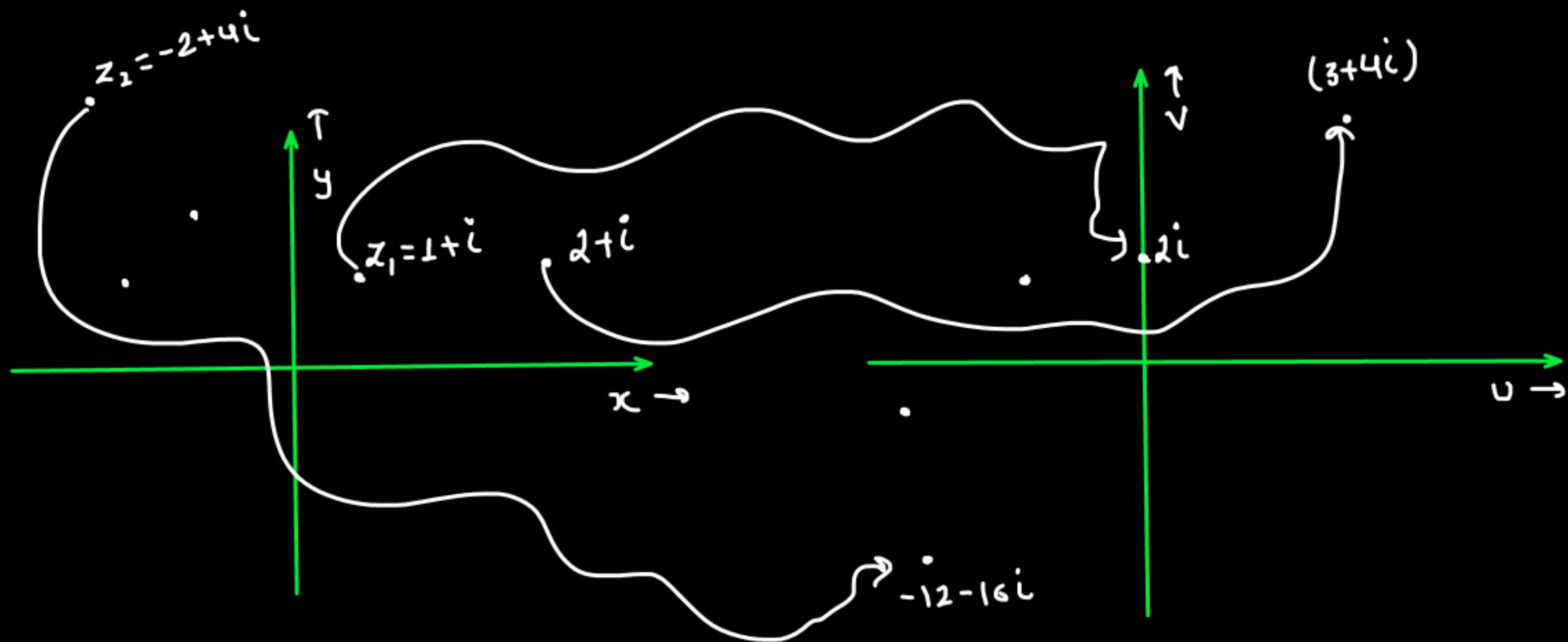
$$z = -2+4i$$

$$\boxed{z = x + iy}$$

$$, u = 0, \quad v = 2$$

$$; u = -12, \quad v = -16$$

$$w = u + iv$$



z -plane

$f(z)$ -plane

w -plane

mapping from z -plane to w -plane

Question:

Let $\omega = f(z) = z + 3$, where $z = x + iy$ and $\omega = u + iv$. If the line $x + 2y = 4$ is given in the z -plane, then the image of the line in the ω -plane is:

- ✓ (A) $u + 2v = 7$
(B) $u + 2v = 4$
(C) $u - 2v = 7$
(D) $2u + v = 7$

4 $\omega = z + 3$

$$z = \omega - 3$$

$$z = \omega - 3$$

$$x + iy = u + iv - 3$$

$$x + iy = u - 3 + i(v)$$

$$x = u - 3, \quad y = v$$

Relatⁿ $x + 2y = 4$

$$u - 3 + 2v = 4$$

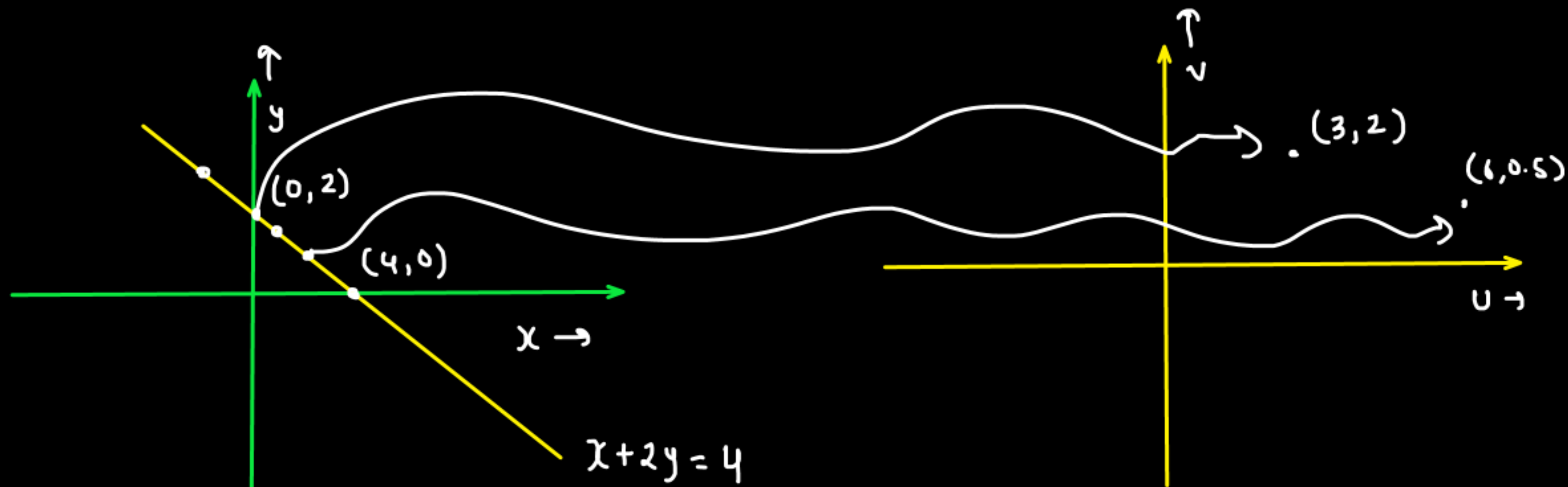
$$u + 2v = 7$$

$$z = g(\omega)$$

$$f(z) = z + 3$$

$$z = 0 + 2i; \quad f(z) = 2i + 3 = 3 + 2i \quad , \quad z = 3 + i0.5; \quad f(z) = 6 + i0.5$$

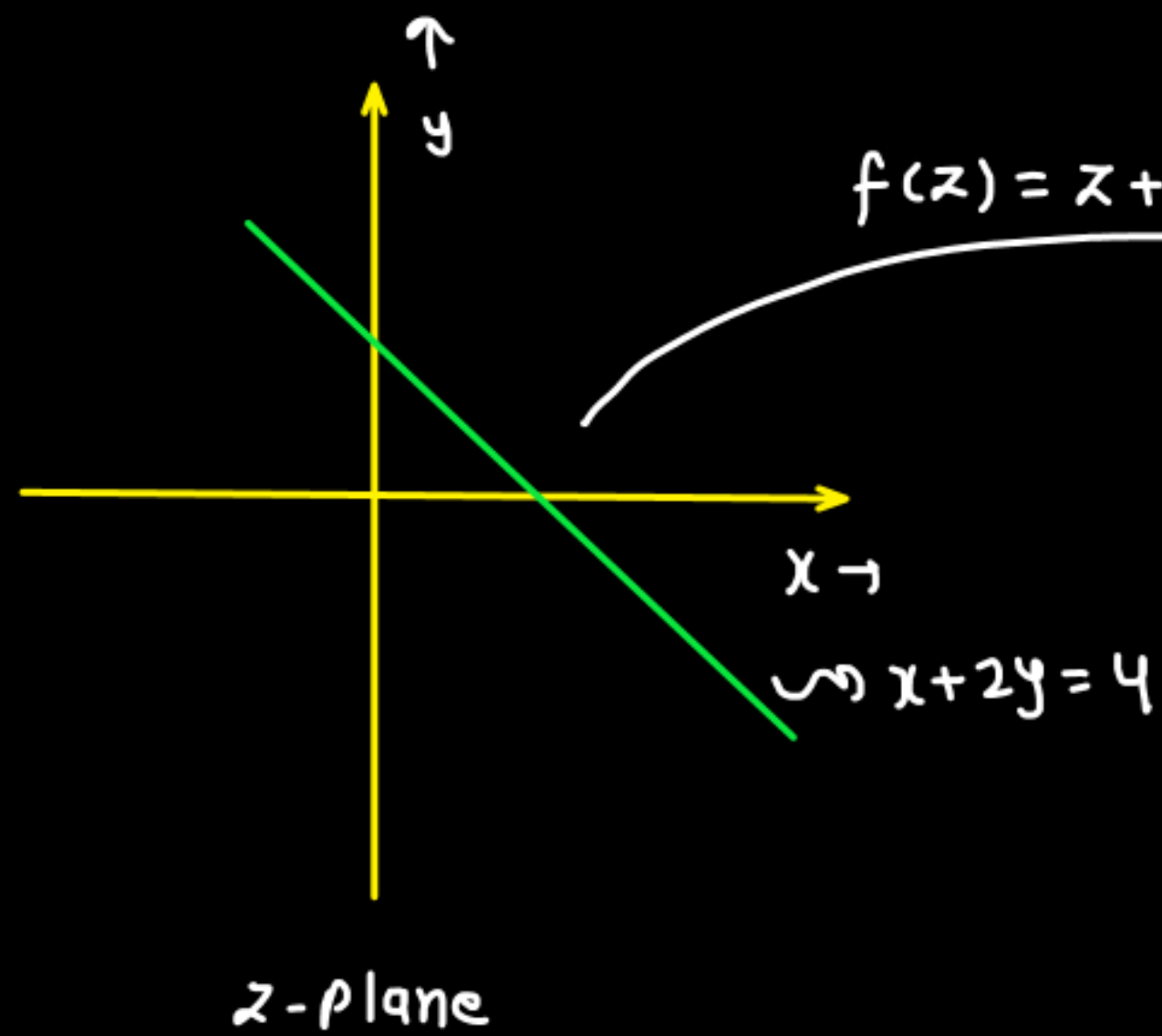
$$u = 3, \quad v = 2$$



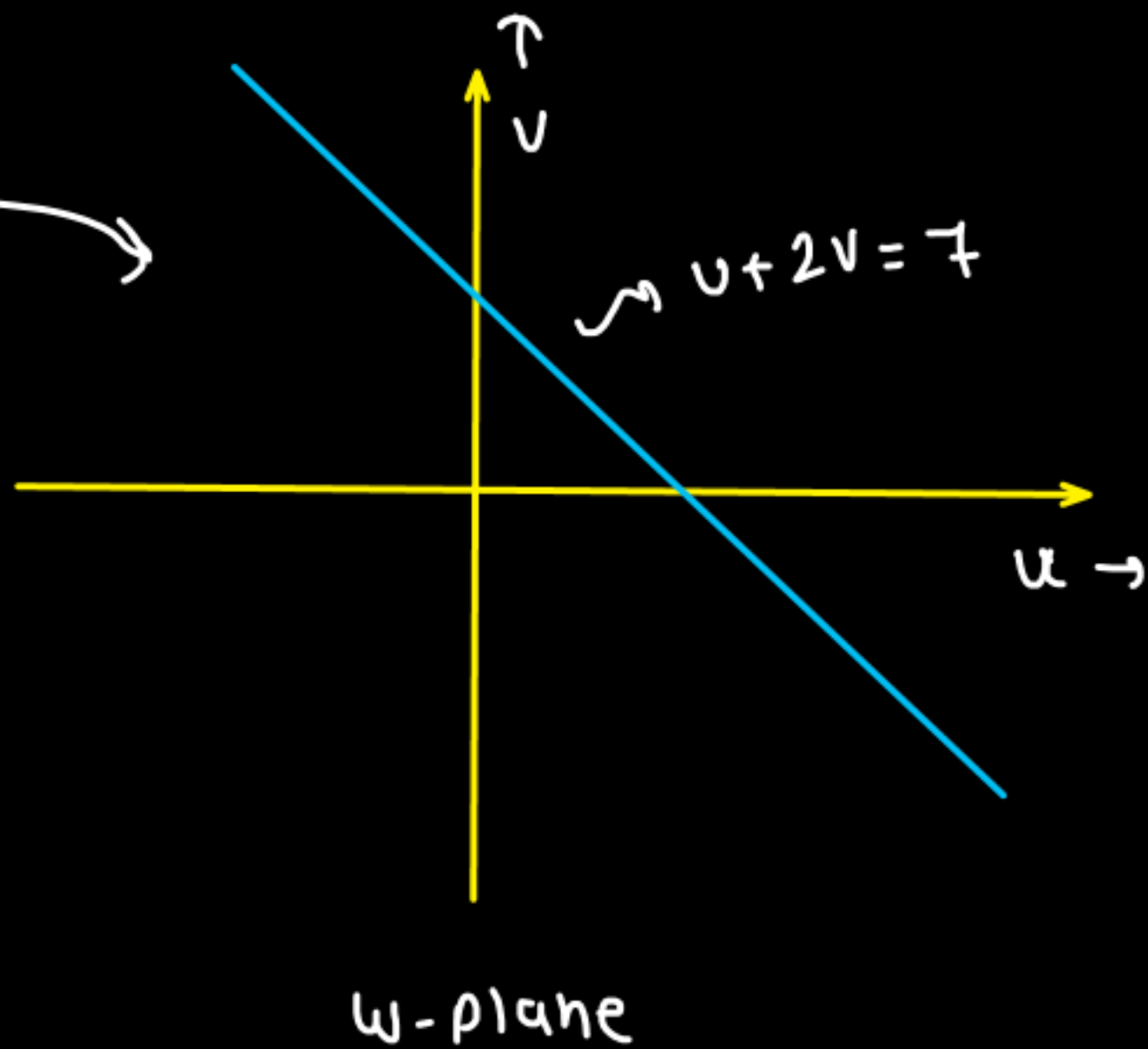
z-plane

w-plane

$$z = 0 + \checkmark 2i, \quad 4 + \checkmark 0i, \quad 3 + \checkmark i0.5, \quad 1 + \checkmark i3/2, \quad -1 + \checkmark i5/2$$



$$f(z) = z + 3$$



Q.

Question:

Consider the complex transformation $w = f(z) = ze^{i\pi/4}$. Let R be the region in the z -plane bounded by the lines $x = 0$, $y = 0$, and $x + y = 1$. Which of the following best describes the corresponding image region R' in the w -plane?

- (A) A triangular region with vertices at $(0, 0)$, $(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$, and $(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$
- (B) A triangular region with vertices at $(0, 0)$, $(1, 1)$, and $(-1, 1)$
- (C) A rectangular region with vertices at $(0, 0)$, $(\frac{1}{\sqrt{2}}, 0)$, $(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$, and $(0, \frac{1}{\sqrt{2}})$
- (D) A triangular region with vertices at $(0, 0)$, $(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}})$, and $(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$

$$w = ze^{i\pi/4}$$

$$z = we^{-i\pi/4}$$

$$z = g(w)$$

$$x + iy = (u + iv) [\cos \pi/4 - i \sin \pi/4]$$

$$\underline{x + iy} = (u + iv) \left(\frac{1}{\sqrt{2}} - i \frac{1}{\sqrt{2}} \right) = \frac{u}{\sqrt{2}} - i \frac{u}{\sqrt{2}} + \frac{iv}{\sqrt{2}} + \frac{v}{\sqrt{2}} = \frac{u+v}{\sqrt{2}} + i \left(\frac{v-u}{\sqrt{2}} \right)$$

$$x = \frac{u+v}{\sqrt{2}}$$

$$; y = \frac{v-u}{\sqrt{2}}$$

$$x > 0$$

$$\frac{u+v}{\sqrt{2}} > 0$$

$$\Rightarrow u > -v$$

$$y > 0$$

$$\frac{v-u}{\sqrt{2}} > 0$$

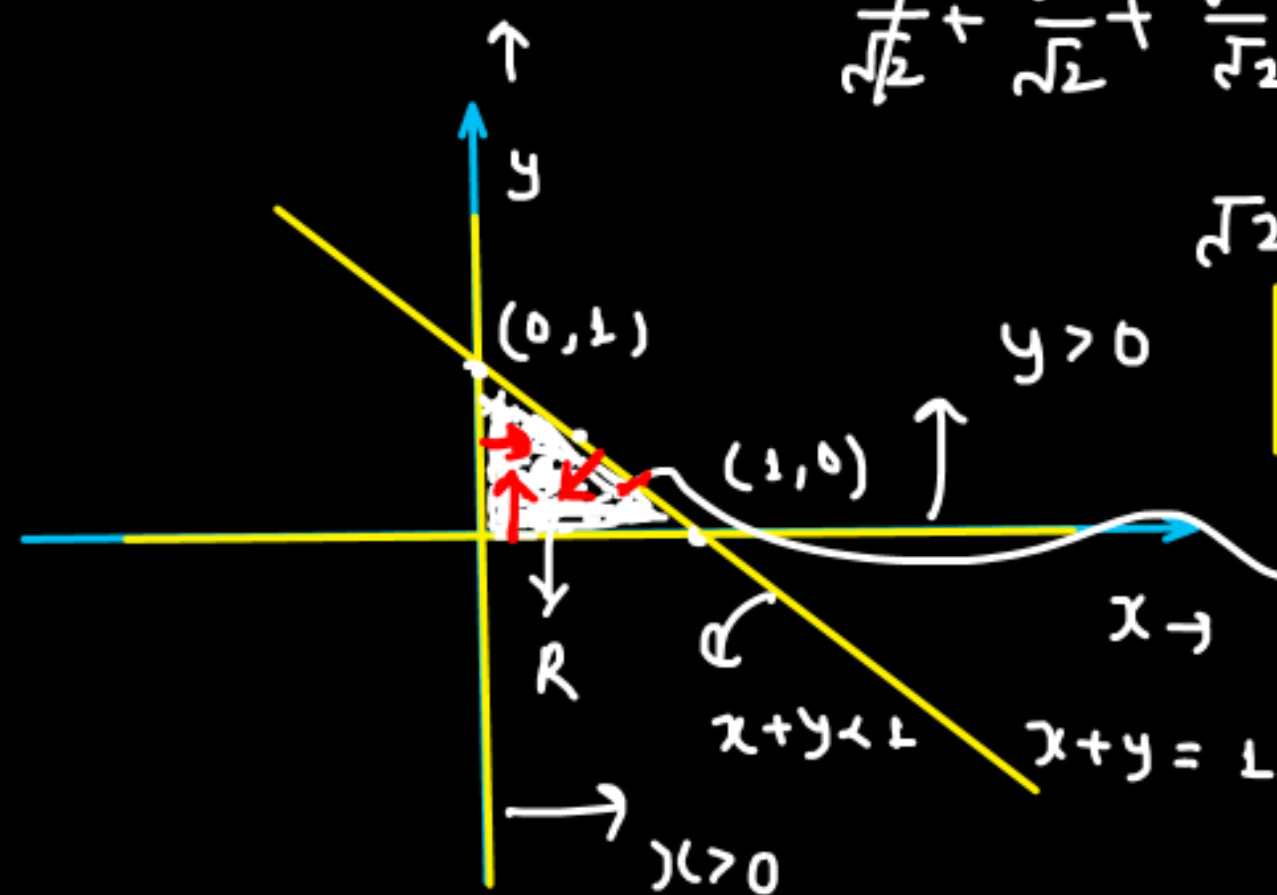
$$v > u$$

$$x+y < 1$$

$$\frac{u}{\sqrt{2}} + \frac{v}{\sqrt{2}} + \frac{v}{\sqrt{2}} - \frac{u}{\sqrt{2}} < 1$$

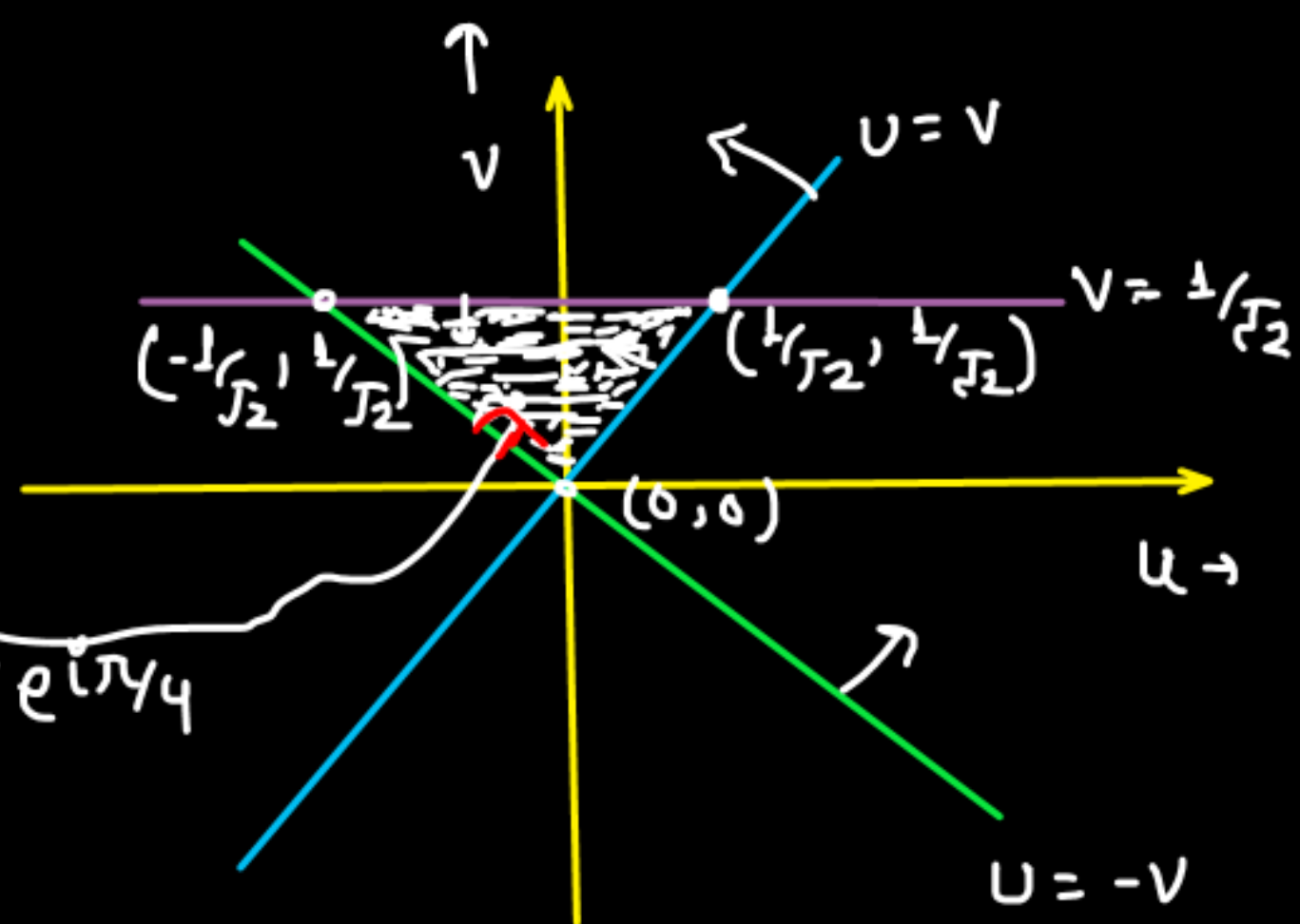
$$\sqrt{2}v < 1$$

$$v < \frac{1}{\sqrt{2}}$$



$$z = 0.5 + 0.4i$$

$$f(z) = z e^{i\pi/4}$$



w -plane

Q.

$$\rightarrow x^2 + (y-3)^2 = (3)^2 \rightarrow c(0,3), r=3$$

Question:

Consider the circle given by the equation $|z - 3i| = 3$ in the complex z -plane. Which of the following describes the image of this circle under the transformation $w = \frac{1}{z}$ in the w -plane (where $w = u + iv$)?

- (A) A circle centered at $(0, -\frac{1}{6})$ with radius $\frac{1}{6}$
- (B) A straight line parallel to the v -axis given by $u = -\frac{1}{6}$
- ☒ (C) A straight line parallel to the u -axis given by $v = -\frac{1}{6}$
- (D) A circle centered at $(0, 3)$ with radius 3

↳

$$f(z) = w = \frac{1}{z}$$

$$z = g(w)$$

$$w = \frac{1}{z}$$

$$z = \frac{1}{w} \rightarrow x + iy = \frac{1}{u + iv} \times \frac{u - iv}{u - iv} = \frac{u - iv}{u^2 + v^2} = \frac{u}{u^2 + v^2} - i \frac{v}{u^2 + v^2}$$

$$x = \frac{u}{u^2 + v^2}$$

$$y = -\frac{v}{u^2 + v^2}$$

$$x^2 + (y-3)^2 = (3)^2$$

$$\left(\frac{u}{u^2+v^2}\right)^2 + \left(\frac{-v}{u^2+v^2} - 3\right)^2 = (3)^2$$

$$\frac{u^2}{(u^2+v^2)^2} + \frac{v^2}{(u^2+v^2)^2} + \cancel{9} + \frac{6v}{u^2+v^2} = \cancel{9}$$

$$\frac{\cancel{u^2+v^2}}{(u^2+v^2)^2} + \frac{6v}{u^2+v^2} = 0$$

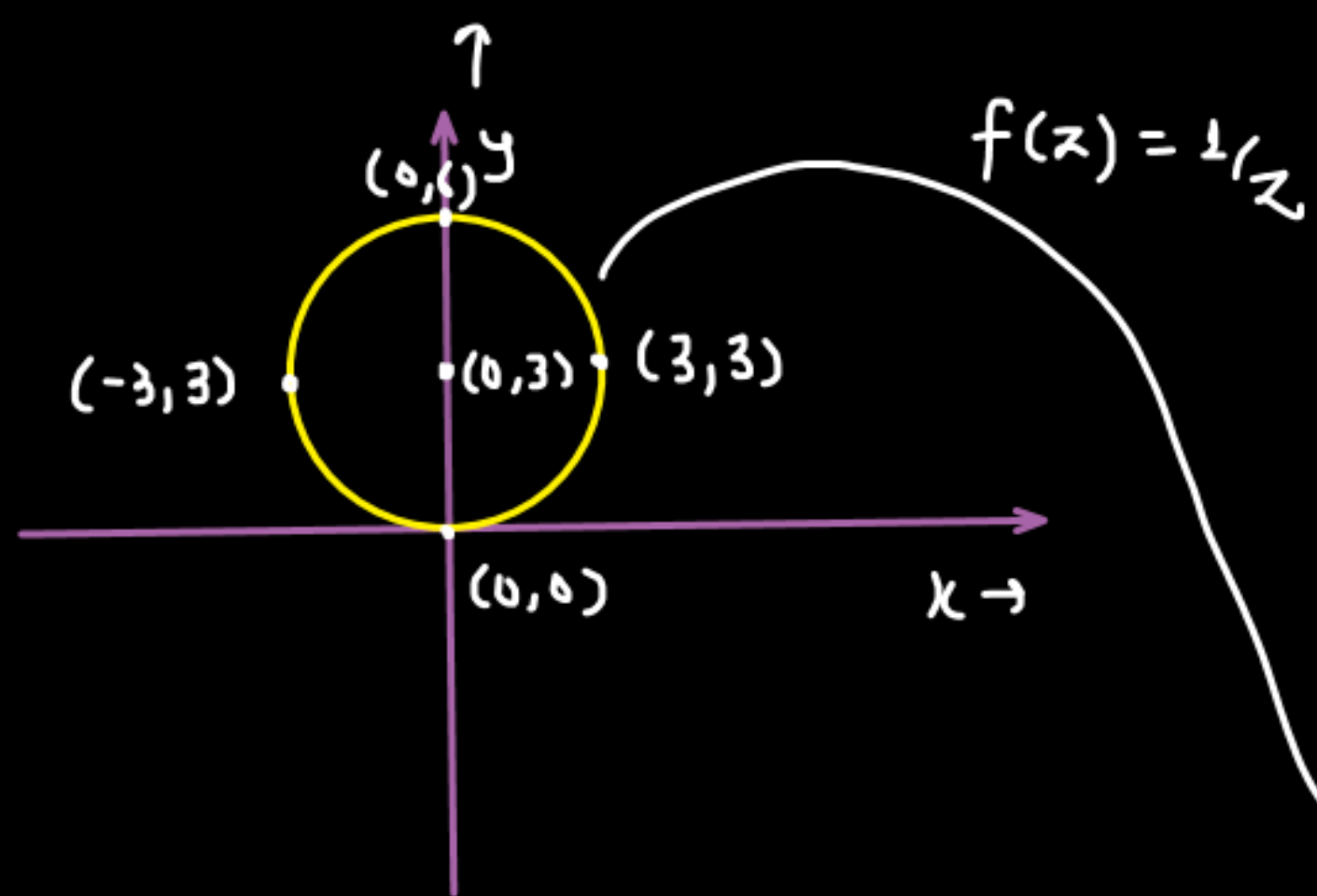
$$\frac{1+6v}{u^2+v^2} = 0$$

 $\Rightarrow$

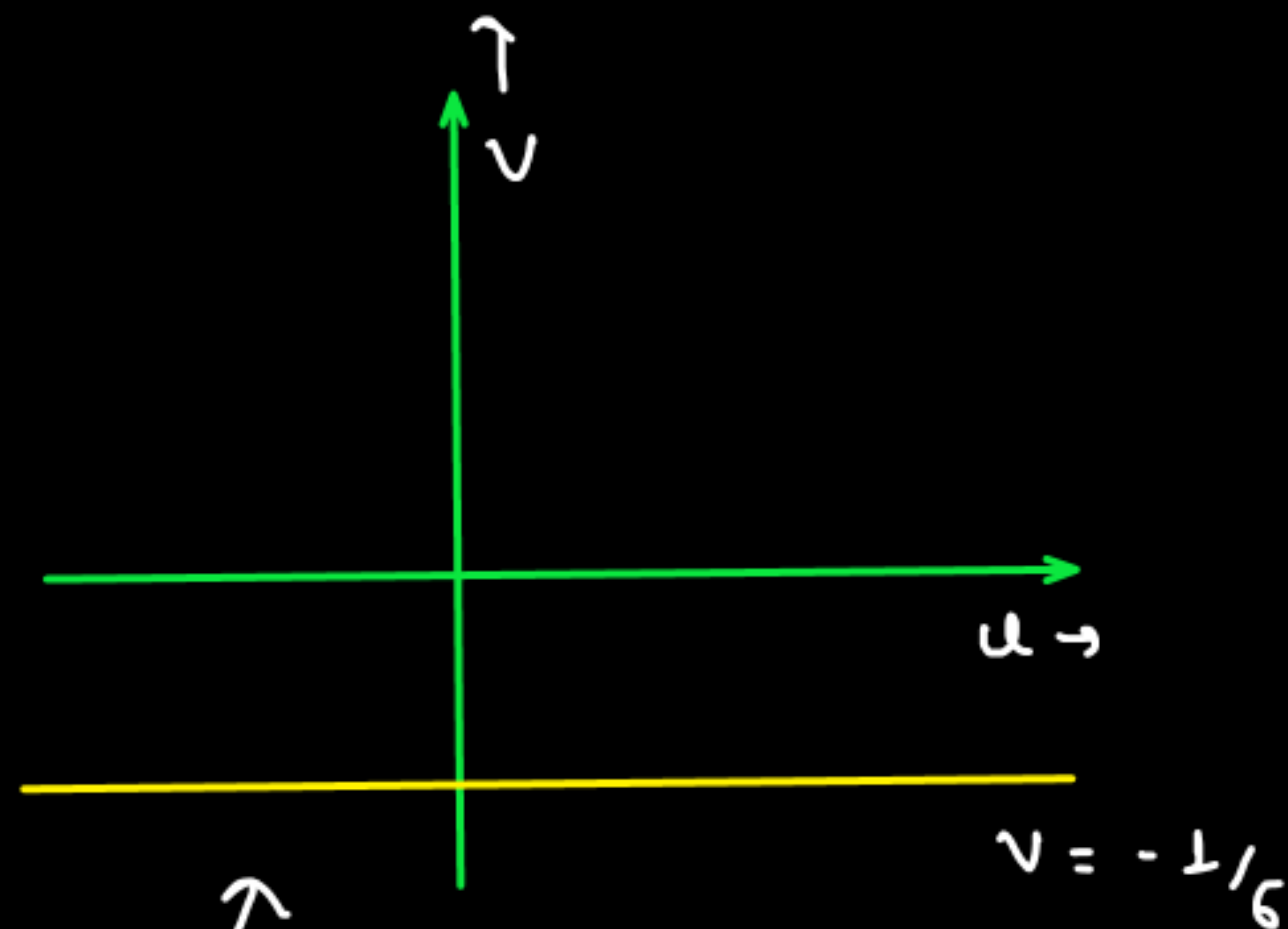
$$1+6v=0$$

$$\Rightarrow v = -1/6$$

↻



$$f(z) = 1/z$$



Q.

Question:

Consider the region R in the complex z -plane defined by the horizontal strip $\frac{1}{4} \leq y \leq \frac{1}{2}$, $-\infty < x < \infty$

Under the transformation $w = \frac{1}{z}$, the region R is mapped onto a region R' in the w -plane. If the area of the region R' is A , then the value of $\frac{A}{\pi}$ is 3.

(Provide an integer answer)

$$w = f(z)$$

$$z = g(w)$$

$$\hookrightarrow w = 1/z$$

$$z = 1/w$$

$$x = \frac{u}{u^2 + v^2}, \quad y = \frac{-v}{u^2 + v^2}$$

$$-\infty \leq \frac{u}{u^2 + v^2} \leq \infty$$

$$\Rightarrow -\infty \leq u \leq \infty$$

$$\frac{1}{4} \leq y \leq \frac{1}{2}$$

$$\frac{1}{4} \leq \frac{-v}{u^2 + v^2} \leq \frac{1}{2}$$

$$\hookrightarrow \frac{-v}{u^2 + v^2} \leq \frac{1}{2}, \quad \frac{-v}{u^2 + v^2} > \frac{1}{4}$$

$$-2v \leq u^2 + v^2$$

$$-4v > u^2 + v^2$$

$$u^2 + v^2 + 2v > 0$$

$$u^2 + v^2 + 2v + (1)^2 > 1^2$$

$$u^2 + (v+1)^2 > 1^2$$

$$\downarrow$$

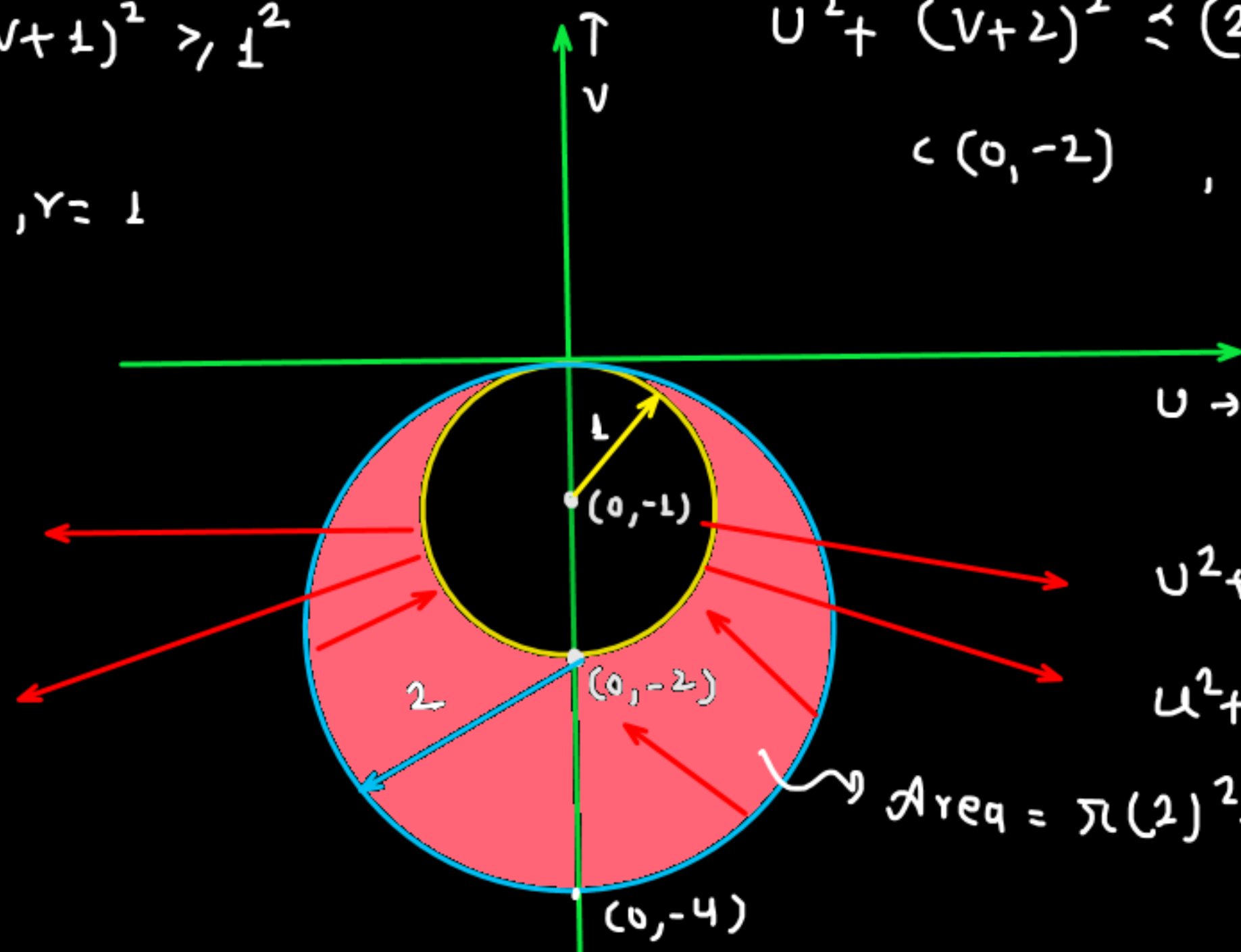
$$C(0, -1), r = 1$$

$$u^2 + v^2 + 4v \leq 0$$

$$u^2 + v^2 + 2(2)v + (2)^2 \leq (2)^2$$

$$u^2 + (v+2)^2 \leq (2)^2$$

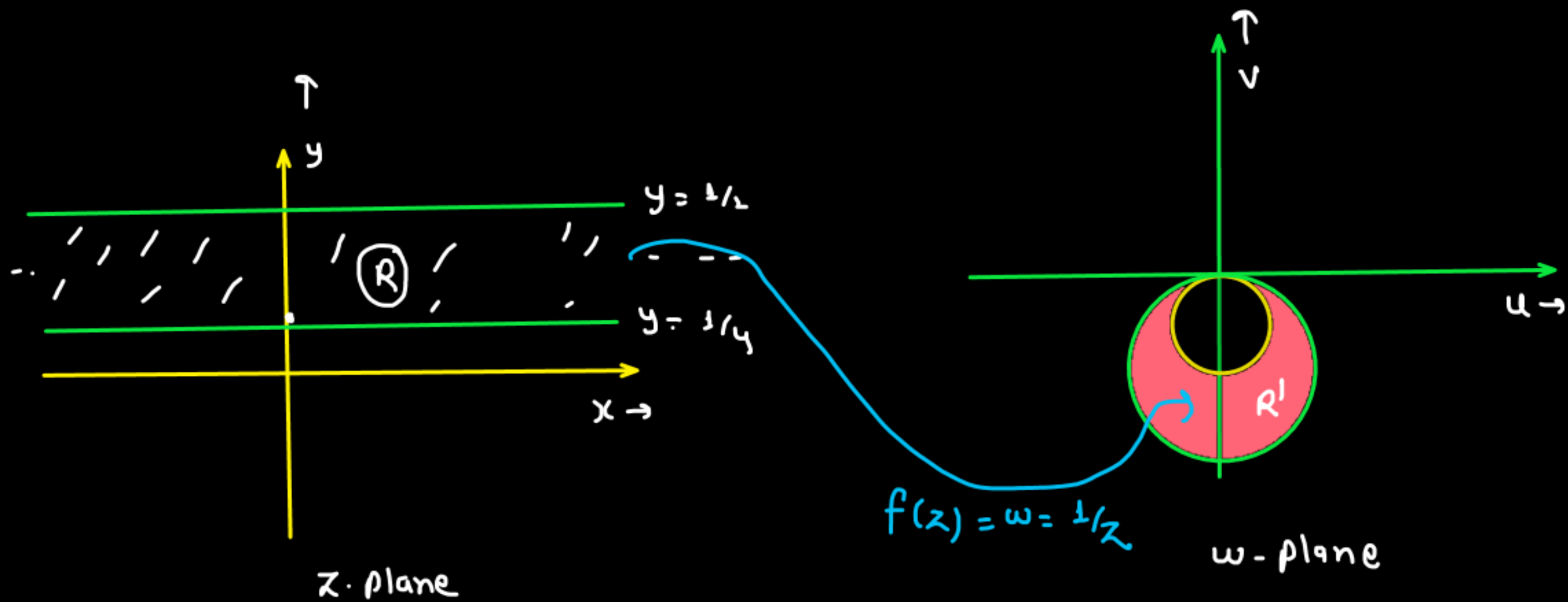
$$C(0, -2), r = 2$$



$$u^2 + (v+1)^2 > 1^2$$

$$u^2 + (v+2)^2 \leq 2^2$$

$$Area = \pi(2)^2 - \pi(1)^2 = 3\pi = Ans.$$



MSQ
Q.

Question:

Let

$$P(z) = z^3 + (1+i)z^2 + (2+i)z + 3,$$

where z is a complex number.

Which of the following statement is **true**?

☒ A. The sum of the roots of $P(z) = 0$ is a real number.

☒ B. All the roots cannot be real.

☒ C. $\overline{P(z)} = P(\bar{z})$ for all z .

☒ D. The complex roots of the equation $P(z) = 0$ come in conjugate pairs.

studied

$$[f(z)]^* = f(z^*)$$

↓

$$[P(z)]^* = (z^*)^3 + (1-i)(z^*)^2 + (2-i)z^* + 3$$

$$P(z^*) = (z^*)^3 + (1+i)(z^*)^2 + (2+i)z^* + 3$$

← NOT same

$$p(z) = z^3 + (1+i)z^2 + (2+i)z + 3 = 0$$

$$\boxed{\text{sum of roots} = -(1+i)} \neq \text{REAL number}$$

$$p(z) = 0 \begin{cases} \rightarrow \alpha + i\beta \\ \rightarrow \alpha - i\beta \end{cases}$$

↓
polynomial in z with real co. efficient

↳ If (0) is correct:-

$$z^3 + (1+i)z^2 + (2+i)z + 3 = 0 \begin{cases} \rightarrow \alpha + i\beta \\ \rightarrow \alpha - i\beta \\ \rightarrow \gamma \end{cases}$$

↓
max^m 3 roots

$$\text{sum of roots} = \alpha + i\beta + \alpha - i\beta + \gamma$$

$$= 2\alpha + \gamma : \text{REAL number} \neq -(1+i)$$

$$ax^2 + bx + c = 0$$

$$\alpha + \beta = -b/a$$

$$\alpha\beta\gamma = c/a$$

$$ax^3 + bx^2 + cx + d = 0$$

$$\alpha + \beta + \gamma = -b/a$$

$$\alpha\beta\gamma = -d/a$$

(A) $z^3 + (1+i)z^2 + (2+i)z + 3 = 0$

$\rightarrow \alpha$
 $\rightarrow \beta$
 $\rightarrow \gamma$

If roots were real $\uparrow$

Sum of roots = $\alpha + \beta + \gamma$: REAL number $\neq -(1+i)$

$\downarrow$
all roots can't be real

* Finding the limit for a single variable f^n :-

$$1. \lim_{x \rightarrow 2} \frac{x^2 + 2}{x + 2} = \frac{6}{4} = \frac{3}{2}$$

$$2. \lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} \xrightarrow{L.H.L.} \lim_{x \rightarrow 2} \frac{2x}{1} = 4$$

$$3. \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$4. \lim_{x \rightarrow 0} \frac{|x|}{x} = \text{D.N.E.} \begin{cases} \lim_{x \rightarrow 0^+} \frac{x}{x} = 1 = \text{R.H.L.} \\ \lim_{x \rightarrow 0^-} \frac{-x}{x} = -1 = \text{L.H.L.} \end{cases}$$

$$\vec{z} = \underbrace{x + iy}_{\text{Two variable}}$$

$$\lim_{x \rightarrow a} f(x) \checkmark$$

$$\lim_{\substack{x \rightarrow a \\ y \rightarrow b}} f(x, y) = ?$$

$\text{R.H.L.} \neq \text{L.H.L.}$
limit DNE

* finding limits for Two variable functions:-

$$z = x + iy$$

Case-1:- $\lim_{\substack{x \rightarrow a \\ y \rightarrow b}} f(x, y) = ?$

$$a \neq 0, b \neq 0 \quad \left\{ \begin{array}{l} x \rightarrow 0 \text{ or } z \rightarrow 0 \\ \underbrace{y \rightarrow 0}_{\text{origin}} \end{array} \right\}$$

Eg:- $\lim_{\substack{x \rightarrow 2 \\ y \rightarrow -1}} \frac{x^2 + y - 2x}{2y^2 + 3} = ?$

$$h_1 = h_2 = \text{limit exist} = -\frac{1}{5}$$

$$L_1 = \lim_{y \rightarrow -1} \left\{ \lim_{x \rightarrow 2} \frac{x^2 + y - 2x}{2y^2 + 3} \right\} = \lim_{y \rightarrow -1} \frac{4 + y - 4}{2y^2 + 3} = \frac{-1}{5}$$

$$h_2 = \lim_{x \rightarrow 2} \left\{ \lim_{y \rightarrow -1} \frac{x^2 + y - 2)(}{2y^2 + 3} \right\} = \lim_{x \rightarrow 2} \frac{x^2 - 1 - 2x}{5} = \frac{4 - 1 - 4}{5} = -\frac{1}{5}$$

Question:

Evaluate the limit of the following two-variable function $f(x, y)$:

$$L = \lim_{(x,y) \rightarrow (1,2)} \frac{xy - 2x - y + 2}{\sqrt{xy - 2x - y + 3} - 1}$$

$$L_1 = \lim_{x \rightarrow 1} \left\{ \lim_{y \rightarrow 2} \frac{xy - 2x - y + 2}{\sqrt{xy - 2x - y + 3} - 1} \right\} = \lim_{x \rightarrow 1} \left\{ \lim_{y \rightarrow 2} \frac{\cancel{x} - \cancel{y} - 1}{\frac{1 \times (\cancel{x} - 1)}{2\sqrt{xy - 2x - y + 3}}} - 0 \right\}$$

$0/0 \rightarrow L \cdot H$ for the limit of y
 x : constant

$$= \lim_{x \rightarrow 1} \left\{ \lim_{y \rightarrow 2} 2\sqrt{xy - 2x - y + 3} \right\} = \lim_{x \rightarrow 1} 2\sqrt{2x - 2x - 2 + 3} = \lim_{x \rightarrow 1} 2$$

$L_1 = 2$

$$L_2 = \lim_{y \rightarrow 2} \left\{ \lim_{x \rightarrow 1} \frac{xy - 2x - y + 2}{\sqrt{xy - 2x - y + 3} - 1} \right\}$$

0/0 form: h.t. w.r.t. x , y constant

$$= \lim_{y \rightarrow 2} \left\{ \lim_{x \rightarrow 1} \frac{\cancel{y}/2}{\frac{1(\cancel{y}/2)}{2\sqrt{xy - 2x - y + 3}} - 0} \right\} = \lim_{y \rightarrow 2} \left\{ \lim_{x \rightarrow 1} 2\sqrt{xy - 2x - y + 3} \right\}$$

$$= \lim_{y \rightarrow 2} 2\sqrt{y - 2 - y + 3}$$

$$= \lim_{y \rightarrow 2} 2 = 2$$

$$\boxed{L_1 = L_2 = 2}$$

Case-2:-

$$\lim_{\substack{x \rightarrow a \\ y \rightarrow b}} f(x, y) = ?$$

$$a = b = 0$$

$$\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} f(x, y) = ?$$

$$\underbrace{h_1, h_2, h_3, h_4}$$

$$\text{eg:- } \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^2 y^2}{x^2 + y^2} = ?$$

$$h_1 = \lim_{x \rightarrow 0} \left\{ \lim_{y \rightarrow 0} \frac{x^2 y^2}{x^2 + y^2} \right\} = \lim_{x \rightarrow 0} \{0\} = 0$$

$$h_2 = \lim_{y \rightarrow 0} \left\{ \lim_{x \rightarrow 0} \frac{x^2 y^2}{x^2 + y^2} \right\} = \lim_{y \rightarrow 0} \{0\} = 0$$

$$h_3: y = mx, \quad x \rightarrow 0, \quad y \rightarrow 0$$

$$h_3 = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^2 (mx)^2}{x^2 + (mx)^2} \Rightarrow h_3 = \lim_{x \rightarrow 0} \frac{m^2 x^4}{x^2 + m^2 x^2} = \lim_{x \rightarrow 0} \frac{m^2 x^4}{(1 + m^2) x^2}$$

$$L_3 = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{m^2 x^2}{1+m^2} = 0 \quad L_3 = 0$$

$$L_4: y = mx^2$$

$$\begin{aligned} L_4 &= \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^2 (mx^2)^2}{x^2 + (mx^2)^2} = \lim_{x \rightarrow 0} \frac{m^2 x^6}{x^2 + m^2 x^4} = \lim_{x \rightarrow 0} \frac{m^2 x^6}{x^2 (1 + m^2 x^2)} \\ &= \lim_{x \rightarrow 0} \frac{m^2 x^4}{1 + m^2 x^2} = 0 \end{aligned}$$

$$\boxed{L_1 = L_2 = L_3 = L_4 = 0} \quad \rightarrow \text{Limit exist} = 0$$

$$\underbrace{L_3 = g(m)}, \quad \underbrace{L_4 = h(m)} \rightarrow \text{Limit doesn't exist}$$

2.

$$L = \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 + y^2}{\sqrt{x^2 + y^2 + 9} - 3}$$

$$h_1 = \lim_{y \rightarrow 0} \left\{ \lim_{x \rightarrow 0} \frac{x^2 + y^2}{\sqrt{x^2 + y^2 + 9} - 3} \right\} = \lim_{y \rightarrow 0} \left\{ \frac{y^2}{\sqrt{y^2 + 9} - 3} \right\}$$

0/0 : L.H.

$$h_2 = \lim_{x \rightarrow 0} \left\{ \lim_{y \rightarrow 0} f(x, y) \right\}$$

$$= \lim_{x \rightarrow 0} \left\{ \frac{x^2}{\sqrt{x^2 + 9} - 3} \right\} = 6$$

$$= \lim_{y \rightarrow 0} \frac{\cancel{y^2}}{\frac{1(2y)}{2\sqrt{y^2 + 9}}} = \lim_{y \rightarrow 0} 2\sqrt{y^2 + 9}$$

$$= 2 \times \sqrt{9} = 6$$

$$h_3: y = mx \rightarrow$$

$$h_3 = \lim_{x \rightarrow 0} \frac{x^2 + m^2 x^2}{\sqrt{x^2 + m^2 x^2 + 9} - 3} = \lim_{x \rightarrow 0} \frac{\cancel{(1+m^2)} 2x}{\frac{1(2x + \cancel{m^2 2x})}{2\sqrt{x^2 + m^2 x^2 + 9}}} =$$

$$L_3 = \lim_{x \rightarrow 0} 2\sqrt{x^2 + m^2 x^4 + g} = 2\sqrt{g} = 6$$

$$h_4: y = mx^2$$

$$h_4 = \lim_{x \rightarrow 0} \frac{x^2 + m^2 x^4}{\sqrt{x^2 + m^2 x^4 + g} - 3}$$

$$= \lim_{x \rightarrow 0} \frac{2x + \cancel{4m^2 x^3}}{\frac{1(2x + \cancel{4m^2 x^3})}{2\sqrt{x^2 + m^2 x^4 + g}}}$$

$$= \lim_{x \rightarrow 0} 2\sqrt{x^2 + m^2 x^4 + g}$$

$$= 6$$

$$h_1 = L_2 = L_3 = L_4 = 6$$

Q. $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x y^2}{x^2 + y^4} = ?$

~~(a)~~ 0 (b) (c) ~~(d)~~ D.N.E.

$L_3 = g(m) = \text{Limit D.N.E.}$

$L_3: y = mx$

$$L_3 = \lim_{x \rightarrow 0} \frac{x m^2 x^2}{x^2 + m^4 x^4} = \lim_{x \rightarrow 0} \frac{m^2 x^3}{x^2 + m^4 x^4} = \lim_{x \rightarrow 0} \frac{m^2 x^3}{x^2 (1 + m^4 x^2)}$$

$L = 0$ Ans

$L_1 = \lim_{x \rightarrow 0} \{0\} = 0$

$L_2 = \lim_{y \rightarrow 0} \{0\} = 0$

$L_4: y = mx^2$

$$\lim_{x \rightarrow 0} \frac{m^2 x^5}{x^2 + m^4 x^8} = \lim_{x \rightarrow 0} \frac{m^2 x^5}{x^2 (1 + m^4 x^6)} = 0$$

$$= \lim_{x \rightarrow 0} \frac{m^2 x}{1 + m^4 x^2} = 0$$

$$Q. \lim_{z \rightarrow 0} \frac{z}{\bar{z}}$$

$$z \rightarrow 0 \quad x+iy \rightarrow 0 \quad \leadsto \quad \begin{matrix} x \rightarrow 0 \\ y \rightarrow 0 \end{matrix}$$

$$= \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x+iy}{x-iy}$$

$$\frac{z}{\bar{z}} = \frac{x+iy}{x-iy}$$

$$h_3: y = mx$$

$$= \lim_{x \rightarrow 0} \frac{x+imx}{x-imx} = \lim_{x \rightarrow 0} \frac{x(1+im)}{x(1-im)} = \lim_{x \rightarrow 0} \frac{1+im}{1-im} = \frac{1+im}{1-im}$$

$$\boxed{h_3 = \frac{1+im}{1-im} = g(m)} : \text{limit DNE} =$$

$$Q. \lim_{z \rightarrow 0} z \bar{z} = \boxed{0}$$

$$z \bar{z} = |z|^2 = x^2 + y^2$$

$$= \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} x^2 + y^2$$

$$h_3: y = mx$$

$$= \lim_{x \rightarrow 0} x^2 + m^2 x^2 = 0$$

$$h_4: y = mx^2$$

$$= \lim_{x \rightarrow 0} x^2 + m^2 x^4 = 0$$

$$h_1 = \lim_{x \rightarrow 0} \left\{ \lim_{y \rightarrow 0} x^2 + y^2 \right\}$$

$$= \lim_{x \rightarrow 0} \{ x^2 \} = 0$$

$$h_2 = \lim_{y \rightarrow 0} \left\{ \lim_{x \rightarrow 0} x^2 + y^2 \right\}$$

$$= \lim_{y \rightarrow 0} \{ y^2 \} = 0$$

$$\boxed{h_1 = h_2 = h_3 = h_4 = 0}$$

Ans.

$$Q. \lim_{z \rightarrow 0} \frac{z}{|z|} \rightarrow \text{DNE}$$

$$= \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x+iy}{\sqrt{x^2+y^2}}$$

$$L_3: y = mx$$

$$L_3 = \lim_{x \rightarrow 0} \frac{x+imx}{\sqrt{x^2+m^2x^2}} = \lim_{x \rightarrow 0} \frac{(1+im)x}{\sqrt{(1+m^2)x^2}} = \lim_{x \rightarrow 0} \frac{1+im}{\sqrt{1+m^2}} \frac{x}{|x|}$$

$$L_3 = \frac{1+im}{\sqrt{1+m^2}} \underbrace{\lim_{x \rightarrow 0} \frac{x}{|x|}}_{\text{DNE.}} \quad \begin{array}{l} \nearrow \lim_{x \rightarrow 0^+} \frac{x}{x} = 1 \\ \searrow \lim_{x \rightarrow 0^-} \frac{x}{-x} = -1 \end{array}$$

$$L_3: \text{DNE}$$

$$Q. \lim_{z \rightarrow i} \frac{z^3 - iz^2 + z - i}{z - i} = ?$$

$$z \rightarrow i$$

$$x + iy \rightarrow 0 + i1$$

$$x \rightarrow 0$$

$$y \rightarrow 1$$

$$\hookrightarrow \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 1}} \frac{(x + iy)^3 - i(x + iy)^2 + (x + iy) - i}{x + iy - i}$$

$$L_1 = \lim_{y \rightarrow 1} \left\{ \lim_{x \rightarrow 0} f(x, y) \right\} = \lim_{y \rightarrow 1} \frac{(iy)^3 - i(iy)^2 + iy - i}{iy - i} = 0/0$$

$$= \lim_{y \rightarrow 1} \frac{(i)^3 (3y^2) - i3(2y) + i}{i}$$

$$= \frac{3(i)^3 - 2i^3 + i}{i} = \frac{i^3 + i}{i} = \frac{-i + i}{i} = \boxed{0} \text{ Ans.}$$

$$\begin{aligned}
 \hookrightarrow \lim_{z \rightarrow i} \frac{z^3 - iz^2 + z - i}{z - i} &= \lim_{z \rightarrow i} \frac{z^2(z-i) + 1(z-i)}{(z-i)} \\
 &= \lim_{z \rightarrow i} \frac{(z/i)(z^2+1)}{(z/i)} = \lim_{z \rightarrow i} z^2 + 1 \\
 &= (i)^2 + 1 = -1 + 1 = 0
 \end{aligned}$$

Ans = 0

$$\hookrightarrow \lim_{z \rightarrow i} \frac{z^3 - iz^2 + z - i}{z - i} = 0/0$$

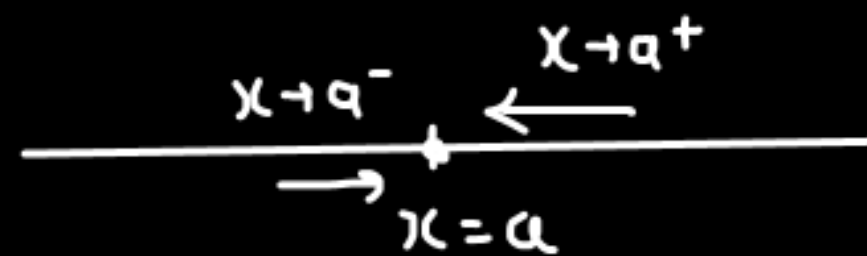
$\hookrightarrow$ L.H. w.r.t. z

$$\begin{aligned}
 \therefore \lim_{z \rightarrow i} \frac{3z^2 - 2zi + 1}{1} &= 3(i)^2 - 2(i)(i) + 1 = -3 - 2(-1) + 1 = -3 + 2 + 1 \\
 &= 0
 \end{aligned}$$

* Continuity of a single variable f^n :-

$f(x)$ is continuous at $x=a$ if:-

$$\lim_{x \rightarrow a} f(x) = f(a)$$

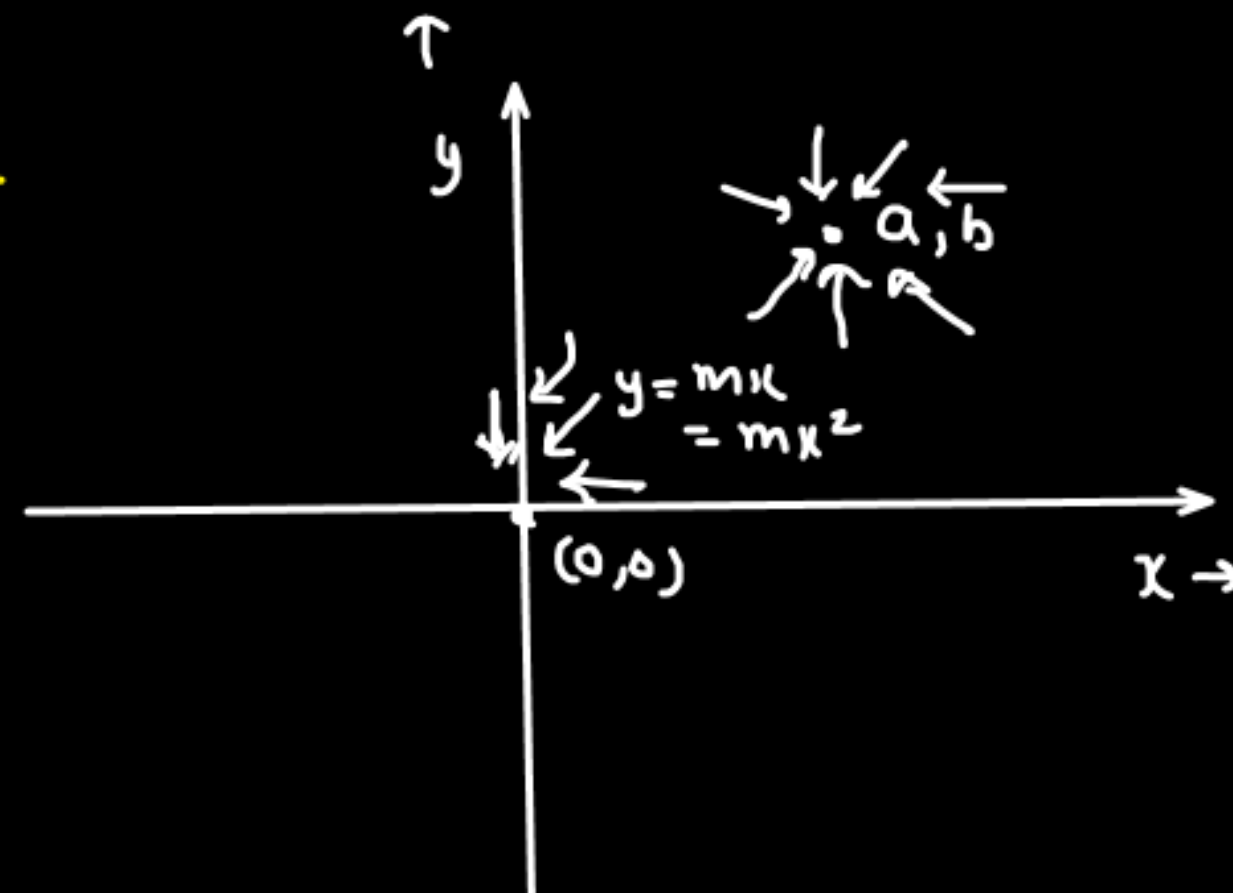


* Continuity of a two variable f^n :-

$f(x,y)$ is continuous at (a,b) if

$$\lim_{\substack{x \rightarrow a \\ y \rightarrow b}} f(x,y) = f(a,b)$$

continuous



Q.

Question:

Consider the complex function $f(z)$ defined as:

$$f(z) = \begin{cases} \frac{\operatorname{Re}(z^2)}{|z|^2} & \text{for } z \neq 0 \\ 0 & \text{for } z = 0 \end{cases}$$

$$z^2 = (x+iy)^2$$

$$z^2 = \underbrace{x^2 - y^2}_{\operatorname{Re}(z^2)} + i(2xy)$$

$$\operatorname{Re}(z^2) = x^2 - y^2$$

$$|z|^2 = x^2 + y^2$$

Which of the following statements correctly describes the behavior of $f(z)$ at $z = 0$?

(A) $f(z)$ is continuous at $z = 0$ because $\lim_{z \rightarrow 0} f(z) = 0$.

(B) $f(z)$ is continuous at $z = 0$ because both the numerator and denominator approach zero at the same rate.

✓ (C) $f(z)$ is discontinuous at $z = 0$ because the limit as $z \rightarrow 0$ does not exist.

(D) $f(z)$ is discontinuous at $z = 0$ because the limit as $z \rightarrow 0$ exists but is equal to 1. ✗

$$f(x, y) = \begin{cases} \frac{x^2 - y^2}{x^2 + y^2} & ; (x, y) \neq (0, 0) \\ 0 & ; (x, y) = (0, 0) \end{cases}$$

$$\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} f(x, y) = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^2 - y^2}{x^2 + y^2}$$

$$h_3: y = mx$$

$$h_3 = \lim_{x \rightarrow 0} \frac{x^2 - m^2 x^2}{x^2 + m^2 x^2} = \lim_{x \rightarrow 0} \frac{1 - m^2}{1 + m^2} \times \frac{\cancel{x^2}}{\cancel{x^2}} = \frac{1 - m^2}{1 + m^2}$$

$$h_3 = \frac{1 - m^2}{1 + m^2} = g(m) = \text{DNE}$$

$$\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} f(x, y) = \text{DNE} \neq f(x=0, y=0) \\ \neq 0$$

Q.

Question:

Consider the complex function $f(z)$ defined purely in terms of the complex variable z and its conjugate $\bar{z}$:

$$f(z) = \begin{cases} \frac{z^2}{\bar{z}} & \text{for } z \neq 0 \\ 0 & \text{for } z = 0 \end{cases}$$

$$z = x + iy$$

$$z^2 = x^2 - y^2 + i(2xy)$$

$$\bar{z} = x - iy$$

Which of the following statements correctly describes the behavior of $f(z)$ at $z = 0$?

~~(A)~~ $f(z)$ is discontinuous at $z = 0$ because the limit depends on the argument (angle) of approach.

✓ (B) $f(z)$ is continuous at $z = 0$ because $\lim_{z \rightarrow 0} f(z) = 0$, independent of the path.

~~(C)~~ $f(z)$ is discontinuous at $z = 0$ because $\bar{z}$ is not an analytic function.

(D) $f(z)$ is continuous only along the real and imaginary axes, but discontinuous elsewhere.

$$f(x, y) = \begin{cases} \frac{x^2 - y^2 + i(2xy)}{x - iy} & ; (x, y) \neq (0, 0) \\ 0 & ; (x, y) = (0, 0) \end{cases}$$

$$\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^2 - y^2 + i(2xy)}{x - iy} = ?$$

$$\rightarrow \boxed{\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} f(x, y) = 0 = f(0, 0)}$$

$$L_3: y = mx$$

$$L_3 = \lim_{x \rightarrow 0} \frac{x^2 - m^2 x^2 + i(2x \cdot mx)}{x - imx} = \lim_{x \rightarrow 0} \frac{x^2 [1 - m^2 + i2m]}{x [1 - im]} = 0$$

$$L_4: y = mx^2$$

$$L_4 = \lim_{x \rightarrow 0} \frac{x^2 - m^2 x^4 + i(2x \cdot mx^2)}{x - imx} = \lim_{x \rightarrow 0} \frac{x^2 [1 - m^2 x^2 + i2mx]}{x [1 - im]} = 0$$

$$L_1 = \lim_{x \rightarrow 0} \left\{ \lim_{y \rightarrow 0} f(x, y) \right\} = \lim_{x \rightarrow 0} \left\{ \frac{x^2}{x} \right\} = 0$$

$$L_2 = \lim_{y \rightarrow 0} \left\{ \lim_{x \rightarrow 0} f(x, y) \right\} = \lim_{y \rightarrow 0} \left\{ \frac{-y^2}{-iy} \right\} = 0$$

* Differentiability:-

$$f(z) = z^2 = (x^2 - y^2) + i(2xy) \\ = u(x, y) + i v(x, y)$$

If BOTH conditions (continuity of partial derivatives AND the Cauchy-Riemann equations) are satisfied at (x_0, y_0) : The function $f(z)$ is differentiable at z_0 .

$$f(z) = u(x, y) + i v(x, y) = u + iv$$

$$u_x = \frac{\partial u}{\partial x}$$

$$v_x = \frac{\partial v}{\partial x}$$

$$u_y = \frac{\partial u}{\partial y}$$

$$v_y = \frac{\partial v}{\partial y}$$



$$\begin{bmatrix} u_x & v_x \\ u_y & v_y \end{bmatrix}$$

$$u_x = v_y$$

$$u_y = -v_x$$

$$\Rightarrow \boxed{u_x = v_y} \quad \boxed{v_x = -u_y} : \text{CR equation} =$$

$$\Rightarrow \hookrightarrow u_x, u_y, v_x, v_y \neq \text{undefined}, \infty \downarrow$$

Q. $f(z) = z^2 - 2z \Rightarrow$ Always differentiable $\forall z$
check differentiability at all z . over entire z -plane

$\hookrightarrow f(z) = z^2 - 2z = x^2 - y^2 + i2xy - 2[x + iy]$
 $= (x^2 - y^2 - 2x) + i(2xy - 2y) = u + iv$

$$u = x^2 - y^2 - 2x \quad ; \quad v = 2xy - 2y$$

$$\left. \begin{array}{l} u_x = 2x - 2 \\ v_x = 2y \end{array} \right\} \quad \left. \begin{array}{l} u_y = -2y \\ v_y = 2x - 2 \end{array} \right\} \text{continuous}$$

$$\hookrightarrow \underbrace{u_x = v_y}, \quad \underbrace{v_x = -u_y} \quad \text{for } \forall x, y =$$

Q. $f(z) = |z|^2 \rightarrow$ Differentiable only at $(0,0)$

Differentiability at $z=0$, at $z=1+i$, at $z=3-2i$
 $\hookrightarrow x=0, y=0$ $\hookrightarrow x=1, y=1$ $\hookrightarrow x=3, y=-2$

$$\hookrightarrow f(x,y) = (x^2+y^2) + i(0) = u + iv$$

$$u = x^2 + y^2, \quad v = 0$$

$$u_x = 2x$$

$$u_y = 2y$$

$$v_x = 0$$

$$v_y = 0$$

} continuous =

$$\underbrace{u_x = v_y, \quad u_y = -v_x}_{\hookrightarrow (0,0)} \Rightarrow 2x=0, \quad 2y=0$$

$$\hookrightarrow x=0, y=0$$

$$\text{at } (x,y) = (1,1)$$

$$u_x = 2$$

$$v_x = 0$$

$$u_y = 2$$

$$v_y = 0$$

$$\hookrightarrow u_x \neq v_y, \quad u_y \neq -v_x$$

CR eqn x x x

Q.

Question:

Let $f(z) = z \cdot \operatorname{Re}(z)$. Determine all points in the complex plane where this function is differentiable.

$$\hookrightarrow f(x, y) = (x + iy) \cdot x = x^2 + ixy = u + iv$$

$$u = x^2, \quad v = xy$$

$$\left. \begin{array}{l} u_x = 2x \\ u_y = 0 \end{array} \right\} \text{continuous}$$

$$\left. \begin{array}{l} v_x = y \\ v_y = x \end{array} \right\}$$

C-R eqn

$$u_x = v_y, \quad u_y = -v_x$$

$$2x = x, \quad 0 = -y$$

$$\hookrightarrow x = 0, \quad y = 0$$

$$f(z) = z \cdot \operatorname{Re}(z)$$

↓

Differentiable only at $(0, 0)$

Q.

Question:

Analyze the differentiability of the function $f(z) = e^{\bar{z}}$ across the entire complex plane.

$$\begin{aligned}
 \hookrightarrow f(z) = e^{\bar{z}} &= e^{x-iy} = u + iv \\
 &= e^x \cdot e^{-iy} = e^x [\cos y - i \sin y] = e^x \cos y + i(-e^x \sin y) \\
 &= u + iv
 \end{aligned}$$

$$u = e^x \cos y, \quad v = -e^x \sin y$$

$$\left. \begin{aligned}
 u_x &= e^x \cos y \\
 u_y &= -e^x \sin y \\
 v_x &= -e^x \sin y \\
 v_y &= -e^x \cos y
 \end{aligned} \right\} \text{continuous}$$

$$\text{CA eqn: } u_x = v_y, \quad u_y = -v_x$$

$$\begin{aligned}
 e^x \cos y &= -e^x \cos y & ; & \quad -e^x \sin y = +e^x \sin y \\
 2e^x \cos y &= 0 & \text{---} & \quad 2e^x \sin y = 0 \text{ ---}
 \end{aligned}$$

$$\underbrace{e^x}_{\substack{e^x=0 \\ x=-\infty \\ \text{---} \\ \text{xx}}} \sin y = 0 \quad \rightarrow \quad \underbrace{\sin y = 0}$$

$$e^x \underbrace{\cos y = 0} \quad \checkmark \quad \underbrace{e^x = 0}_{\substack{x=-\infty \\ \text{---} \\ \text{xx}}}$$

CA eqn will be satisfied if

$$\underbrace{\sin y = 0} \quad \text{and} \quad \underbrace{\cos y = 0}$$



Both Together NOT POSSIBLE

$f(z) = e^{\bar{z}} : \text{NEVER Differentiable}$

Q. $f(z) = \sin z$; check differentiability.

$$\hookrightarrow f(x,y) = \sin(x+iy) = u+iv$$
$$= \sin x \cos(iy) + \cos x \sin(iy)$$

$$f(x,y) = \sin x \cosh y + i \cos x \sinh y$$
$$= u + iv$$

$$u = \sin x \cosh y, \quad v = \cos x \sinh y$$

$$\frac{d}{dy} \cosh y = \frac{d}{dy} \left[\frac{e^y + e^{-y}}{2} \right] = \frac{e^y - e^{-y}}{2} = \sinh y$$

$$\frac{d}{dy} \sinh y = \frac{d}{dy} \left[\frac{e^y - e^{-y}}{2} \right] = \frac{e^y + e^{-y}}{2} = \cosh y$$

$$\cos iy = \frac{e^{i(iy)} + e^{-i(iy)}}{2}$$

$$= \cosh y$$

$$\sin iy = \frac{e^{i(iy)} - e^{-i(iy)}}{2i}$$

$$= \frac{e^{-y} - e^y}{2i}$$

$$= i \sinh y$$

$$u = \sin x \cosh y, \quad v = \cos x \sinh y$$

$$u_x = \cos x \cosh y$$

$$v_x = -\sin x \sinh y$$

$$u_y = \sin x \sinh y$$

$$v_y = \cos x \cosh y$$

} continuous

$$u_x = v_y, \quad u_y = -v_x$$

Always

$f(z) = \sin z$: Differentiable over entire z -plane

$$= \cos z$$

"

"

"

"

* C-R Equation with r, θ :-

$$u(x,y), v(x,y)$$

$$f(z) = u + iv = u(r, \theta) + i v(r, \theta)$$

$$u_r = \frac{\partial u}{\partial r}$$

$$u_\theta = \frac{\partial u}{\partial \theta}$$

$$v_r = \frac{\partial v}{\partial r}$$

$$v_\theta = \frac{\partial v}{\partial \theta}$$

$$r u_r = v_\theta$$

$$r v_r = -u_\theta$$

$$\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta}$$

$$\frac{\partial v}{\partial r} = -\frac{1}{r} \frac{\partial u}{\partial \theta}$$

$$\begin{matrix} u_x & u_y \\ v_x & v_y \end{matrix}$$

$$u_x = v_y$$

$$v_x = -u_y$$

Q. $f(z) = z^2$; $z = re^{i\theta}$; check differentiability.

$$\hookrightarrow f(r, \theta) = (re^{i\theta})^2 = r^2 e^{i2\theta} = r^2 [\cos 2\theta + i \sin 2\theta]$$
$$u + iv = r^2 \cos 2\theta + i r^2 \sin 2\theta$$

$$u(r, \theta) = r^2 \cos 2\theta, \quad v(r, \theta) = r^2 \sin 2\theta$$

$$u_r = 2r \cos 2\theta$$

$$u_\theta = -2r^2 \sin 2\theta$$

$$v_r = 2r \sin 2\theta$$

$$v_\theta = 2r^2 \cos 2\theta$$

} $\rightarrow$ continuous

$$ru_r = v_\theta, \quad rv_r = -u_\theta ; \text{ irrespective of } r \text{ and } \theta$$

$\Rightarrow f(z) = z^2$: Always differentiable over complex plane. $\nearrow$ Entire fn.

Q. $f(z) = |z|$. check differentiability.

$$z = re^{i\theta}$$

$$|z| = r$$

$$\hookrightarrow f(r, \theta) = r + i0 = u + iv$$

$$u = r, \quad v = 0$$

$$\left. \begin{array}{l} u_r = 1, \quad u_\theta = 0 \\ v_r = 0, \quad v_\theta = 0 \end{array} \right\} \rightarrow \text{continuous}$$

$$ru_r = v_\theta$$

$$rv_r = -u_\theta \quad \checkmark$$

$$\hookrightarrow \boxed{r = 0}$$

$$0 = -0 \quad \checkmark$$

$$z = re^{i\theta}$$

$$\text{if } r = 0$$

$$\hookrightarrow z = 0 \quad \downarrow$$

$f(z) = |z|$ is only differentiable at $z = 0$

Q. $f(z) = \ln z$ · check differentiability

$$f(x, y) = \ln(x + iy) = u(x, y) + i v(x, y)$$

$$f(r, \theta) = \ln(re^{i\theta}) = \ln r + i\theta$$

$$u + i v = \ln r + i\theta$$

$$u(r, \theta) = \ln r, \quad v(r, \theta) = \theta$$

$$\left\{ \begin{array}{l} u_r = \frac{1}{r}, \quad u_\theta = 0 \\ v_r = 0, \quad v_\theta = 1 \end{array} \right\} \rightarrow u_r \text{ will be } \rightarrow \infty \text{ (discontinuous) at } \underbrace{r=0 \Rightarrow z=0}$$

$$\Rightarrow \begin{array}{l} r u_r = v_\theta \checkmark, \quad r v_r = -u_\theta \checkmark \\ 1 = 1, \quad 0 = -0 \end{array}$$

$$z = r e^{i\theta}$$

$$r=0 \Rightarrow z=0$$

$$\{ \ln(ab) = \ln a + \ln b \}$$

$$\{ \ln e = \log_e e = 1 \}$$

$\ln(z)$ is Diff. $\forall z \in \mathbb{C} \setminus \{0\}$

$f(z) = \ln z$ is differentiable $\forall z \in \mathbb{C} - \{0\}$

↳ $\ln z$ is non-differentiable at $z=0$ $\hookrightarrow$ NOT Entire f^n

* Other way of thinking:-

Eg-1 $f(z) = \ln z$

at $z=0$; $f(z) = \ln(0) = -\infty$

$\downarrow$
NOT continuous

$z=0$: NOT Differentiable

NOT continuous

$\Downarrow$

NOT Differentiable

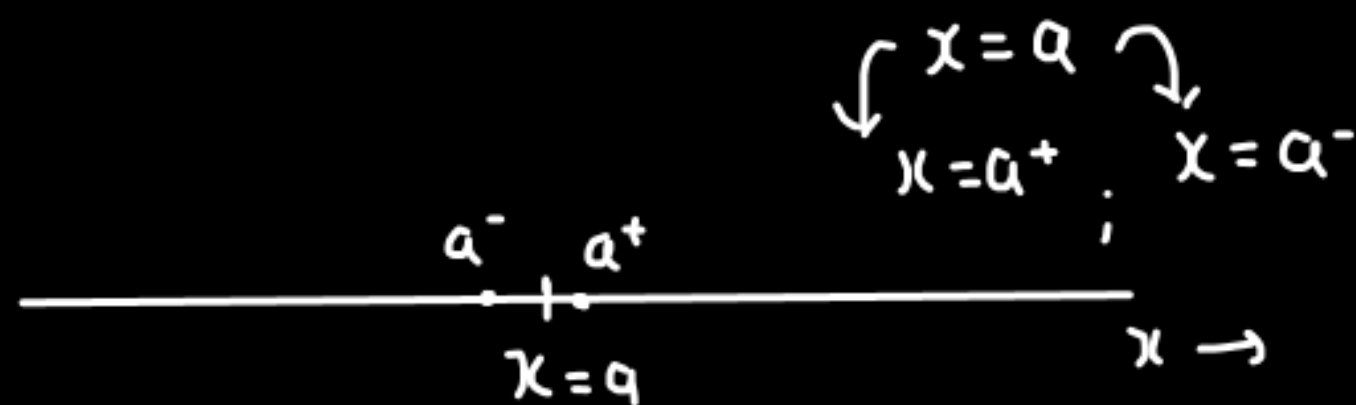
Eg 2 $f(z) = e^{1/z}$; $\hookrightarrow$ NOT conti
 $z=0 \Rightarrow f(z) = e^{1/0} = e^\infty = \infty \rightarrow$ NOT Entire f^n
 $\hookrightarrow$ NOT Diff. at $z=0 \leftarrow$

* Understanding the meaning of Nearby points:-

↳ For a real Variable:-

$$x \rightarrow a$$

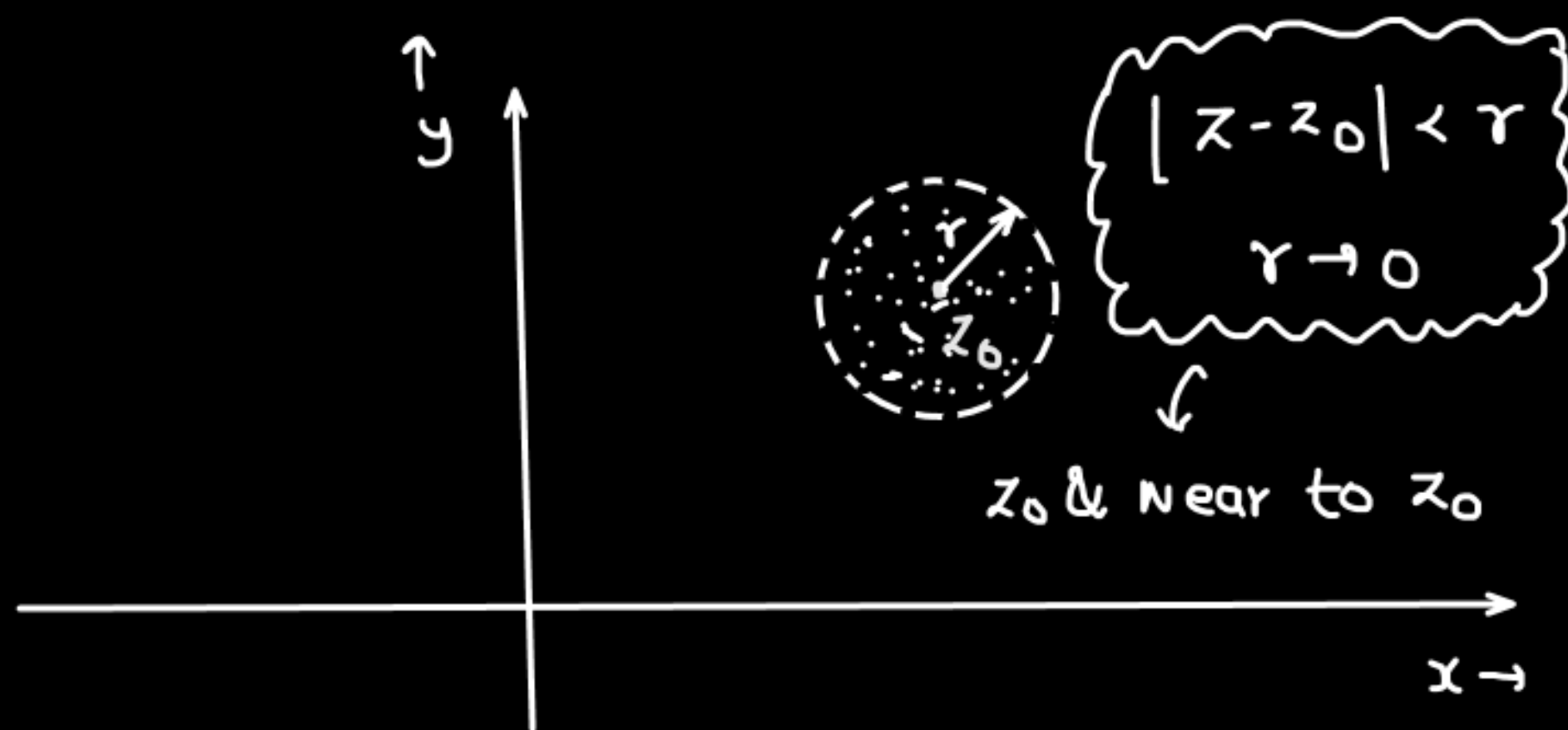
↳ Independent



↳ For a complex variable:-

$$z_0 = x_0 + iy_0$$

independent :- x_0, y_0



Question:

If a complex function $f(z)$ is differentiable at $z = z_0$, which of the following statements is true?

✓ A. $\lim_{z \rightarrow z_0} f(z)$ exists and is finite.

✓ B. $f(z)$ is continuous at $z = z_0$.

C. $\lim_{z \rightarrow z_0} f(z)$ may exist and be finite.

D. $f(z)$ may be continuous at $z = z_0$.

Differentiable ✓

↓
continuous ✓

↓
limit exist ✓

limit DNE
↓
conti X
↓
diff. X

★ Analytic function:-

A function $f(z)$ is analytic at z_0 if there exists a radius $r > 0$ such that $f(z)$ is differentiable for every z satisfying

$$|z - z_0| < r. \quad r \rightarrow 0$$

$f(z)$ will be analytic at $z = z_0$ if it's differentiable at $z = z_0$ and it's nearby points.



Entire Function

A complex function $f(z)$ is said to be an entire function if it is analytic at every point of the complex plane $\mathbb{C}$.

Differentiable $\forall z \in \mathbb{C}$

Analytic $\rightarrow z = z_0 \Rightarrow$ Differentiable $\rightarrow z = z_0$

Q. $f(z) = |z|^2$; Differentiability at $z=0$ and $z=1+i$?

↓
NOT polynomial Analytic at $z=0$ and $z=1+i$?

in z Entire function?

↳ $f(x,y) = x^2 + y^2 + i0$

$$u = x^2 + y^2, \quad v = 0$$

$$u_x = 2x$$

$$v_x = 0$$

$$u_y = 2y$$

$$v_y = 0$$

} Continuous $\checkmark$

$$u_x = v_y, \quad u_y = -v_x$$

$$2x = 0$$

$$2y = -0$$



$$x=y=0$$

CR is satisfied only at

$$(0,0)$$

$\checkmark$

$$\Rightarrow z=0$$

$\downarrow$ $f(z) = |z|^2$ is not Differentiable at $z = 1+i$
 $\downarrow$
 NOT Entire f^n NOT Analytic at $z = 1+i$

$f(z) = |z|^2$ is differentiable at $z = 0$

$z = 0$

$f(z)$ is Analytic or not at $z = 0$?? $\rightarrow$ NOT Analytic at $z = 0$

$\downarrow$
 Differentiable at 1. $z = 0$

2. nearby points of $z = 0$

$u_x = 0.0002$, $u_y = 0.0002$

$v_x = 0$, $v_y = 0$

NO

NOT Satisfied

$x = 0.0001$

$y = 0.0001$

$z = 0.0001 + j0.0001$

Q. $f(z) = \sin z$. Comment

$$f(x,y) = \sin(x+iy) = \underbrace{\sin x \cosh y}_u + i \underbrace{\cos x \sinh y}_v$$

↳ $\sin z$ is differentiable $\forall z \in \mathbb{C}$

↳ $\sin z$ is analytic $\forall z \in \mathbb{C}$

↳ $\sin z$ is entire f^n .

} Same for $\cos z$

NOT Entire

$$f(z) = \frac{\sin z}{\cos z} \rightarrow \begin{matrix} \sin z \rightarrow \text{entire } f^n \\ \cos z \rightarrow \text{entire } f^n \end{matrix}$$

↳ $f(z) = \tan z$ is entire f^n ?

NOT Entire $f^n \Rightarrow$ You need only one point where f^n is not differ.

$$f(z) = \tan z = \frac{\sin z}{\cos z}$$

$$\cos z = 0 \rightarrow z = \pi/2 ; f(z) \rightarrow \infty$$

$$z = \pi/2, 3\pi/2, \dots \quad \pi/2 \pm n\pi \Rightarrow f(z) \rightarrow \infty \Rightarrow f(z) \text{ is disconti} \Rightarrow f(z) \text{ is non diff.}$$

✓ NOT Cont.

⇓
 $f(z) \rightarrow \infty$ NOT Differ.
 or undefined

Q. $f(z) = \underbrace{z^2 - 3z + 2}_{\text{polynomial in } z}$. Analytic? Entire?

⇓
 NOT Entire fⁿ

$$\hookrightarrow f(x,y) = x^2 - y^2 + i(2xy - 3x - i3y + 2)$$

$$= \underbrace{(x^2 - y^2 - 3x + 2)} + i \underbrace{(2xy - 3y)}$$

$$u = x^2 - 3x - y^2 + 2, \quad v = 2xy - 3y$$

$$\left. \begin{array}{ll} u_x = 2x - 3 & v_x = 2y \\ u_y = -2y & v_y = 2x - 3 \end{array} \right\} \rightarrow \text{conti}$$

$$u_x = v_y ; u_y = -v_x \Rightarrow \text{Always}$$

$$f(z) = z^2 - 3z + 2 \text{ is } \underline{\text{differentiable}} \quad \forall z \in \mathbb{C}$$

$$\Downarrow$$

$$\text{analytic } \forall z \in \mathbb{C}$$

$$\Downarrow$$

$$\text{Entire } f^n$$

$$f(z) = \sin z, \cos z, \text{ polynomial in } z, e^z, e^{-z}, a^z_{a>0} \quad \left. \vphantom{f(z)} \right\} \text{Entire } f^n$$

$$e^z = (2.7)^z$$

$$e^{-z} = \left(\frac{1}{e}\right)^z$$

$$= (0.37)^z$$

$$a^z = (\underbrace{2.5}_{>0})^z$$

$$\downarrow$$

$$\underline{\underline{a > 0}}$$

Q. $f(z) = e^z$ or $f(z) = e^{-z}$. Analytic or Entire $f^n = ?$

$$\hookrightarrow f(x, y) = e^{x+iy} = e^x \cdot e^{iy} = e^x (\cos y + i \sin y) \\ = \underbrace{e^x \cos y}_u + i \underbrace{e^x \sin y}_v$$

$$\left. \begin{array}{l} u_x = e^x \cos y, \quad v_x = e^x \sin y \\ u_y = -e^x \sin y, \quad v_y = e^x \cos y \end{array} \right\} \rightarrow \text{cont.}; \quad u_x = v_y, \quad u_y = -v_x$$

Entire f^n

$$e \approx 2.7$$

$$1/e \approx 0.37$$

$$(2) f(z) = e^{-z} = (e^{-1})^z = (1/e)^z$$

$$f(x, y) = e^{-x-iy} = e^{-x} [\cos y - i \sin y] = \underbrace{e^{-x} \cos y}_u + i \underbrace{(-e^{-x} \sin y)}_v$$

$$\left. \begin{array}{l} u_x = -e^{-x} \cos y, \quad v_x = e^{-x} \sin y \\ u_y = -e^{-x} \sin y, \quad v_y = -e^{-x} \cos y \end{array} \right\} \Rightarrow \text{Entire } f^n$$

Q. $f(z) = e^{1/z}$. comment

$$f(z) = e^{1/z} \neq e^{-z}$$

$\downarrow$
 $e^{1/x+iy}$ $\hookrightarrow e^{-x-iy}$

$\hookrightarrow f(z) = e^{1/z}$

at $z=0 \Rightarrow f(z) \rightarrow \infty$

$\hookrightarrow f(z)$: disconti and non diff.

$e^{1/z} \rightarrow$ NOT Entire f^n

Q. $f(z) = \ln z$. comment

at $z=0 \Rightarrow f(z) \rightarrow -\infty$

$\hookrightarrow f(z)$: disconti or non-diff.

NOT Entire f^n

$\hookrightarrow$ polynomial in z

$\hookrightarrow \sin z, \cos z$

$\hookrightarrow e^z, e^{-z}$

$\hookrightarrow a^z; a > 0$

Q. $f(z) = e^{\bar{z}}$ → NOT Entire fⁿ
 Comment

↳ $f(x, y) = e^{x-iy} = e^x \cdot e^{-iy}$

$= e^x [\cos y - i \sin y]$

$= e^x \cos y + i(-e^x \sin y) = u + iv$

polynomial
in z

$\begin{cases} z = x + iy \\ -z = -x - iy \end{cases}$

↳ $\bar{z} = x - iy$

NOT polynomial in z

$u_x = e^x \cos y$ $v_x = -e^x \sin y$

$u_y = -e^x \sin y$ $v_y = -e^x \cos y$

$e^x \cos y = -e^x \cos y$, $-e^x \sin y = e^x \sin y$

↳ $\cos y = \sin y = 0 \rightarrow$ NEVER possible

↳ $f(z) = e^{\bar{z}}$: NEVER Differentiable

polynomial fⁿ in x

$p(x) = ax^{(n)} + bx^{(n-1)} + cx^{(n-2)}$

$+ \dots + d$ power $\in \mathbb{I}^+$
 $p(x) = -x^2 + 2$ ✓

$p(x) = x^{-1} + 2 \rightarrow$ NOT polynomial
 $\sqrt{x} + 2 \rightarrow$

* Examples of Entire functions:-

power of $z \in \mathbb{I}^+$

✓ 1. Polynomial in $z \rightarrow f(z) = z^2 + 3z - 2$ ✓

↓
with real or
Imag. coeffe.

$iz, -iz$ ✓

$$f(z) = (1+i)z^3 - 2z + (3-i) \quad \checkmark$$

$$f(z) = (\bar{z})^2 + 3z + 4 \quad \times \times$$

$$f(z) = z^2 + 3|z| + 4 \quad \times \times$$

$$f(z) = \sqrt{z} + 3 \quad \times \times$$

$\bar{z}$
 $|z|$
 $\sqrt{z}$ } $\rightarrow$ NOT
polyno.
in z

$z, -z \rightarrow$ polyno.
in z

✓ 2. $e^{\text{(polynomial)}}$ $\rightarrow e^z, e^{-z}, e^{z^2+3z+2}, e^{z^3+(1+i)z-4}$

3. Addition, subtraction or Multiplication of Entire f^n : (Division $\times$)

$$4. \underbrace{\sin z = \frac{e^{iz} - e^{-iz}}{2}}_{\text{Entire}}, \underbrace{\cos z = \frac{e^{iz} + e^{-iz}}{2}}_{\text{Entire}}, \underbrace{\sinh z = \frac{e^z - e^{-z}}{2}}_{\text{Entire}}, \underbrace{\cosh z = \frac{e^z + e^{-z}}{2}}_{\text{Entire}}$$

$$\hookrightarrow z^2 e^z \checkmark, \underbrace{\{z^2 + (1+i)z\}} \cdot \underbrace{\sin z} \cdot \underbrace{e^{-z}} + \underbrace{\sinh z} \cdot \underbrace{e^z} \checkmark$$

3. Compositions of Entire Functions

If you plug one entire function into another, the chain rule guarantees that the resulting composition is also differentiable everywhere.

Examples: e^{z^2} , $\sin(e^z)$, or $\cos(\sin(z))$, $\sinh(\cos(ze^z))$ ✓

4. Integrals of Entire Functions ↪ Running integration

If $f(z)$ is an entire function, then its antiderivative $F(z) = \int_0^z f(w)dw$ is also an entire function. This leads to some famous special functions used in engineering and physics:

○ The Error Function: $\operatorname{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z e^{-w^2} dw$

$$f(z) = \underbrace{e^{-z^2}}_{\text{Entire}}$$

$$F(z) = \int_{-L}^z e^{-w^2} \cdot dw$$

↪ Entire f'n

~~Imp.~~
Question:

If $f(z) = u(x, y) + iv(x, y)$ is analytic in a domain D , then which of the following statements are correct?

✓ ~~(A)~~ $u_x = v_y$ and $u_y = -v_x$: NECESSARY CONDⁿ for Analytic

✓ ~~(B)~~ $\nabla^2 u = 0$ and $\nabla^2 v = 0$

✓ ~~(C)~~ $\nabla u \cdot \nabla v = 0$

✓ ~~(D)~~ $u_x v_x + u_y v_y = 0$

✓ ~~(E)~~ The families of curves $u(x, y) = C_1$ and $v(x, y) = C_2$, where C_1 and C_2 are constants, are orthogonal to each other.

✓ ~~(F)~~ u and v are harmonic conjugates of each other.

→ (A) $u_x = v_y$ and $u_y = -v_x$
→ (B) $\nabla^2 u = 0 = \nabla^2 v \rightarrow$ Harmonic f^h

↳ $f(z) = u + iv$: Analytic

$$u_x \quad u_y$$

$$v_x \quad v_y$$

$$(A) \quad u_x = v_y \quad ; \quad v_x = -u_y$$

$$(B) \quad \nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \cancel{\frac{\partial^2}{\partial z^2}}$$

$$\nabla^2 u = \frac{\partial^2}{\partial x^2} u + \frac{\partial^2}{\partial y^2} u = \frac{\partial}{\partial x} \left[\frac{\partial u}{\partial x} \right] + \frac{\partial}{\partial y} \left[\frac{\partial u}{\partial y} \right]$$

$$= \frac{\partial}{\partial x} v_y + \frac{\partial}{\partial y} [-v_x] = v_{yx} - v_{xy} = 0$$

$f(z) = z^2$: Analytic

↓

$$f(x,y) = \underbrace{(x^2 - y^2)}_u + i \underbrace{(2xy)}_v$$

$$u = g(x, y)$$

$$v = h(x, y)$$

$$\nabla^2 u = 0$$

$$\nabla^2 v = 0$$

$\nabla^2 f = 0$: f is a Harmonic fn

$$\left\{ \begin{array}{l} \nabla^2 u = 0, \nabla^2 v = 0 \Rightarrow u \text{ and } v \text{ are Harmonic fn} \\ + u_x = v_y, v_x = -u_y \end{array} \right.$$

$\Downarrow$
 $f(z)$: Analytic

$$(c) (\nabla u) \cdot (\nabla v) = ? = 0$$

$$= \left(\frac{\partial u}{\partial x} \hat{i} + \frac{\partial u}{\partial y} \hat{j} \right) \cdot \left(\frac{\partial v}{\partial x} \hat{i} + \frac{\partial v}{\partial y} \hat{j} \right)$$

$$= (u_x \hat{i} + u_y \hat{j}) \cdot (v_x \hat{i} + v_y \hat{j}) = u_x v_x + u_y v_y$$
$$= v_y v_x - v_x v_y = 0$$

$u = g(x, y) \rightarrow$ Scalar
fn

$v = h(x, y) \rightarrow$ scalar
fn

$$u_x = v_y$$
$$u_y = -v_x$$

$$(\text{grad } u) \cdot (\text{grad } v) = 0$$

$$\begin{array}{l} \text{Surface 1: } u(x, y) = c_1 \\ \text{Surface 2: } v(x, y) = c_2 \end{array} \quad \left. \begin{array}{l} \nearrow \text{Surface 1} \\ \searrow \text{Surface 2} \end{array} \right\} \begin{array}{l} \phi_1 = u(x, y) - c_1 \\ \phi_2 = v(x, y) - c_2 \end{array}$$

$$\begin{array}{l} \nabla \phi_1 = \nabla(u(x, y) - c_1) = \nabla u(x, y) - \underbrace{\nabla c_1}_{=0} = \nabla u(x, y) \\ \nabla \phi_2 = \nabla v(x, y) \end{array}$$

perpendicular vectors of their respective surface

$$(\nabla \phi_1) \cdot (\nabla \phi_2) = 0$$

$\text{Surface 1} \perp \text{Surface 2}$

$$\vec{A} \cdot \vec{B} = 0$$

$$\vec{A} \perp \vec{B}$$

Q. Question:

If

$$f(z) = u(x, y) + iv(x, y),$$

where

$$\underline{\nabla^2 u = 0} \quad \text{and} \quad \underline{\nabla^2 v = 0},$$

then which of the following statements are correct?

~~(A)~~ $f(z)$ is analytic.

✓ (B) $f(z)$ may be analytic.

✓ (C) $f(z)$ is analytic if and only if

$$u_x = v_y \quad \text{and} \quad u_y = -v_x. \quad \checkmark$$

* What Milne and Thomson said:- (MT said)

If $f(z)$ is Entire f^n :-

$$x \leftrightarrow z, \quad y \leftrightarrow 0$$

Eg. $\rightarrow f(z) = (x^2 - y^2) + i(2xy)$: Entire^u

Find $f(z)$ in terms of z .

$$\hookrightarrow f(z) = [z^2 - (0)^2] + i(2z(0))$$

$$\boxed{f(z) = z^2} \text{ Ans.}$$

$$u_x = 2x$$

$$v_x = 2y$$

$$u_y = -2y$$

$$v_y = 2x$$

~~~~~  
Entire  $f^n$

$$f(z) = z^2$$

$$= (x^2 - y^2) + i(2xy)$$

Q.  $f(z)$ : Analytic ;  $u = x^2 - y^2$  ;  $f(z) = u + iv$

Find  $f(z) = ?$

$$\begin{matrix} u_x & \rightarrow & v_x \\ u_y & \rightarrow & v_y \end{matrix}$$

$\hookrightarrow$

$$f(z) = \underbrace{u}_{\checkmark} + i \underbrace{v}_{?} : \text{Analytic} \rightarrow \begin{matrix} u_x = v_y \\ v_x = -u_y \end{matrix}$$

$$u = u_x dx + u_y dy$$

$\parallel$

$$v = h(x, y)$$

$$dv = \frac{\partial v}{\partial x} dx + \frac{\partial v}{\partial y} dy$$

$$dv = v_x dx + v_y dy$$

$$dv = -u_y dx + u_x dy$$

$$u = x^2 - y^2$$

$$u_x = 2x$$

$$u_y = -2y$$

$$dV = 2y dx + 2x dy$$

$$V = \int \underbrace{2y dx}_{y: \text{constant}} + \underbrace{2x dy}_{f(x)=0}$$

→ Differential Eq<sup>n</sup>  
Exact Eq<sup>n</sup> ✓

$$V = 2y \int dx + 0$$

$$V = 2yx + C$$

$$\boxed{V = 2xy + C}$$

$$\leadsto \boxed{f(z) = z^2 + C}$$

$$f(z) = u + iV$$

$$= (x^2 - y^2) + i(2xy + C)$$

$$\boxed{f(z) = x^2 - y^2 + i2xy + C}$$

Analytic;  $\rightarrow x \leftrightarrow z, y \leftrightarrow 0$

$$f(z) = z^2 - 0^2 + i2z(0) + C$$



u.2

$$f(z) = u + iv : \text{Analytic} \rightarrow \underbrace{x \leftrightarrow z}, \underbrace{y \leftrightarrow 0}$$

$$f'(z) = \underbrace{u_x}_{\checkmark} + i \underbrace{v_x}_{?}$$

$$f'(z) = 2x + i2y$$

$$x \leftrightarrow z, y \leftrightarrow 0$$

$$f'(z) = 2z + i2(0)$$

$$f'(z) = 2z$$

$$f(z) = 2 \int z \cdot dz$$

$$f(z) = \frac{2z^2}{2} + c = z^2 + c \Rightarrow \boxed{f(z) = z^2 + c} \text{ Ans.}$$

$$u = x^2 - y^2$$

$$u_x = 2x$$

$$u_y = -2y$$

$$u_x \quad v_x$$

$$u_y \quad v_y$$

$$v_x = -u_y$$

$$v_x = 2y$$



Q. If  $f(z)$  is an analytic function whose imaginary part is

$$\underbrace{v(x, y) = 4xy - 2x^2 + 2y^2,}$$

find the analytic function  $\underbrace{f(z)}$ .

↳ M-I

$$dU = u_x dx + u_y dy$$

$$dU = v_y dx - v_x dy$$

$$dU = 4(x+y)dx - 4(y-x)dy$$

$$U = 4 \underbrace{\int (x+y) \cdot dx}_{y: \text{const}} - 4 \underbrace{\int (y-x) dy}_0$$

$$f(x) = 0$$

$$u_x \quad u_y$$

$$v_x \quad v_y$$

$$v_y = 4x + 4y$$

$$v_x = 4y - 4x$$

$$v = 4 \int x \, dx + 4 \int y \, dx - 4 \int y \, dy$$

$$u = 2x^2 + 4yx - 2y^2 + c$$

$$f(z) = u + iv = (2x^2 + 4yx - 2y^2) + i(4xy - 2x^2 + 2y^2)$$

↓

Analytic

$$x \leftrightarrow z, \quad y \leftrightarrow 0$$

$$= 2z^2 + 4(0)(z) - 2(0)^2 + i[4z(0) - 2z^2 + 2(0)^2]$$

$$= 2z^2 - i2z^2 + c$$

$$f(z) = 2(1-i)z^2 + c$$

11.2

$$f(z) = u + iv$$

$$f'(z) = u_x + i v_x$$

$$f'(z) = 4y + 4x + i(4y - 4x)$$

$$f'(z) = 4z - i4z$$

$$f'(z) = 4(1 - i)z$$

$$f(z) = 4(1 - i) \frac{z^2}{2} + c$$

$$f(z) = 2(1 - i)z^2 + c$$

$$x \leftrightarrow z, y \rightarrow 0$$

$$\begin{array}{cc} u_x & u_y \\ v_x & v_y \end{array}$$

$$u_x = v_y$$

$$v_x = 4y - 4x$$

$$u_x = v_y = 4x + 4y$$

Q. If  $f(z)$  is an analytic function whose real part is

$$u(x, y) = x^3 - 3xy^2,$$

find the value of the imaginary part at  $z = 2 + i$ .

$$\underbrace{\quad}_{v = ?} \quad x = 2, y = 1$$

4. h-I

$$dv = v_x dx + v_y dy$$

$$dv = -u_y dx + u_x dy$$

$$v = \underbrace{\int 6xy dx}_{y: \text{const}} + \underbrace{\int (3x^2 - 3y^2) dy}_0$$

$$v = 6y \left( \frac{x^2}{2} \right) - y^3 = 3yx^2 - y^3$$

$$\begin{array}{cc} u_x & u_y \\ v_x & v_y \end{array}$$

$$u_y = -6xy$$

$$u_x = 3x^2 - 3y^2$$

$$v = 3yx^2 - y^3$$

$$v(2,1) = 3(1)(2)^2 - 1^3 = 12 - 1 = \boxed{11} \text{ Ans.}$$

M.2

$$f(z) = u + iv$$

$$f'(z) = u_x + iv_x$$

$$f'(z) = 3x^2 - 3y^2 + i(6xy)$$

$$f'(z) = 3z^2 - 3(0) + i(6z(0))$$

$$f'(z) = 3z^2$$

$$\boxed{f(z) = z^3 + c}$$

$$f(z) = (x + iy)^3 + c$$

$$\begin{array}{cc} u_x & v_x \\ u_y & v_y \end{array}$$

$$v_x = -u_y$$

$$f(z) = (x+iy)^3 + c$$

$$= x^3 + (iy)^3 + 3x^2(iy) + 3x(iy)^2$$

$$= x^3 - iy^3 + i3x^2y - 3xy^2 + c$$

$$v = 3x^2y - y^3$$

↓

$$v(2, 4) = 11 \quad \text{Ans.} =$$



Q. If

$$f(z) = e^{-kx} (\cos 2y - i \sin 2y)$$

is an analytic function, find the value of  $k$ .

$$\hookrightarrow f(z) = \underbrace{e^{-kx} \cos 2y}_u + i \underbrace{(-e^{-kx} \sin 2y)}_v$$

$$u_x = -k e^{-kx} \cos 2y \quad ; \quad v_x = k e^{-kx} \sin 2y$$

$$u_y = -2 e^{-kx} \sin 2y \quad \quad v_y = -2 e^{-kx} \cos 2y$$

$$u_x = v_y$$

$$-k e^{-kx} \cos 2y = -2 e^{-kx} \cos 2y$$

$$\boxed{k = 2}$$

ANS. =

$$\begin{array}{cc} u_x & u_y \\ v_x & v_y \end{array}$$

$$u_x = v_y$$

$$v_x = -u_y$$



Question:

Consider the complex function

$$f(x, y) = \frac{x - iy}{x^2 + y^2}.$$

Which of the following statements is/are true?

- ☒ (A)  $f(x, y)$  is an entire function.
- ☒ (B)  $f(x, y)$  is not an entire function.
- ☒ (C)  $f(x, y)$  is analytic everywhere except at  $(x, y) = (0, 0)$ .
- ☒ (D)  $f(x, y)$  is nowhere analytic.

$$f(x, y) = \underbrace{\frac{x}{x^2+y^2}}_u + i \underbrace{\left( \frac{-y}{x^2+y^2} \right)}_v$$

$$u_x = \frac{(x^2+y^2) \cdot 1 - x(2x)}{(x^2+y^2)^2} = \frac{-x^2+y^2}{(x^2+y^2)^2}$$

$$u_y = \frac{-x}{(x^2+y^2)^2} \cdot 2y = \frac{-2xy}{(x^2+y^2)^2}$$

$$u_x = v_y, \quad u_y = -v_x$$

$$v_x = \frac{2xy}{(x^2+y^2)^2}$$

$$v_y = \frac{-x^2+y^2}{(x^2+y^2)^2}$$

NOT  
cont.  
(0,0)

NOT Diff.  
at (0,0)

NOT Analytic  
at (0,0)

Q.  $f(z) = \underbrace{(x^2 + ay^2)}_u + i \underbrace{bxy}_v$  is analytic; Find  $a$  and  $b$

↳

$$u_x = 2x$$

$$u_y = 2ay$$

$$v_x = by$$

$$v_y = bx$$

$$2x = by$$

$$\boxed{b = 2}$$

$$by = -2ay$$

$$a = -b/2$$

$$\boxed{a = -1}$$

Q.  $f(z) = u + iv$  : Analytic.  $u = x^2 + ay^2$  ; find  $a = ?$

↳

$$f(z) = (x^2 + ay^2) + iv$$

$f(z)$  : Analytic

$$\nabla^2 u = 0$$

$$\frac{\partial^2}{\partial x^2} u + \frac{\partial^2}{\partial y^2} u = 0$$

$$2 + 2a = 0$$

$$\boxed{a = -1}$$

uv

$$dv = v_x dx + v_y dy$$

$$dv = -u_y dx + u_x dy$$

$$dv = -2ay dx + 2x dy$$

$$v = -2ayx + C$$

$$v = -2axy + C$$

⇓

Solve

$$u_x = v_y \quad u_y = -v_x \quad \checkmark$$

Q  $f(z) = u + iv$  : Analytic.  $v = bxy$  ; Find  $b = ?$

$f(z)$  : Analytic

$$\nabla^2 v = 0$$

$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0$$

$$0 + 0 = 0$$

$$0 = 0$$

nothing  
conclusive

M-I ✓

$$u_x = v_y$$

$$v_x = -u_y$$

$$du = u_x dx + u_y dy$$

$$v = bxy$$

$$du = v_y dx - v_x dy$$

$$du = bx dx - by dy$$

$$u = \frac{bx^2}{2} - \frac{by^2}{2} = \frac{b}{2}(x^2 - y^2)$$

$$u_x = bx \quad u_y = -by$$

$$v_x = by \quad v_y = bx$$

Q.  $f(z) = 2z^3 + b|z|^3$  : Analytic

Find  $b = ?$

↳

$$f(z) = \underbrace{2z^3}_{\text{polynomial in } z} + \underbrace{b|z|^3}_{\text{NOT Analytic}}$$

↓  
polynomial

in  $z \Rightarrow$  Analytic

$$\Rightarrow \boxed{b=0} \text{ Ans.}$$

$$f(z) = 2z^3 : \text{Analytic}$$

---

$$\hookrightarrow f(x, y) = 2(x+iy)^3 + b(x^2+y^2)^{3/2}$$

$$|z| = \sqrt{x^2+y^2}$$

$$= 2(x^2 - y^2 + 2ixy) + b(x^2+y^2)^{3/2}$$

$$= \underbrace{2x^2 - 2y^2 + b(x^2+y^2)^{3/2}}_{u} + i(\underbrace{4xy}_{v}) = u + iv$$

$\begin{matrix} u_x & u_y \\ v_x & v_y \end{matrix}$



$$\hookrightarrow f(r, \theta) = 2(re^{i\theta})^3 + br^3$$

$$= 2r^3 e^{i3\theta} + br^3$$

$$= 2r^3 [\cos 3\theta + i \sin 3\theta] + br^3$$

$$= 2r^3 \cos 3\theta + br^3 + i 2r^3 \sin 3\theta = u + iv$$

$$u_r = 6r^2 \cos 3\theta + 3br^2$$

$$v_r = 6r^2 \sin 3\theta$$

$$u_\theta = -6r^3 \sin 3\theta$$

$$v_\theta = 6r^3 \cos 3\theta$$

$$ru_r = v_\theta$$

$$6r^3 \cancel{\cos 3\theta} + 3br^3 = 6r^3 \cancel{\cos 3\theta}$$

$$3br^3 = 0 \quad \Rightarrow \quad \boxed{b=0}$$



Question:

Let

$$f(x, y) = x^2 - y^2, \quad g(x, y) = i 2xy,$$

and

$$h(x, y) = f(x, y) + g(x, y).$$

Which of the following statements is/are true?

☒ (A)  $f(x, y)$  is differentiable only at  $(0, 0)$ .

☒ (B)  $g(x, y)$  is differentiable only at  $(0, 0)$ .

☒ (C)  $h(x, y)$  is differentiable everywhere.

(D) All of the above.

$$h(x, y) = (x^2 - y^2) + i 2xy$$

$$h(x, y) = z^2 \quad \checkmark$$

↳ Entire f<sup>n</sup>

$$(a) \quad f(x, y) = (x^2 - y^2) + i(0) \quad \leadsto u = x^2 - y^2, \quad v = 0$$

$$\left. \begin{array}{ll} u_x = 2x & v_x = 0 \\ u_y = -2y & v_y = 0 \end{array} \right\} \text{differentiable only at } (0, 0)$$

$$(b) \quad g(x, y) = 0 + i(2xy) \quad \leadsto u = 0, \quad v = 2xy$$

$$\left. \begin{array}{ll} u_x = 0 & v_x = 2y \\ u_y = 0 & v_y = 2x \end{array} \right\} \rightarrow \text{differentiable only at } (0, 0)$$

MSQ

Question:

Let

$\curvearrowright z = z_0$

$$f(z) = \underbrace{g(z)} + \underbrace{h(z)},$$

where  $f$ ,  $g$ , and  $h$  are complex-valued functions of the complex variable  $z$ .

Which of the following statements is/are true?

~~(A)~~ If  $f(z)$  is differentiable at  $z = z_0$ , then  $g(z)$  and  $h(z)$  are also differentiable at  $z = z_0$ .

☒ (B) If  $g(z)$  and  $h(z)$  are differentiable at  $z = z_0$ , then  $f(z)$  is also differentiable at  $z = z_0$ .

~~(C)~~ If  $f(z)$  is continuous at  $z = z_0$ , then it is differentiable at  $z = z_0$ .

~~(D)~~ If  $f(z)$  is differentiable at  $z = z_0$ , then its real and imaginary parts are also differentiable (as real-valued functions) at  $z = z_0$ .

$\curvearrowright$  Addition / subtr /  
Multiplication of  
Entire  $f^h$  will  
be Entire  $f^h$   
only

$$(a) \quad g(z) = \underbrace{1/z}_{z=0}, \quad h(z) = \underbrace{z^2 - 1/z}_{z=0}$$

$$f(z) = 1/z + z^2 - 1/z$$

$$f(z) = z^2 : \text{Differentiable at } z=0$$

$$(d) \quad f(z) = z^2 \rightsquigarrow \text{Diff.} \quad \checkmark \quad z=1 \quad \checkmark$$

$$= \underbrace{(x^2 - y^2)}_{z=1} + i \underbrace{(2xy)}_{z=1}$$

$$\underbrace{z=1}_x$$

$$\underbrace{z=1}_x$$

Diff  $x$

Q. Question:  $f(z) = \frac{N(z)}{D(z)} ; D(z) \neq 0$   $\cap (0,0) ; r=1$

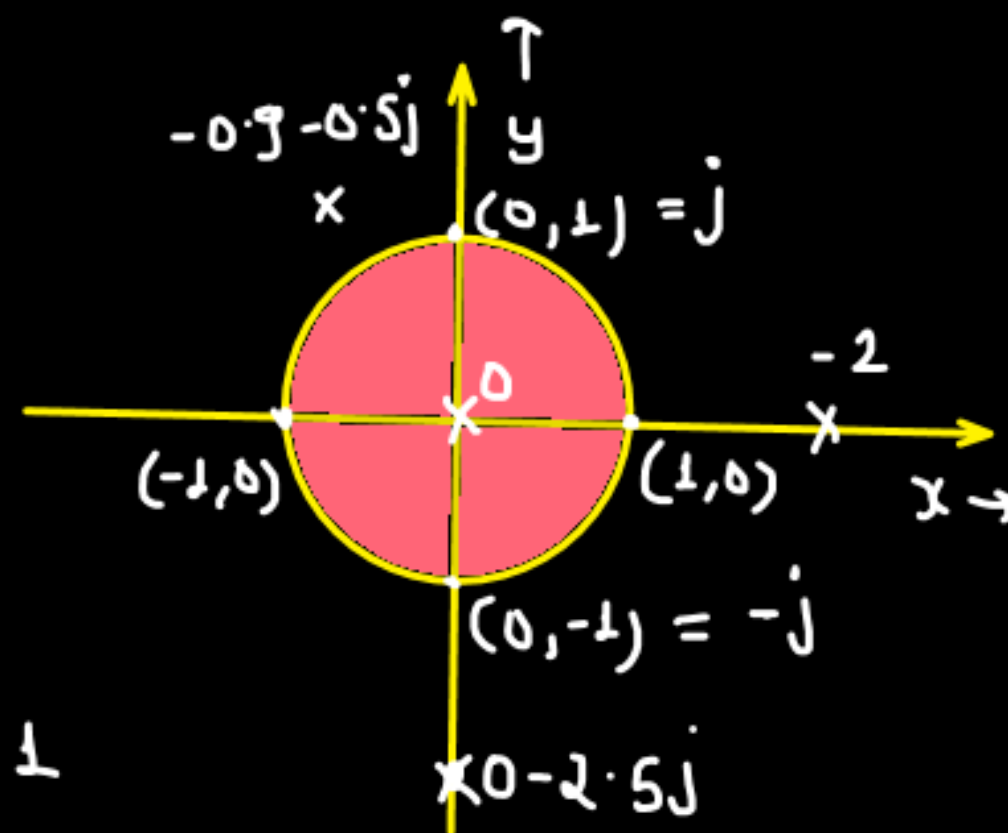
Which of the following functions is analytic in the region  $|z| \leq 1$ ?

~~(A)~~  $\frac{z^2 - 1}{z}$   $\leadsto z=0 \Rightarrow$  NOT analytic in  $|z| \leq 1$

~~(B)~~  $\frac{z^2 - 1}{z + 0.9 + 0.5j}$   $\leadsto z = -0.9 - 0.5j$

~~(C)~~  $\frac{z^2 - 1}{z + 2.5j}$   $\leadsto z = -2.5j = 0 + (-2.5)j$

~~(D)~~  $\frac{z^2 - 1}{z + 2}$   $\leadsto z = -2 + i0 \Rightarrow$  analytic in  $|z| \leq 1$



analytic:  $f(z) = f(x,y)$

$= u + iv$

NOT conti  $\Rightarrow$  NOT diff.  $\Rightarrow$  NOT analytic

$$\left. \begin{array}{cc} u_x & u_y \\ v_x & v_y \end{array} \right\}$$

For the point  $z = z_0$  lying inside the circle:

$$|z_0| < 1$$

outside the circle:-  $|z_0| > 1$

$$| -0.9 - 0.5j | = \sqrt{0.81 + 0.25} = \sqrt{1.06} > 1$$

↓  
lying outside of unity circle

(a)  $|z| = 1$  ,  $|0| = 0 < 1 \rightarrow \text{Inside}$

(b)  $|z| = 1$  ,  $| -0.9 - 0.5j | = \sqrt{1.06} > 1 \rightarrow \text{outside}$

(c)  $|z| = 1$  ;  $| -2.5j | = 2.5 > 1 \rightarrow \text{outside}$

(d)  $|z| = 1$  ,  $| -2 | = 2 > 1$   
outside



Q.

Question:

$$z = x + iy = re^{i\theta} \quad r = \sqrt{x^2 + y^2}$$

For the complex transformation

$$w = \ln z, \quad f(z) = \ln z$$

where

$$w = u + iv, \quad z = x + iy,$$

the family of curves  $u = \text{constant}$  gets mapped in the  $z$ -plane as

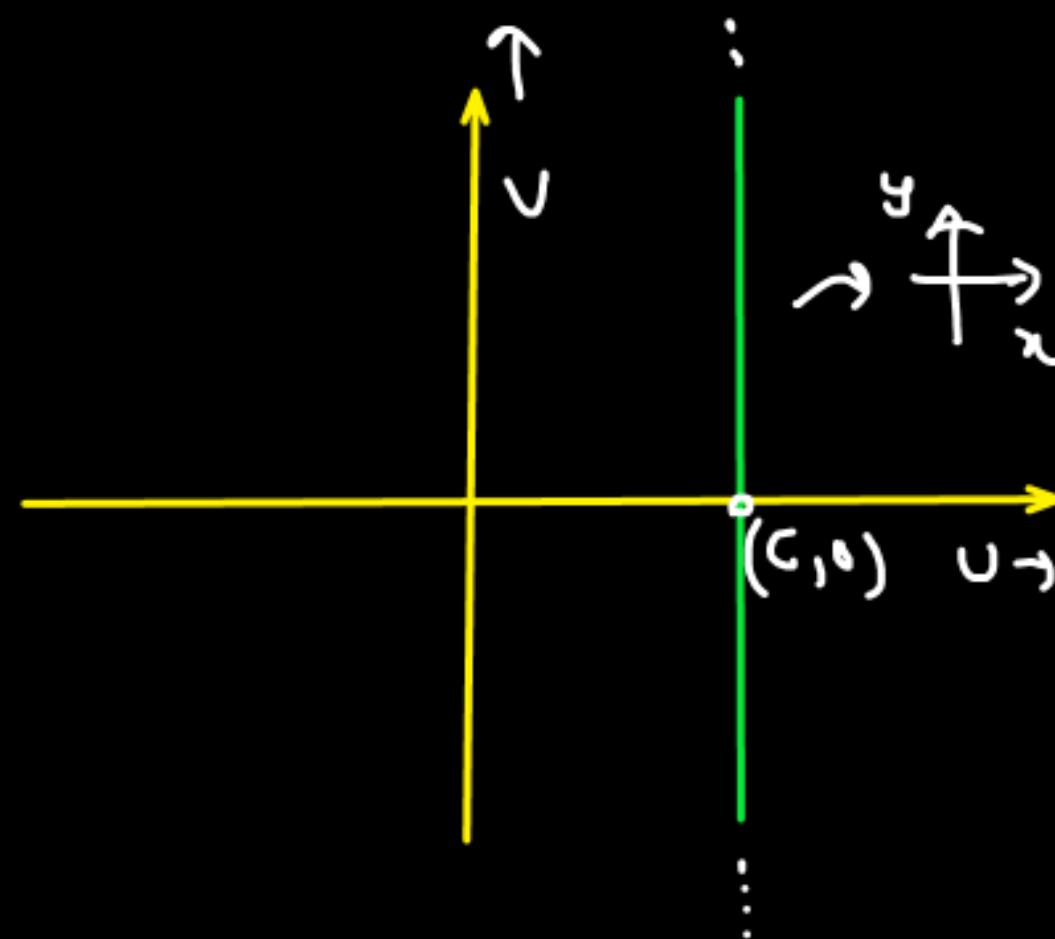
(A) A set of radial straight lines

✓ (B) A set of concentric circles

(C) A set of confocal hyperbolas

(D) A set of confocal ellipses

$$u + iv = \ln(re^{i\theta}) = \ln r + \ln e^{i\theta} = \ln r + i\theta$$



$$u = c, \quad -\infty < v < \infty$$



$$\underbrace{u + i \underbrace{v}} = \underbrace{\ln r} + i \underbrace{\theta}$$

$$u = \ln r$$

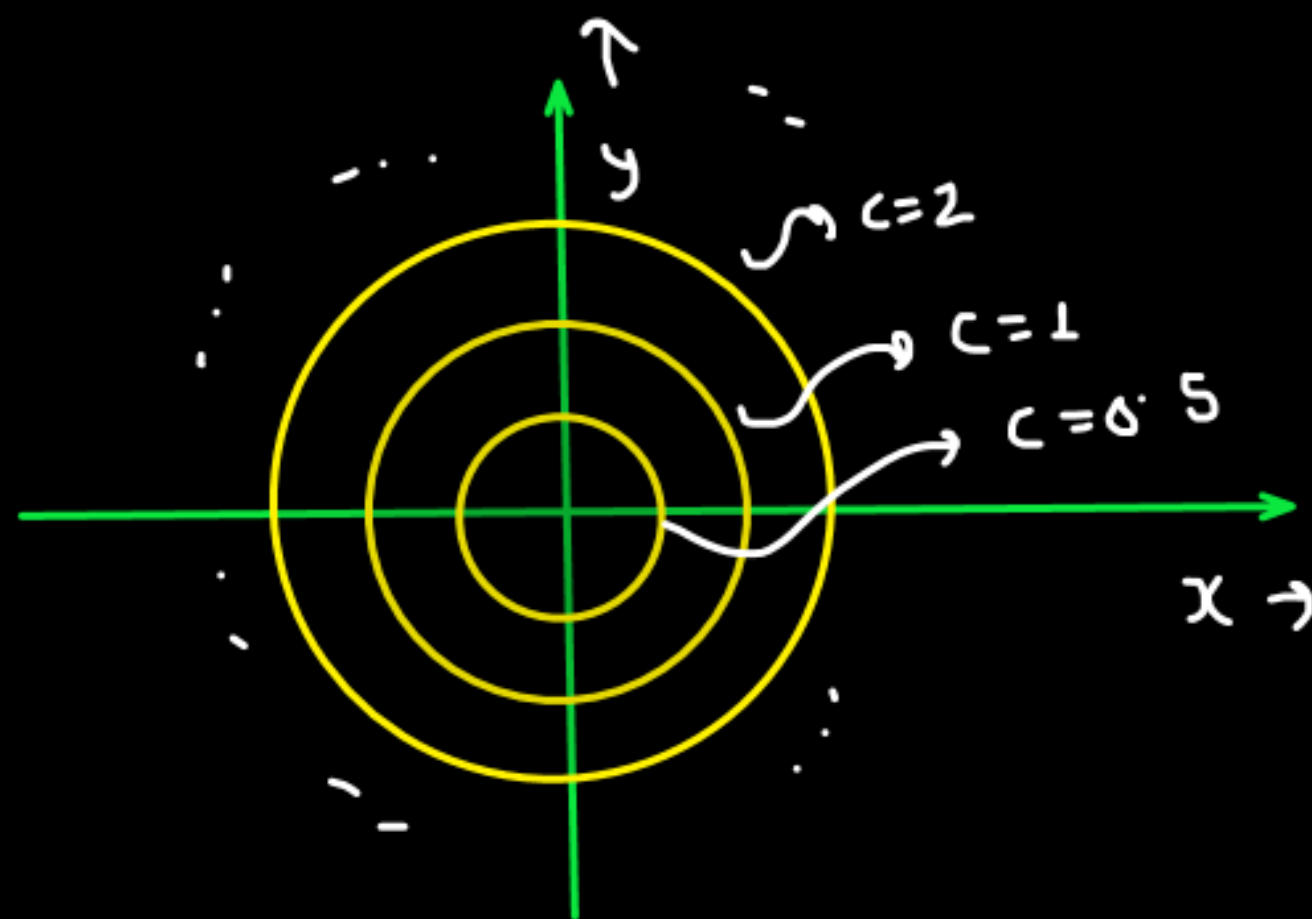
$$u = c = \ln r$$

$$r = e^c = c$$

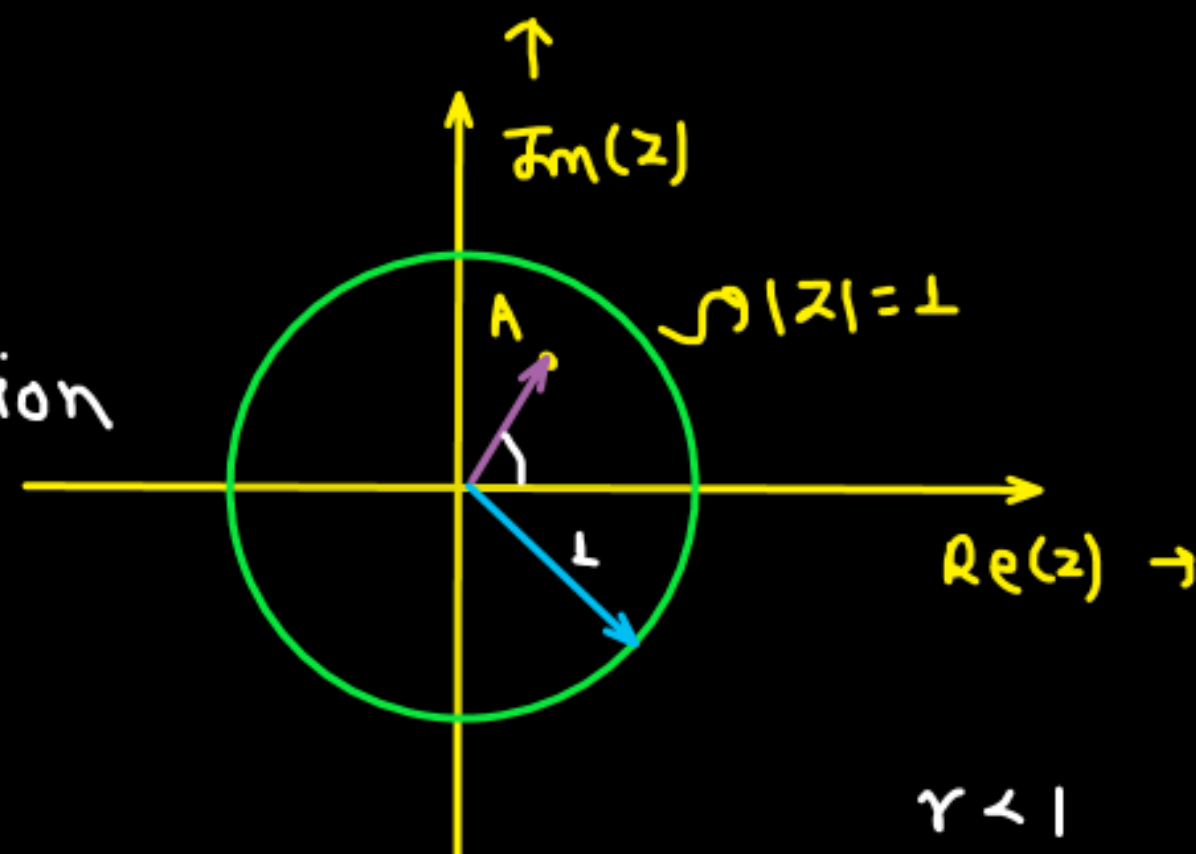
$$\sqrt{x^2 + y^2} = c$$

$$x^2 + y^2 = c^2$$

$$x^2 + y^2 = c$$



Q.  $w = 1/z$   
Transformation



$$z = re^{i\theta} \quad r < 1$$

$$0 < \theta < \pi/2$$

$$r < 1$$

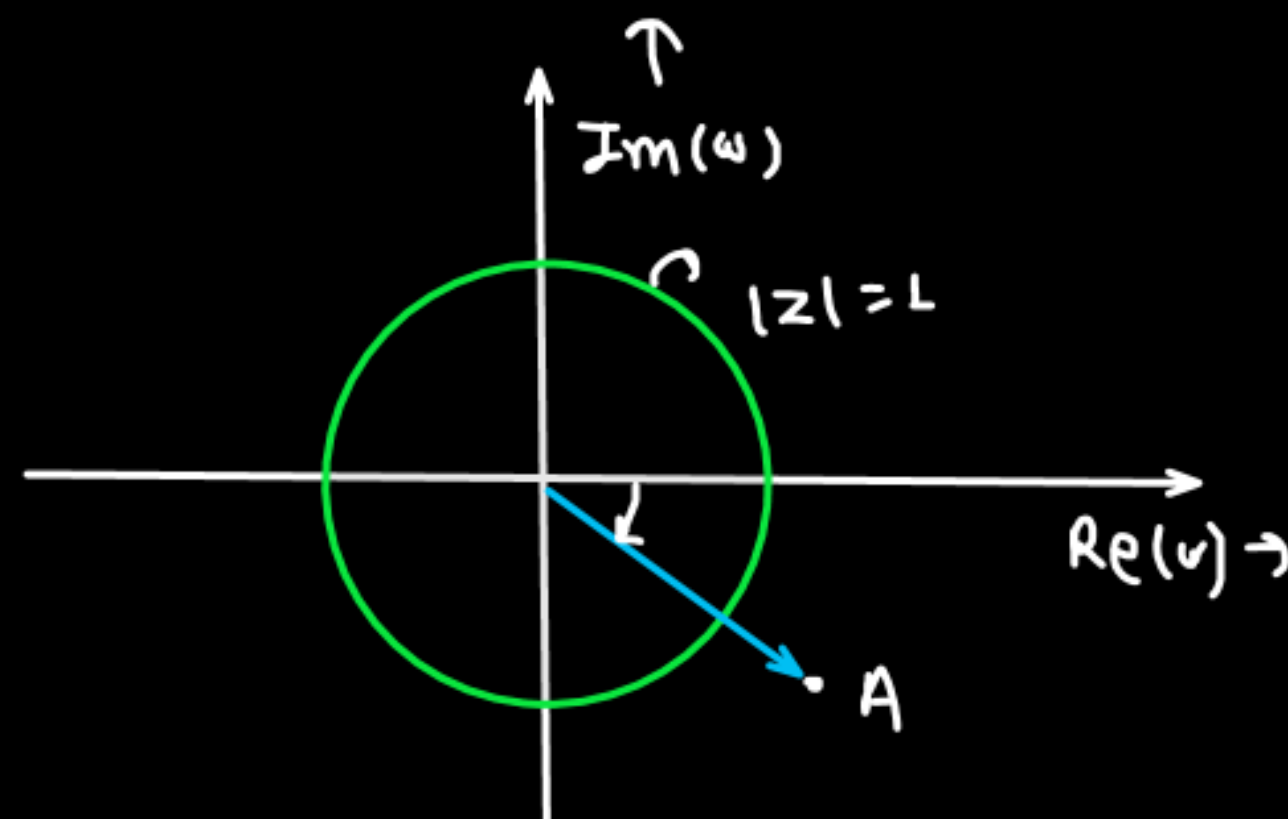
$$\frac{1}{r} > 1$$

$$w = \frac{1}{z} = \frac{1}{re^{i\theta}} = \frac{1}{r} e^{-i\theta}$$

$$\boxed{w = \frac{1}{r} e^{-i\theta}}$$

$\hookrightarrow$  magnitude  $= 1/r > 1$   
 angle  $= -\theta \Rightarrow -\pi/2 < -\theta < 0$

Map point A in w-plane.





Question:

Let

such that

$$S = \{z : |z| = 1\}$$

be the set of points in the complex plane corresponding to the unit circle. Consider the function

$$f(z) = z\bar{z},$$

where  $\bar{z}$  denotes the complex conjugate of  $z$ . Then  $f(z)$  maps  $S$  to which of the following in the complex plane?

(A) The unit circle

(B) The horizontal line segment from the origin to  $(1, 0)$

✓ (C) The point  $(1, 0)$

(D) The entire horizontal axis

$$v = 0$$

$$u = x^2 + y^2 = 1$$

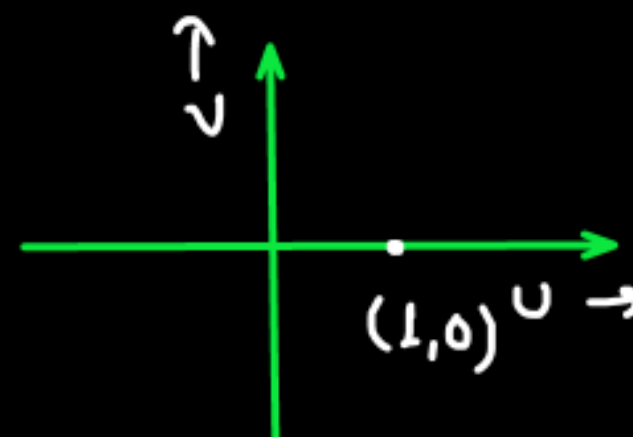
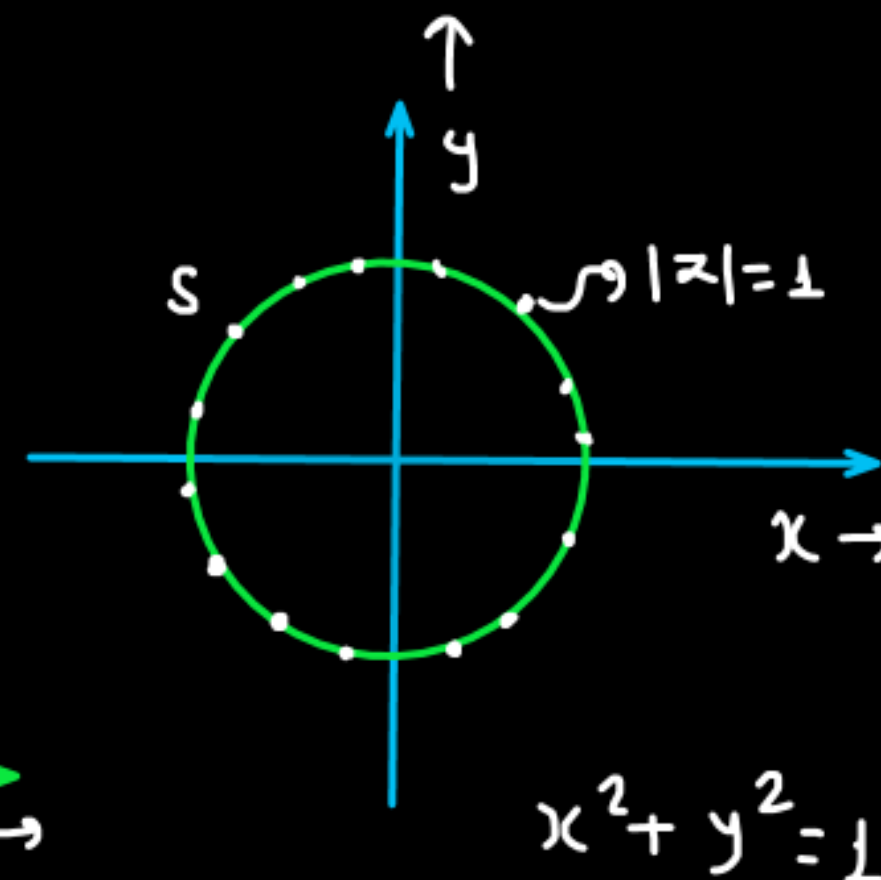
$$f(z) = w = z\bar{z} = x^2 + y^2$$

$$u + iv = (x^2 + y^2) + i0$$

$$v = 0$$

$$u = 1$$

$$z\bar{z} = |z|^2$$



Q. Question:

The bilinear transformation

$$\omega = \frac{z-1}{z+1}$$

unity circle:  $|z|=1$

Inside of unity circle

$$|z| < 1 \Rightarrow x^2 + y^2 < 1$$

outside :-

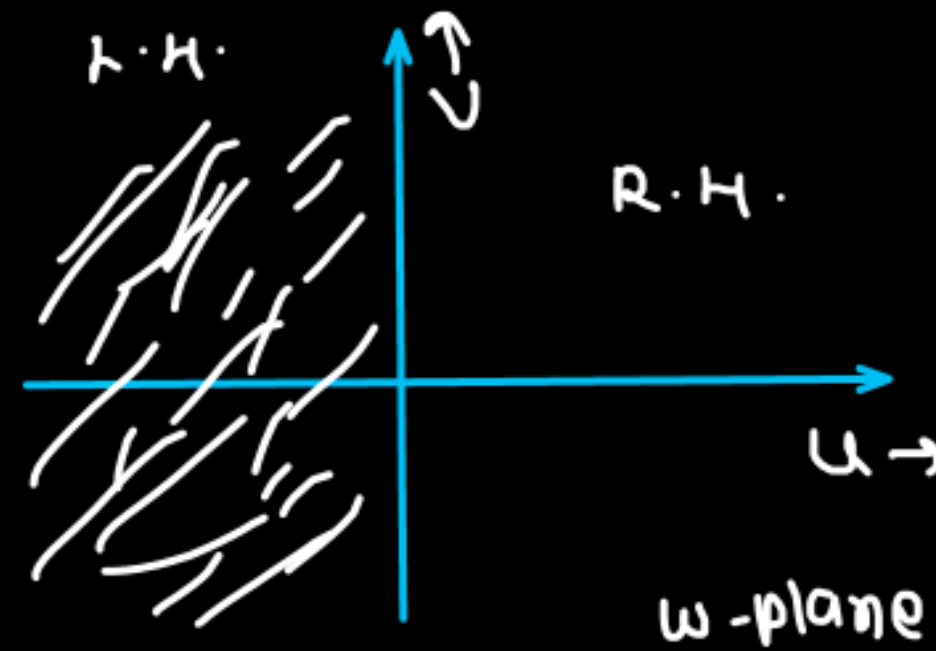
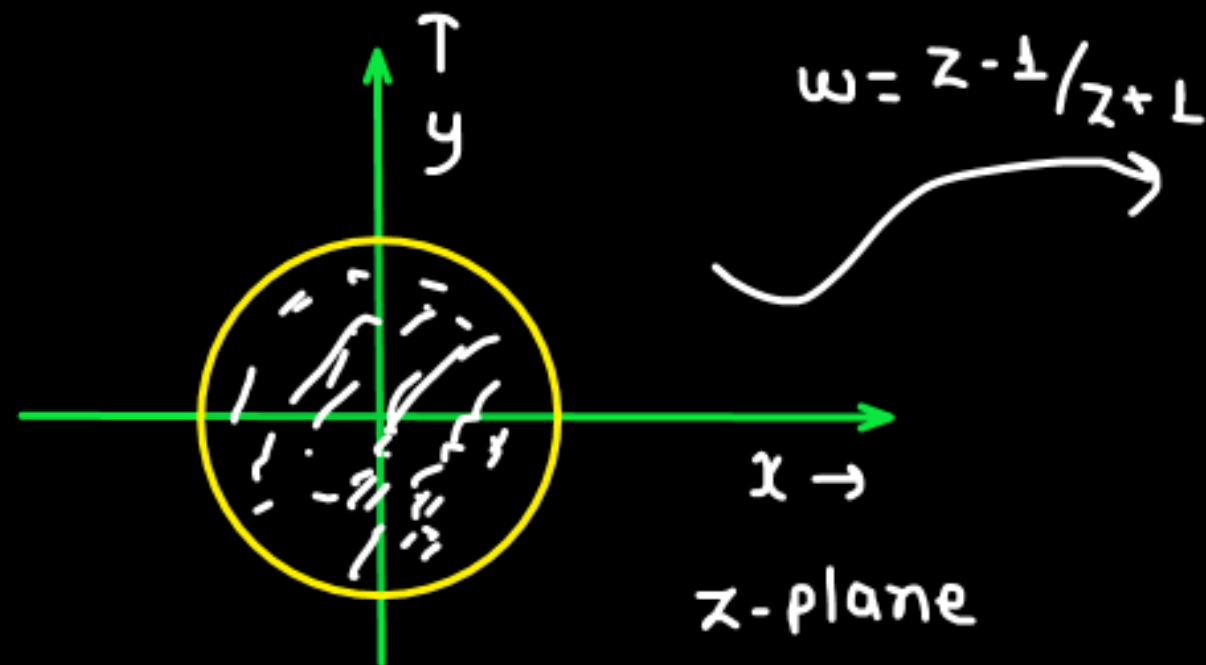
$$|z| > 1 \Rightarrow x^2 + y^2 > 1$$

(A) maps the inside of the unit circle in the  $z$ -plane to the left half of the  $\omega$ -plane.

(B) maps the outside of the unit circle in the  $z$ -plane to the left half of the  $\omega$ -plane.

(C) maps the inside of the unit circle in the  $z$ -plane to the right half of the  $\omega$ -plane.

(D) maps the outside of the unit circle in the  $z$ -plane to the right half of the  $\omega$ -plane.



$$\omega = f(z)$$
$$z = g(\omega)$$

$$w = \frac{z-1}{z+1}$$

$$wz + w = z - 1$$

$$w + 1 = z - wz$$

$$z = \frac{1+w}{1-w}$$

$$|z| = \left| \frac{1+w}{1-w} \right| = \frac{|1+w|}{|1-w|}$$

$$|z| < 1 \Rightarrow \frac{|1+w|}{|1-w|} < 1$$

$$|1+w| < |1-w|$$

$$x+iy = \frac{1+u+iv}{1-u-iv}$$

$\Downarrow$   
Difficult

$$w = u+iv$$

2M Q  $\rightarrow$  70%  $\downarrow$   
1M Q.  $\Rightarrow$  NEVER  
COMPLEX  
CALC.

$$|z_1 z_2| = |z_1| |z_2|$$

$$\left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|}$$

$$|z_1 + z_2| \neq |z_1| + |z_2|$$

$$||z_1| - |z_2|| \leq |z_1 \pm z_2| \leq |z_1| + |z_2|$$

$$|1+u+iv| < |1-u-iv|$$

$$\sqrt{(1+u)^2+v^2} < \sqrt{(1-u)^2+v^2}$$

$$1+u^2+2u+v^2 < 1+u^2-2u+v^2$$

$$4u < 0$$

$$u < 0$$

$$|z| < 1 \Rightarrow u < 0$$

$$|z| > 1 \Rightarrow u > 0$$



# \* Singularity:-

$$f(z) = \frac{N(z)}{D(z)} \Rightarrow D(z) = 0$$

Definition:

A singularity (or singular point) of a complex function is a point where the function fails to be analytic.

Eg.  $\rightarrow 1. f(z) = e^{1/z} \Rightarrow z=0 \rightsquigarrow \text{Essential}$

$$f(z) = \frac{z^2 - 1}{z - 1} = \frac{(z+1)(z-1)}{\cancel{z-1}}, z \neq 1$$

2.  $f(z) = \frac{z^2 - 1}{z - 1} \Rightarrow z = 1 \rightsquigarrow \text{Removable}$

$$f(z) = z + 1; z \neq 1$$

3.  $f(z) = \frac{z+3}{(z-2)(z+1)} \Rightarrow z = 2, -1$    
  $\swarrow \searrow$    
 simple poles

$$f(1) = 2 \leftarrow$$

$$\lim_{z \rightarrow 1} f(z) = 2 \checkmark$$

4.  $f(z) = \frac{z-2}{(z+2-j)^2}, z = \underbrace{-2+j, -2+j}_{\text{multiple pole}}$

$$f(1) = \frac{0}{0} : \text{undefined}$$



↪ Singularity ↩

Isolated

- Simple pole
- Multiple pole
- Removable
- Essential

↳ Non-Isolated ↩

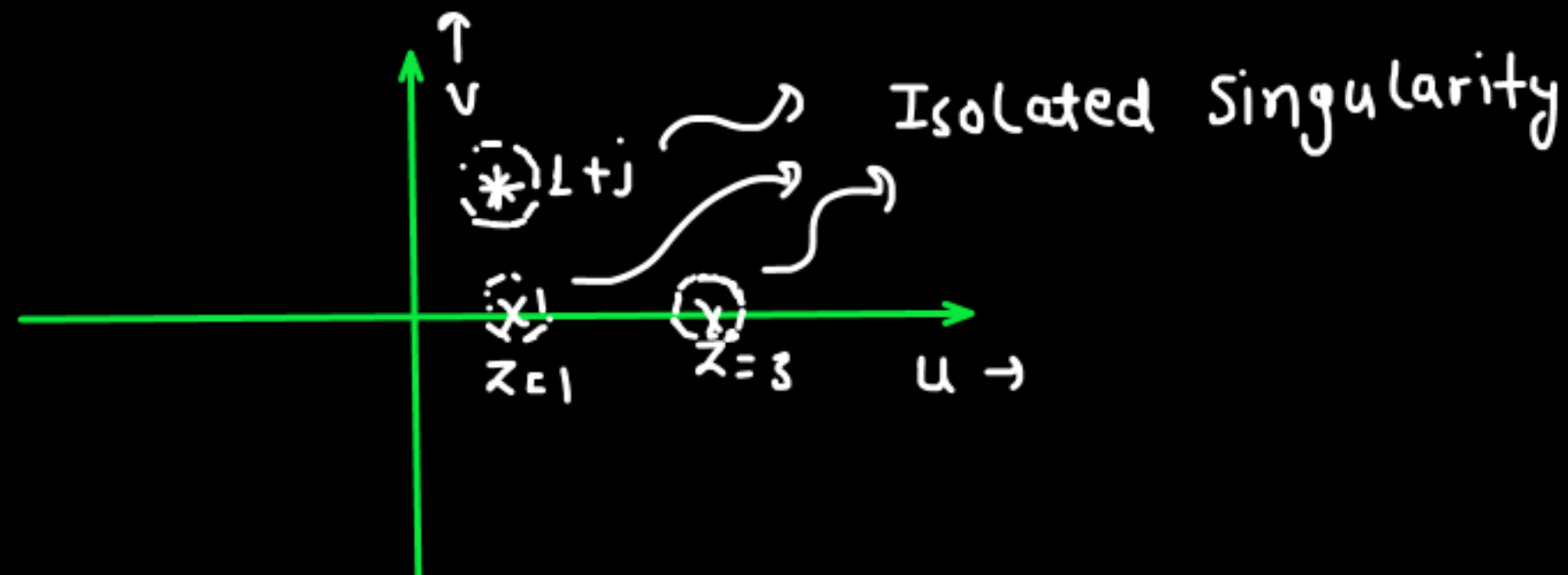
## Difference between Isolated and Non-Isolated Singularities

### Isolated Singularity

- A singularity is called an **isolated singularity** if there exists a neighborhood around the singular point in which there is no other singularity.
- In other words, it is the **only singular point** in some small region around it.

eg.  $\rightarrow f(z) = \frac{z}{(z-1)(z-3)(z-1-j)^2}$

$$z = 1, 3, 1+j, 1-j$$



## Non-Isolated Singularity

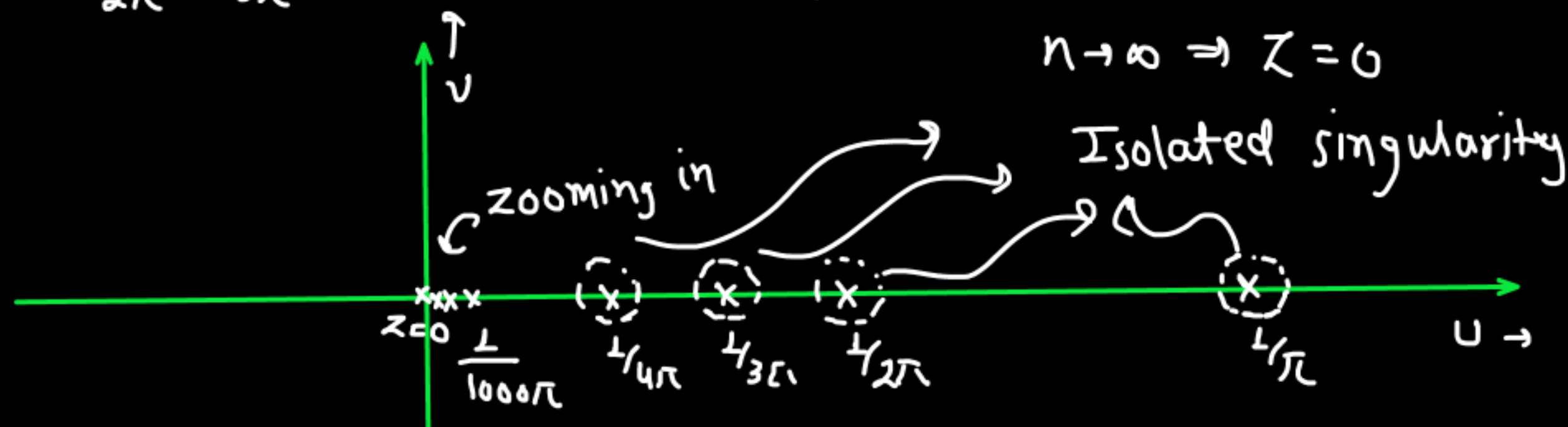
A singularity is called a non-isolated singularity if every neighborhood of the singular point contains at least one other singularity. In other words, there is no neighborhood around the singular point that is free from other singularities.

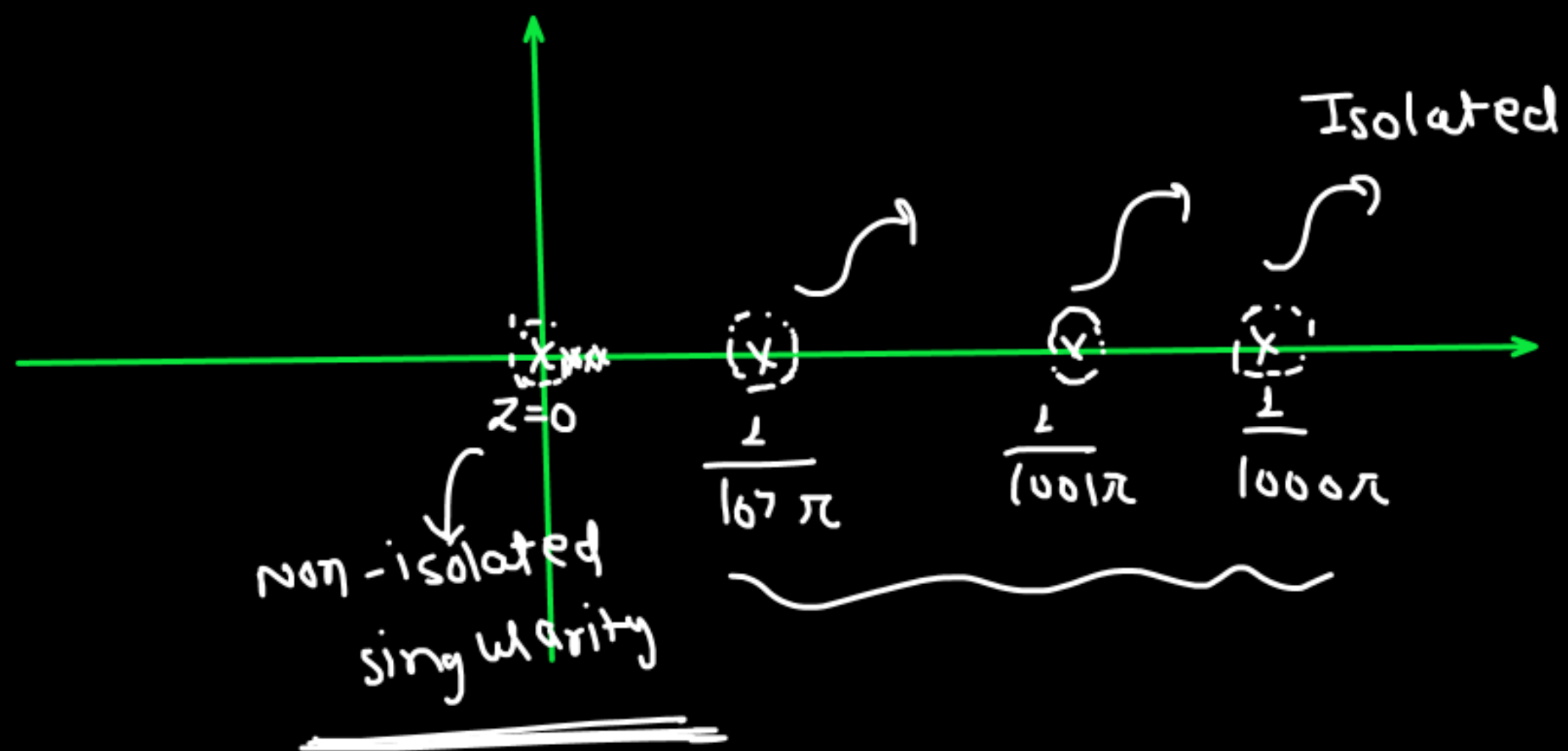
Eg.  $\rightarrow f(z) = \frac{1}{\sin(z/2)}$  ;  $\sin(z/2) = 0$

$$z/2 = n\pi \Rightarrow z = \frac{1}{n}\pi \quad n \in \mathbb{I}$$

$$z = \frac{1}{\pi}, \frac{1}{2\pi}, \frac{1}{3\pi}, \dots, \frac{1}{-1}, \frac{1}{-2\pi}, \frac{1}{-3\pi}, \dots$$

$$n \rightarrow \infty \Rightarrow z = 0$$





## 1. Simple Pole (Pole of Order 1) / single pole

A singularity at  $z = a$  is called a **simple pole** (or pole of order 1) if the function can be written in the form

$$f(z) = \frac{\phi(z)}{z - a}, \quad = \frac{N(z)}{D(z)} = 0 \Rightarrow z = a$$

where  $\phi(z)$  is analytic at  $z = a$  and  $\phi(a) \neq 0$ .

In other words, the denominator contains only one factor of  $(z - a)$ , and the function tends to infinity as  $z \rightarrow a$ .

$$D(z) = 0 \Rightarrow f(z) \rightarrow \infty$$

$z = z_0$ : pole

$$\text{eg. } f(z) = \frac{(z+2)(z-3)}{(z+1)} = \frac{N(z)}{D(z)}$$

$$D(z) = 0$$

$$z+1=0$$

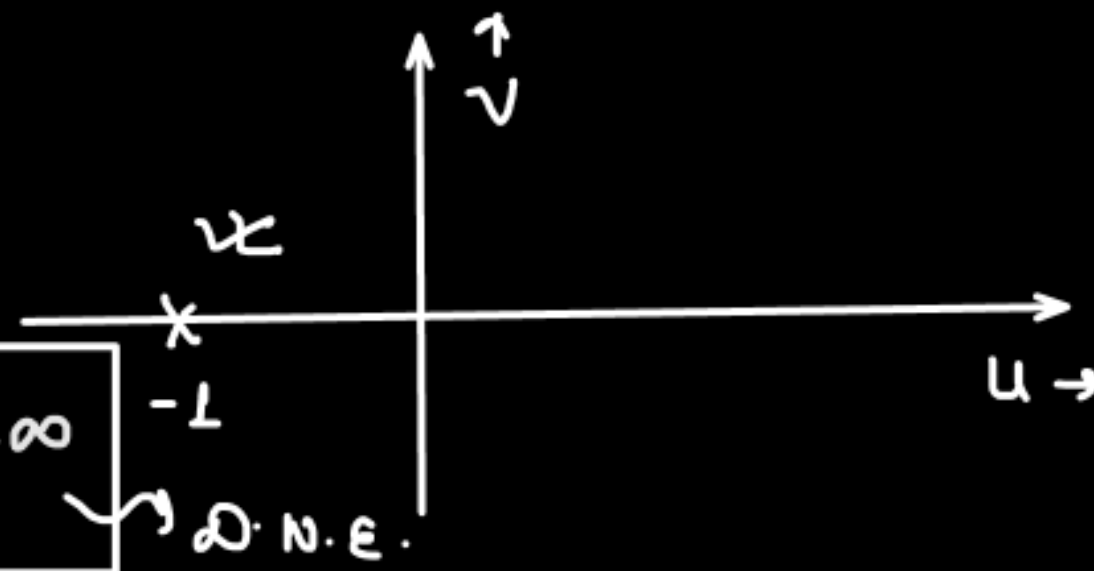
$$\hookrightarrow z = -1$$

$$\lim_{z \rightarrow -1} f(z) = -\infty$$

$$z \rightarrow -1$$

D.N.E.

$$z = -1 \Rightarrow f(z): \text{NOT analytic}$$



$$2. \quad f(z) = \frac{z+3}{(z+1)(z-1)(z^2+3z+4)} = \frac{N(z)}{D(z)}$$

$$D(z) = 0$$

$$(z+1)(z-1)(z^2+3z+4) = 0$$

$$z+1=0, \quad z-1=0, \quad z^2+3z+4=0$$

$$z = -1$$

$$z = 1$$

$$z = \frac{-3 \pm \sqrt{9-16}}{2} =$$

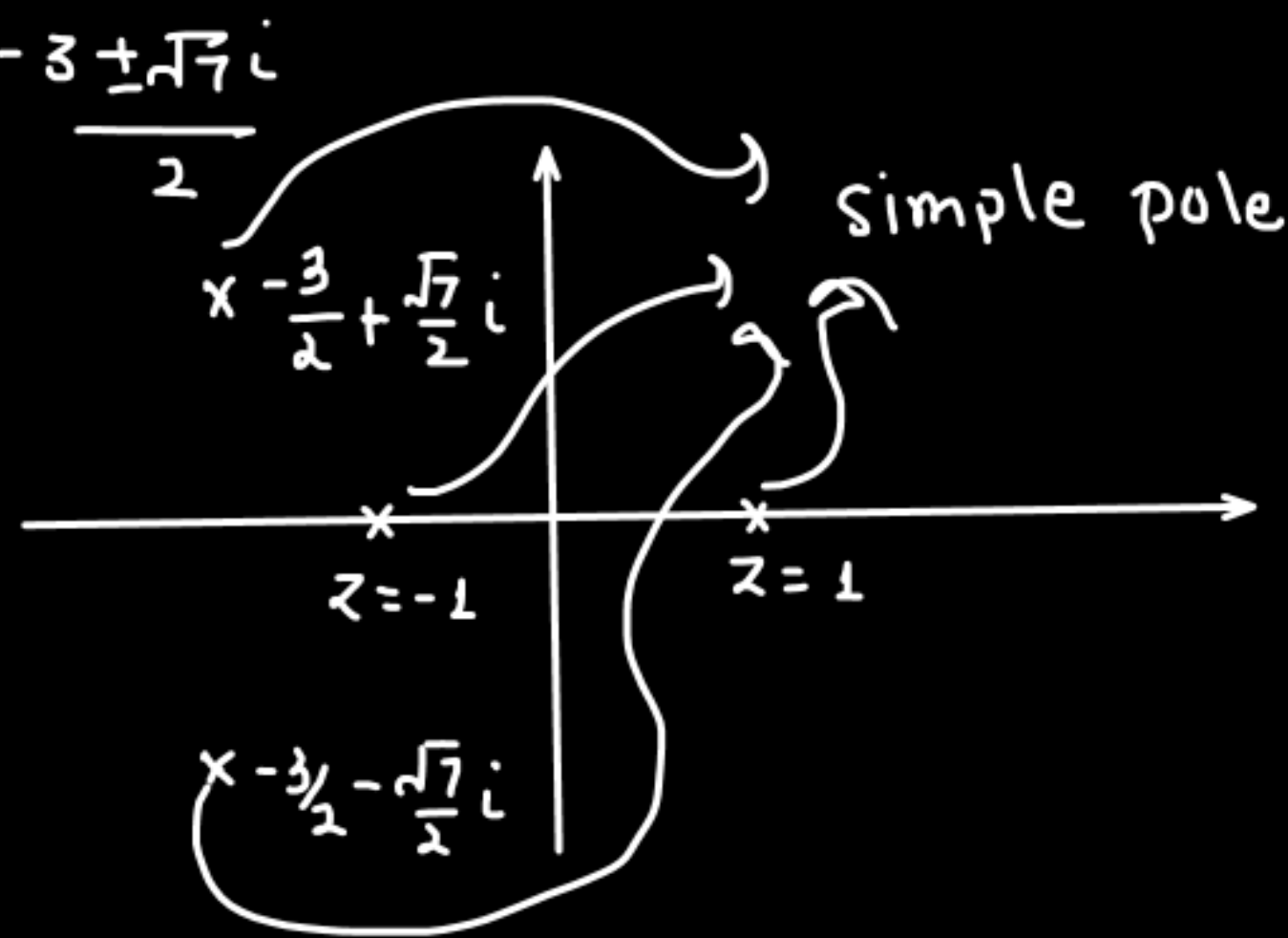
$$\frac{-3 \pm \sqrt{7}i}{2}$$

$$z = -1, 1, \frac{-3 - \sqrt{7}i}{2}, \frac{-3 + \sqrt{7}i}{2}$$



simple poles

$$\lim_{z \rightarrow z_0} f(z) = DNE$$





## 2. Multiple Pole (Pole of Order $n$ ) / Repeated pole

A singularity at  $z = a$  is called a **multiple pole** (or pole of order  $n$ , where  $n \geq 2$ ) if the function can be written in the form

$$f(z) = \frac{\phi(z)}{(z-a)^n},$$

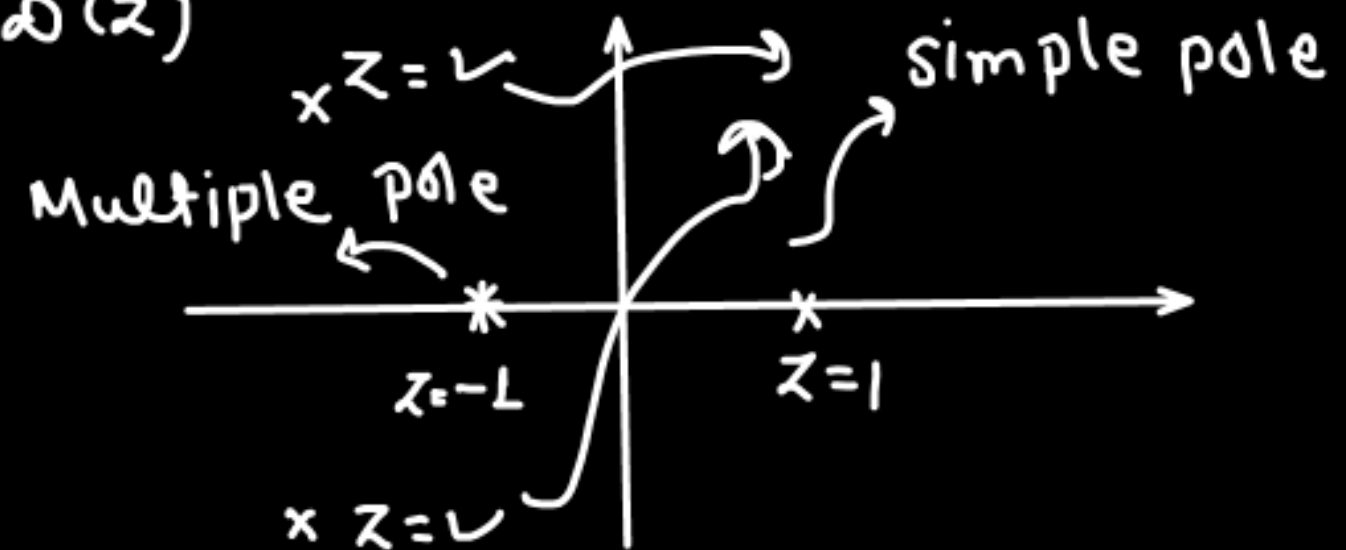
where  $\phi(z)$  is analytic at  $z = a$  and  $\phi(a) \neq 0$ .

In other words, the denominator contains  $n$  factors of  $(z-a)$ , and the function tends to infinity as  $z \rightarrow a$ .

$$\lim_{z \rightarrow z_0} f(z) = \infty \cdot N.E.$$

$$f(z) = \frac{z+3}{(z+1)^2(z-1)(z^2+3z+4)} = \frac{N(z)}{D(z)} \Rightarrow D(z) = 0$$

$$z = \underbrace{-1, -1}_{\text{pole of order } = 2}, \underbrace{\frac{-3-\sqrt{7}i}{2}}, \underbrace{\frac{-3+\sqrt{7}i}{2}}$$





### 3. Removable Singularity

A singularity at  $z = a$  is called a **removable singularity** if the function can be made analytic at  $z = a$  by suitably defining or redefining its value at that point.

Equivalently, if

$$\lim_{z \rightarrow a} f(z)$$

exists and is finite, then  $z = a$  is a removable singularity.

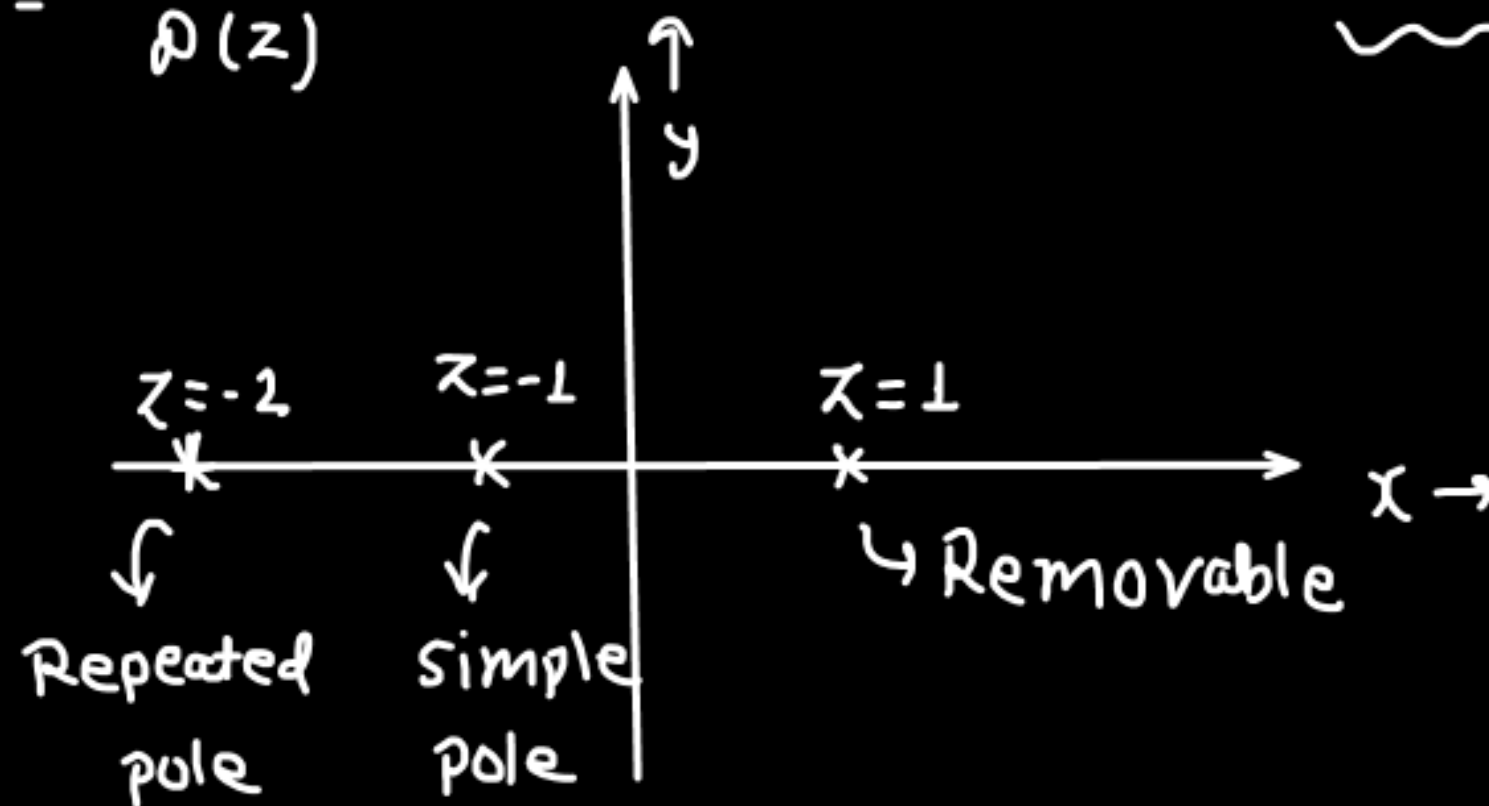
$$\text{Eg.} \rightarrow f(z) = \frac{z-1}{(z^2-1)(z+2)^2} = \frac{N(z)}{D(z)}$$

$$(z^2-1)(z+2)^2=0$$

$$(z+1)(z-1)(z+2)^2=0$$

$$\boxed{z = \underbrace{1}, \underbrace{-1}, \underbrace{-2, -2}}$$

↪



$$\lim_{z \rightarrow -1} f(z) = \frac{-2}{0} = -\infty$$

$$\lim_{z \rightarrow -2} f(z) = -\infty$$

$$\lim_{z \rightarrow 1} \frac{(z \cancel{-1})}{(z+1)(z \cancel{-1})(z+2)^2}$$

$$= \text{finite} = \frac{1}{18}$$

$$2. f(z) = \frac{\sin(z-a)}{(z-a)} = \frac{N(z)}{D(z)}$$

$$D(z) = 0$$

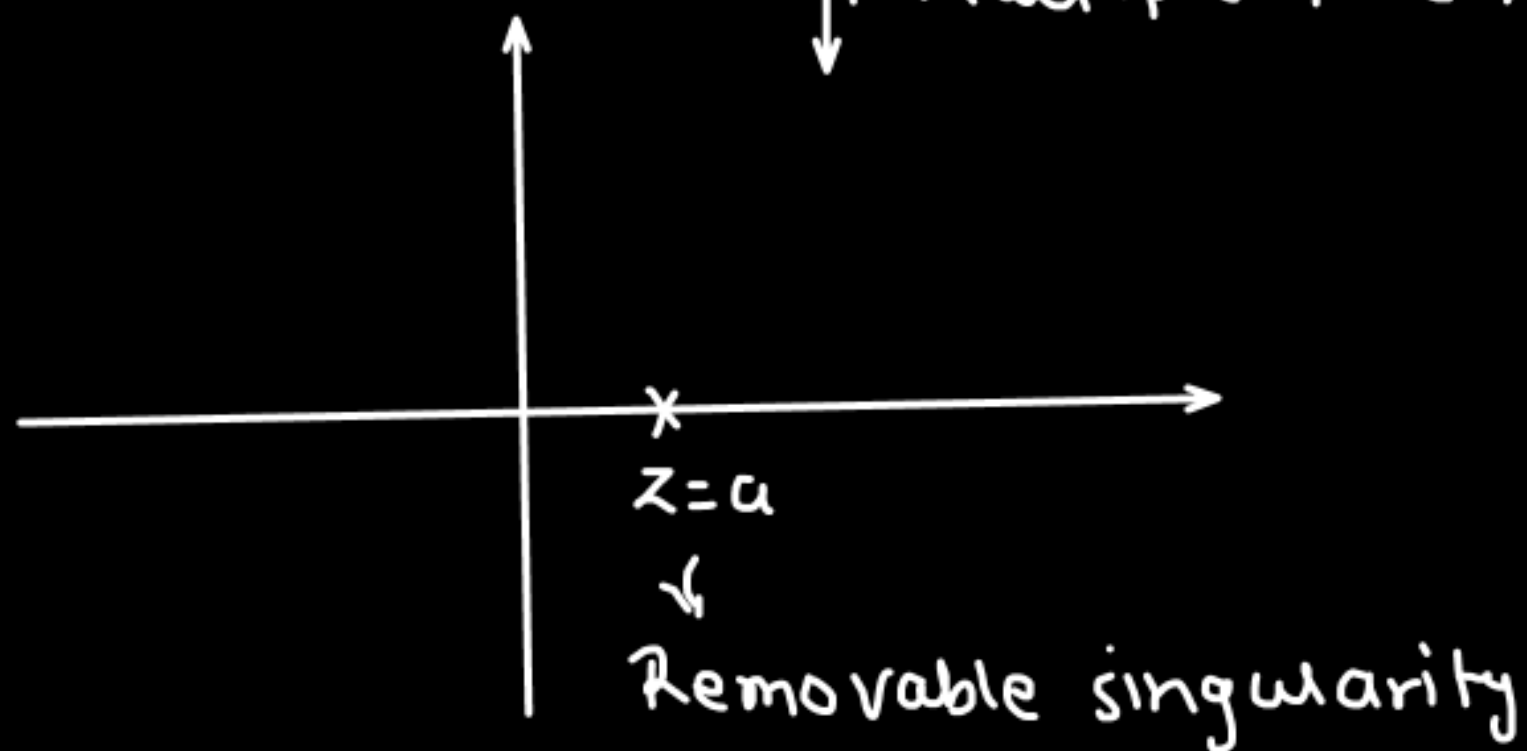
$$z-a=0$$

$$\boxed{z=a} \rightarrow \text{singularity}$$

$$\lim_{z \rightarrow a} \frac{\sin(z-a)}{(z-a)} = \frac{0}{0} = 1 : \text{Finite}$$

$$\lim_{z \rightarrow a} \frac{\cos(z-a)}{1} = 1$$

$\rightarrow$  Removable  $\checkmark$   
 $\rightarrow$  single pole  $\times$   
 $\rightarrow$  Multiple pole  $\times$



## Essential Singularity

A singularity at  $z = a$  is called an essential singularity if it is **neither a removable singularity nor a pole**.

In other words, the singularity cannot be removed by redefining the function, and it is not of finite pole order.

Eg.  $\rightarrow f(z) = e^{1/z} \rightarrow z=0$   
 $\hookrightarrow$  Expand

$$e^x = 1 + x + \frac{x^2}{2!} - - -$$

$$e^{1/z} = 1 + \frac{1}{z} + \left(\frac{1}{z}\right)^2 \times \frac{1}{2!} + \left(\frac{1}{z}\right)^3 \times \frac{1}{3!} + - - -$$

$$e^{1/z} = 1 + \frac{1}{z} + \frac{1}{2z^2} + \frac{1}{6z^3} + - - - - - + \frac{1}{n!} \left(\frac{1}{z}\right)^n$$

Essential  
singularity  $\hookrightarrow \boxed{z=0} \left(\frac{1}{z-0}\right)^\infty$

$z=z_0$ : Essential singular  
 $f(z) \equiv - - - \left(\frac{1}{z-z_0}\right)^\infty$

$$2. \quad f(z) = \sin\left(\frac{1}{z-2}\right)$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$$

$$= \frac{1}{z-2} - \left(\frac{1}{z-2}\right)^3 \times \frac{1}{3!} + \left(\frac{1}{z-2}\right)^5 \times \frac{1}{5!} - \dots - \left(\frac{1}{z-2}\right)^\infty$$

$$z-2=0$$

$$z=2$$

Essential Singularity

$$\frac{1}{z-2}$$

$z=z_0$ : Essential singular  $\rightarrow \frac{1}{z-2}$

## \* Finding Residues:-

{ simple  
Removable  
Multiple  
Essential

Case-1:- For simple pole

eg.- 1.  $f(z) = \frac{z+2}{z-1}$  ;  $z=1 \rightarrow$  simple pole

Residue of  $f(z)$  at  $z=1$  :- ,

$$R_1 = \lim_{z \rightarrow 1} (z-1) f(z) = \lim_{z \rightarrow 1} (\cancel{z-1}) \frac{(z+2)}{(\cancel{z-1})}$$

$$R_1 = 3$$

$$\hookrightarrow R_1 = \lim_{z \rightarrow 1} z+2 = 3$$

$f(z) \rightarrow$  singularity  
 $z=z_1, z_2, z_3, \dots$

$\hookrightarrow$  Residue of  $f(z)$  at

$z=z_1, z=z_2, z=z_3$

$\hookrightarrow R(f, z_1), R(f, z_2),$

$R(f, z_3)$

$\hookrightarrow R_1, R_2, R_3$

$$2. \quad f(z) = \frac{z^2 - 2}{(z+1)(z-2)(z+4)} \quad ; \quad \underbrace{z = -1, 2, -4}_{\text{simple pole}}$$

↳ Residue at  $z = -1$

$$R_1 = \lim_{z \rightarrow -1} (z+1) f(z) = \lim_{z \rightarrow -1} \frac{z^2 - 2}{(z-2)(z+4)} = \frac{-1}{(-3)(3)} = \frac{1}{9}$$

↳ Residue at  $z = 2$

$$R_2 = \lim_{z \rightarrow 2} (z-2) f(z) = \lim_{z \rightarrow 2} \frac{z^2 - 2}{(z+1)(z+4)} = \frac{2}{3 \times 6} = \frac{1}{9}$$

↳ Residue at  $z = -4$

$$R_3 = \lim_{z \rightarrow -4} (z+4) f(z) = \lim_{z \rightarrow -4} \frac{z^2 - 2}{(z+1)(z-2)} = \frac{14}{(-3)(-6)} = \frac{14}{18} = \frac{7}{9}$$



$$3. \quad f(z) = \frac{z-1}{z^2-z-2} ;$$

$$f(z) = \frac{z-1}{(z+1)(z-2)}$$

Residue at  $z = -1$

$$R_1 = \lim_{z \rightarrow -1} \frac{z-1}{z-2} = \frac{-2}{-3} = \frac{2}{3}$$

Residue at  $z = 2$

$$R_2 = \lim_{z \rightarrow 2} \frac{z-1}{z+1} = \frac{1}{3}$$

$$z^2 - z - 2 = 0$$

$$z^2 - 2z + z - 2 = 0 \quad \hookrightarrow z = -1, 2$$

$$(z+1)(z-2) = 0$$

M-2 : for simple pole only

$$f(z) \checkmark ; z = -1, 2$$

$$f(z) = \frac{N(z)}{D(z)} \Rightarrow \phi(z) = \frac{N(z)}{D'(z)} \Big|_{z=\text{sing}}$$

$$\text{Eq.} \rightarrow \phi(z) = \frac{z-1}{2z-1}$$

$$R_1 = \lim_{z \rightarrow -1} \frac{z-1}{2z-1} = \frac{2}{3} ; R_2 = \lim_{z \rightarrow 2} \frac{z-1}{2z-1} = \frac{1}{3}$$



M. 2

$$f(z) = \frac{N(z)}{D(z)}$$

;  $z = z_a$  <sup>Ad \*\*\*</sup> simple pole singularity

$$\phi(z) = \frac{N(z)}{D'(z)}$$

Residue at  $z=a$

$$R = \lim_{z \rightarrow a} \phi(z)$$

 ans =

Q.

## Question

Let  $f(z) = \tan z$ . Find all the singularities lying inside the circle  $|z| = 5$ . Also, find the sum of the residues of all the singularities.

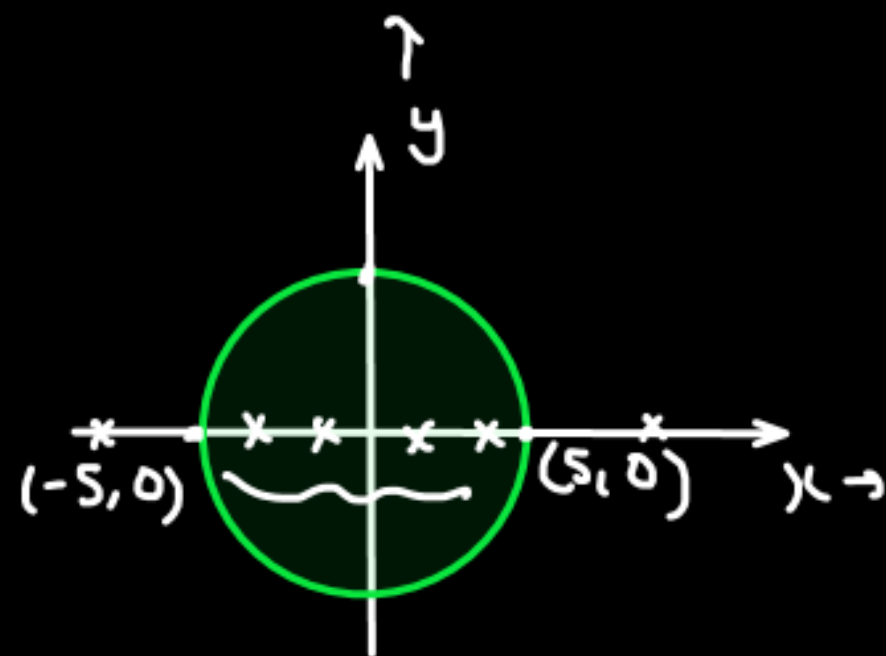
$$\hookrightarrow f(z) = \frac{\sin z}{\cos z};$$

$$\cos z = 0 \Rightarrow z = \pm \pi/2, \pm 3\pi/2, \pm 5\pi/2, \pm 7\pi/2, \dots$$

$$= \pm 1.57, \pm 4.71, \pm 7.85, \dots$$

$$z = -3\pi/2, -\pi/2, \pi/2, 3\pi/2$$

Simple pole singularity



Residue at  $z = -3\pi/2 \Rightarrow R_1 = \lim_{z \rightarrow -3\pi/2} (z + 3\pi/2) f(z) = \lim_{z \rightarrow -3\pi/2} (z + 3\pi/2) \tan z = 0 \times \infty = 0/0$

→ lengthy process

M-2

$$z = -3\pi/2, -\pi/2, \pi/2, 3\pi/2$$

simple pole

$$f(z) = \frac{\sin z}{\cos z}$$

$$\phi(z) = \frac{\sin z}{-\sin z} = -1$$

1. Residue at  $z = -3\pi/2$

$$R_1 = \lim_{z \rightarrow -3\pi/2} \phi(z) = \lim_{z \rightarrow -3\pi/2} (-1) = -1$$

2. Residue at  $z = -\pi/2$

$$R_2 = \lim_{z \rightarrow -\pi/2} -1 = -1$$

3. Residue at  $z = \pi/2$

$$R_3 = \lim_{z \rightarrow \pi/2} -1 = -1$$

4. Residue at  $z = 3\pi/2$  ;  $R_4 = \lim_{z \rightarrow 3\pi/2} (-1) = -1$

$$\sum R = -1 - 1 - 1 - 1 = -4$$

Q.

For the function  $f(z) = \frac{1}{\cos(2z)}$ , find the sum of residues inside the circle  $|z| = 2$ .

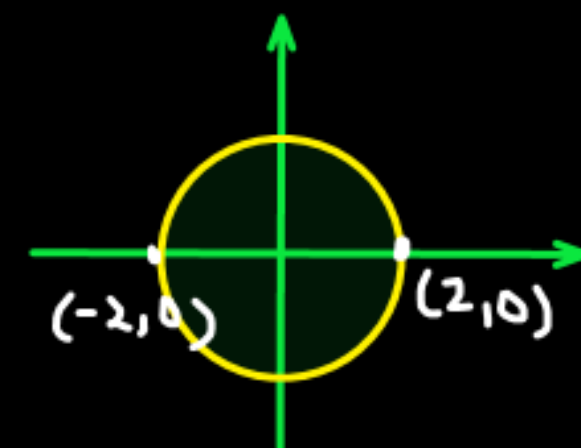
$$\hookrightarrow f(z) = \frac{1}{\cos(2z)}; \quad \cos 2z = 0$$

$$2z = \pm \pi/2, \pm 3\pi/2, \pm 5\pi/2, \dots$$

$$z = \pm \pi/4, \pm 3\pi/4, \pm 5\pi/4, \dots$$

$$= \pm 0.78, \pm 2.35, \dots$$

Residue at  $z = \pi/4, -\pi/4$   
simple pole



$$\underbrace{|z| < 2}$$

$$\phi(z) = \frac{1}{-2 \sin(2z)}; \quad R_1 = \lim_{z \rightarrow \pi/4} \frac{-1}{2 \sin(2z)} = \boxed{-\frac{1}{2}}$$

$$R_2 = \lim_{z \rightarrow -\pi/4} \frac{-1}{2 \sin(2z)} = \boxed{\frac{1}{2}}$$

$$\boxed{\sum R = 0}$$

Q.

What is the sum of residues of  $f(z) = \frac{1}{e^z - 1}$  inside the region  $|z| = 7$ ?

$$\hookrightarrow f(z) = \frac{1}{e^z - 1} \quad \left. \begin{array}{l} e^z - 1 = 0 \\ z = \ln 1 = 0 \end{array} \right\} \Rightarrow \underbrace{z=0} \quad \} \chi$$

$$e^z - 1 = 0$$

$$e^{(z + i2n\pi)} = 1 \quad \{ e^{i2n\pi} = 1 \}$$

$$z + i2n\pi = 0 \quad ; \quad n \in \mathbb{I}$$

$$z = -i2n\pi \Rightarrow z = \dots -i6\pi, -i4\pi, -i2\pi, 0, i2\pi, i4\pi, i6\pi \dots$$

$$\dots -i12.56, -i6.28, 0, i6.28, i12.56 \dots$$

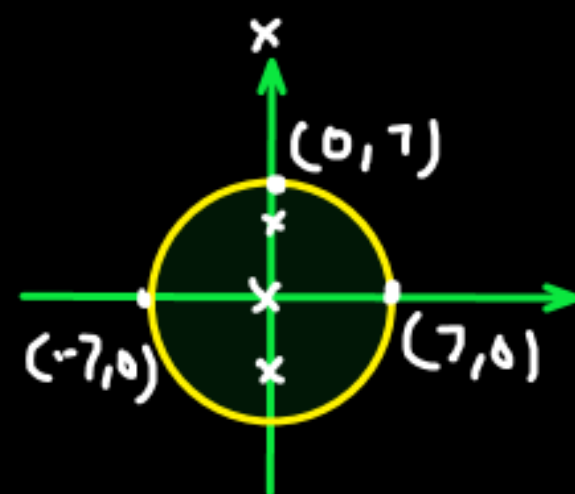
$\times$        $\checkmark$        $\checkmark$        $\checkmark$        $\times$

$|z| < 7$

$$z = 0, +i2\pi, -i2\pi$$

simple pole

$$\phi(z) = \frac{1}{e^z} \quad ; \quad \lim_{z \rightarrow \infty} \phi(z) = ?$$



$$R_1 = \lim_{z \rightarrow 0} \frac{1}{e^z} = 1 \quad ; \quad R_2 = \lim_{z \rightarrow i2\pi} \frac{1}{e^z} = \frac{1}{e^{i2\pi}} = 1$$

$$R_3 = \lim_{z \rightarrow -i2\pi} \frac{1}{e^z} = \frac{1}{e^{-i2\pi}} = 1$$

$$\sum R = 3$$



Q.

Sum the residues of  $f(z) = \frac{1}{\sinh z}$  inside the circle  $|z| = 4$ .

$$f(z) = \frac{1}{\frac{e^z - e^{-z}}{2}} = \frac{2}{e^z - e^{-z}}$$

$$\sinh z = \frac{e^z - e^{-z}}{2}$$

$$e^z - e^{-z} = 0 \Rightarrow e^z = e^{-z}$$

$$e^{(z + i2n\pi)} = e^{-z}$$

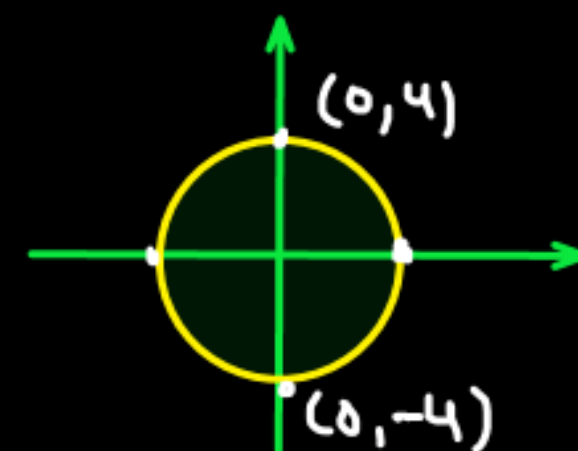
$$z + i2n\pi = -z$$

$$2z = -i2n\pi$$

$$z = -in\pi$$

$$z = -i\pi, 0, i\pi$$

simple pole



$$z = \dots -i2\pi, -i\pi, 0, i\pi, i2\pi \dots$$

$$\underbrace{-i6.28, -i3.14, 0, i3.14, i6.28}_{\text{checkmarks}} \quad \times$$



$$\phi(z) = \frac{2}{e^z + e^{-z}}$$

$$z=0;$$

$$R_1 = \lim_{z \rightarrow 0} \frac{2}{1+1} = 1$$

$$z=i\pi;$$

$$R_2 = \lim_{z \rightarrow i\pi} \frac{2}{-1-1} = -1$$

$$z=-i\pi;$$

$$R_3 = \lim_{z \rightarrow -i\pi} \frac{2}{-1-1} = -1$$

$$e^0 = 1, e^{i2\pi} = 1$$

$$e^{i\pi} = -1, e^{i\pi/2} = i$$

$$e^{-i\pi} = -1, e^{-i\pi/2} = -i$$

$$\Sigma R = 1 - 1 - 1 = -1$$

Case-2:- For Removable singularity:- Same Mtd as simple pole

Eg.  $\rightarrow$  ①  $f(z) = \frac{\sin(z-2)}{z-2}$  ;  $z=2$  : Removable singularity  
 $\lim_{z \rightarrow 2} f(z) = \text{Finite}$

Residue at  $z=2$

$$R = \lim_{z \rightarrow 2} (z-2) f(z) = \lim_{z \rightarrow 2} \cancel{(z-2)} \frac{\sin(z-2)}{\cancel{(z-2)}} = \sin(2-2) = \sin 0 = \boxed{0}$$

$$\boxed{R=0}$$

$$\textcircled{2} \quad f(z) = \frac{z-2}{z^2-4} = \frac{(z-2)}{(z+2)(z-2)} \quad ; \quad \begin{array}{l} z=2, -2 \\ \downarrow \quad \quad \quad \searrow \\ \text{Removable} \quad \text{simple pole} \end{array}$$

$$R_1 = \lim_{z \rightarrow -2} (z+2) f(z) = \lim_{z \rightarrow -2} \frac{z-2}{z-2} = 1$$

$$R_2 = \lim_{z \rightarrow 2} (z-2) f(z) = \lim_{z \rightarrow 2} \frac{(z-2)(z/2)}{(z+2)(z/2)} = \frac{0}{4} = \boxed{0}$$

Residue at Removable singularity = 0

$$\hookrightarrow f(z) = \frac{1}{z+2} \quad ; \quad \underbrace{z=-2}_{\text{Simple pole}} \quad ; \quad R = \lim_{z \rightarrow -2} 1 = 1 \quad ; \quad \boxed{\sum R = 1}$$

Case 3:- For Repeated Poles:-

$$\text{Eg.} \rightarrow 1. f(z) = \frac{e^{2z}}{(z-1)^2} ; (z-1)^2 = 0 \Rightarrow z=1, 1$$

$\underbrace{z=1}_{\hookrightarrow}$  Repeated pole of order 2

Residue at  $z=1$

$$R_1 = \lim_{z \rightarrow 1} \frac{1}{1!} \times \frac{d}{dz} (z-1)^2 \frac{e^{2z}}{(z-1)^2}$$

$$R_1 = \lim_{z \rightarrow 1} \frac{1}{1!} \frac{d}{dz} e^{2z} = \lim_{z \rightarrow 1} e^{2z} (2)$$

$$R = 2e^2$$

Ans.

$z=z_0$ : Repeated pole of order  $n$

$R =$

$$\lim_{z \rightarrow z_0} \frac{1}{(n-1)!} \times \frac{d^{n-1}}{dz^{n-1}} (z-z_0)^n f(z)$$

✓

$$2. \quad f(z) = \frac{\cos z}{(z+2)^3} ; \quad z = \underbrace{-2, -2, -2}_{3 \text{ times}}$$

↳ Residue at  $z = -2$

$$R = \lim_{z \rightarrow -2} \frac{1}{2!} \frac{d^2}{dz^2} \cos z$$

$$= \lim_{z \rightarrow -2} \frac{1}{2} [-\cos z] = -\frac{1}{2} \cos(-2) = \boxed{-\frac{1}{2} \cos 2} \text{ Ans.}$$

$$3. \quad f(z) = \frac{1}{(z-1)(z+2)^3} \rightarrow z = \underbrace{1}_{\text{simple pole}}, \underbrace{-2, -2, -2}_{\text{Multiple pole of order} = 3}$$

$$\cos z \xrightarrow{d/dz} -\sin z \xrightarrow{d/dz} -\cos z$$

↳ Residue at  $z=1$

$$R_1 = \lim_{z \rightarrow 1} \frac{1}{(z+2)^3} = \boxed{\frac{1}{27}}$$

↳ Residue at  $z=-2$

$$R_2 = \lim_{z \rightarrow -2} \left[ \frac{1}{2!} \underbrace{\frac{d^2}{dz^2} \frac{1}{z-1}} \right]$$

$$= \lim_{z \rightarrow -2} \frac{1}{2!} \times \frac{2}{(z-1)^3}$$

$$= \frac{1}{(-2-1)^3} = \boxed{-\frac{1}{27}}$$

$$\begin{aligned} \frac{1}{z-1} &\rightarrow \frac{-1}{(z-1)^2} \rightarrow \frac{2}{(z-1)^3} \\ &= - (z-1)^{-2} \end{aligned}$$



### Case-4:- Essential Singularity

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$\text{eg.} \rightarrow f(z) = e^{1/z} = 1 + x \frac{1}{z} + \frac{1}{z^2} \times \frac{1}{2!} + \frac{1}{z^3} \times \frac{1}{3!} + \dots$$

$z=0$ : Singularity  $\Rightarrow$  (Residue at  $z=0$ ) = coefficient of  $\frac{1}{z}$  in  $f(z)$

$$R = 1$$

$$\hookrightarrow f(z) = \sin\left(\frac{1}{z-2}\right) = \frac{1}{z-2} - \frac{1}{(z-2)^3} \times \frac{1}{3!} + \frac{1}{(z-2)^5} \times \frac{1}{5!} - \dots$$

$z=2$ : Singularity

$R$  = co-efficient of  $\frac{1}{z-2}$  in  $f(z) = \boxed{1}$  Ans. =

(Residue at  $z=z_0$ )  
= (co-efficient of  $\frac{1}{z-z_0}$  in  $f(z)$ )



$$3. f(z) = z^3 e^{1/z}$$

$$= z^3 \left[ 1 + \frac{1}{z} + \frac{1}{z^2 2!} + \frac{1}{z^3 3!} + \frac{1}{z^4 4!} + \frac{1}{z^5 5!} + \frac{1}{z^6 6!} \dots \right]$$

$$f(z) = z^3 + z^2 + \frac{z}{2!} + \frac{1}{3!} + \frac{1}{z \times 4!} + \frac{1}{z^2 \times 5!} + \frac{1}{z^3 \times 6!} \dots \left(\frac{1}{z}\right)^\infty$$

$z=0$  : Essential singularity

$$R = \text{coefficient of } \frac{1}{z} \text{ in } f(z) = \frac{1}{4!} = \boxed{\frac{1}{24}}_{\text{Ans.}}$$

$$4. f(z) = (z-2)^2 \cos\left(\frac{1}{z-2}\right)$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} \dots$$

$$= (z-2)^2 \left[ 1 - \frac{1}{(z-2)^2 2!} + \frac{1}{(z-2)^4 4!} - \frac{1}{(z-2)^6 6!} \dots \right]$$

$$f(z) = (z-2)^2 - \frac{1}{2!} + \frac{1}{4! (z-2)^2} - \frac{1}{6! (z-2)^4} \dots - \frac{1}{(z-2)^\infty}$$

↓  
Laurent

series expansion  $z=2$  : Essential singularity

$$+ 0 \cdot \left(\frac{1}{z-2}\right)$$

Residue = co-efficient of  $\frac{1}{z-2}$  if  $f(z)$

$$R=0$$

Q. The residue of the complex function  $f(z) = \sin(z) \cos\left(\frac{1}{z}\right)$  at the essential singularity  $z = 0$  is:

- (A) 0
- ☒ (B)  $-\sum_{n=0}^{\infty} \frac{1}{(2n+1)!(2n+2)!}$
- (C)  $\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!(2n+2)!}$
- (D)  $-\sum_{n=0}^{\infty} \frac{1}{(2n)!(2n+1)!}$

↳  $f(z) = \sin z \cdot \cos\left(\frac{1}{z}\right)$

$$f(z) = \left[ z - \frac{z^3}{3!} + \frac{z^5}{5!} - \frac{z^7}{7!} - \dots \right] \left[ 1 - \underbrace{\frac{1}{z^2 2!}} + \underbrace{\frac{1}{z^4 4!}} - \underbrace{\frac{1}{z^6 6!}} - \dots \right]$$

$z=0$ : Essential singularity

R = co-efficient of  $\frac{1}{z}$  in  $f(z)$

$$f(z) = \left[ z - \frac{z^3}{3!} + \frac{z^5}{5!} - \frac{z^7}{7!} - \dots \right] \left[ 1 - \underbrace{\frac{1}{z^2 2!}} + \underbrace{\frac{1}{z^4 4!}} - \underbrace{\frac{1}{z^6 6!}} - \dots \right]$$

$$z \left( -\frac{1}{z^2 2!} \right) - \frac{z^3}{3!} \left( \frac{1}{z^4 4!} \right) + \frac{z^5}{5!} \left( -\frac{1}{z^6 6!} \right) - \dots$$

$$= -\frac{1}{2! z} - \frac{1}{3! 4!} \times \frac{1}{z} - \frac{1}{5! 6!} \times \frac{1}{z} - \dots$$

$$= -\frac{1}{z} \left[ \frac{1}{2! 2!} + \frac{1}{3! 4!} + \frac{1}{5! 6!} + \dots \right]$$

$$\rightarrow \text{Residue} = - \sum_{n=0}^{\infty} \frac{1}{(2n+4)! (2n+2)!} \quad \checkmark$$

The commonly used expansions are:

$$\left\{ \begin{array}{l} (1+x)^{-1} = 1 - x + x^2 - x^3 + x^4 - \dots, \\ \hookrightarrow (1-x)^{-1} = 1 + x + x^2 + x^3 + x^4 + \dots, \\ (1+x)^{-2} = 1 - 2x + 3x^2 - 4x^3 + 5x^4 - \dots, \\ (1-x)^{-2} = 1 + 2x + 3x^2 + 4x^3 + 5x^4 + \dots, \end{array} \right. \quad \begin{array}{l} |x| < 1 \\ |x| < 1 \\ |x| < 1 \\ |x| < 1 \end{array}$$

Q.  $f(z) = \frac{1}{2-z}$ ; Expand  $f(z)$  (i)  $|z| < 2$  (ii)  $|z| > 2$

$$\hookrightarrow \text{(i)} \quad f(z) = (2-z)^{-1} = \frac{1}{2} (1 - z/2)^{-1} = \frac{1}{2} \left[ 1 + z/2 + (z/2)^2 + (z/2)^3 + \dots \right]$$

$$|z| < 2 \Rightarrow \underbrace{\frac{|z|}{2}} < 1 \quad = \frac{1}{2} + \frac{z}{4} + \frac{z^2}{8} + \frac{z^3}{16} + \dots$$

$$(ii) |z| > 2 \Rightarrow 1 > \frac{2}{|z|} \Rightarrow \underbrace{\frac{2}{|z|}} < 1 \quad |z| < 1$$

$$f(z) = (2-z)^{-1} = z^{-1} \left( \frac{2}{z} - 1 \right)^{-1} = -z^{-1} \left( 1 - \frac{2}{z} \right)^{-1} = -\frac{1}{z} \left( 1 - \frac{2}{z} \right)^{-1}$$

$$f(z) = -\frac{1}{z} \left( 1 - \frac{2}{z} \right)^{-1} = -\frac{1}{z} \left[ 1 + \frac{2}{z} + \left( \frac{2}{z} \right)^2 + \left( \frac{2}{z} \right)^3 - \dots \right]$$

$$f(z) = -\frac{1}{z} - \frac{2}{z^2} - \frac{4}{z^3} - \frac{8}{z^4} - \dots$$

↪ Laurent's  
series  
expansion

Taylor series ↪  
power always +ve

Laurent series ↪  
power could be  
both +ve and -ve

$$\left\{ (1-x)^{-1} = 1 + x + x^2 + x^3 - \dots \right\}$$



Q. GATC

In the Laurent series expansion of

$$|z| < 1$$

$$f(z) = \frac{1}{(z-1)(z-2)},$$

valid in the region

$$1 < |z| < 2,$$

$$\curvearrowright \begin{aligned} |z| &> 1 \\ |z| &< 2 \end{aligned}$$

the coefficient of  $\frac{1}{z^2}$  is:

$$\hookrightarrow f(z) = \frac{1}{z-2} - \frac{1}{z-1} ; \quad |z| > 1, \quad |z| < 2$$

$$\underbrace{\frac{1}{|z|} < 1}$$

$$\underbrace{\frac{|z|}{2} < 1}$$

$$\frac{1}{z-2} = (z-2)^{-1} = 2^{-1} \left( \frac{z}{2} - 1 \right)^{-1} = -\frac{1}{2} \left( 1 - \frac{z}{2} \right)^{-1}$$



$$\frac{1}{z-2} = -\frac{1}{2} (1 - z/2)^{-1} = -\frac{1}{2} [1 + z/2 + (z/2)^2 + (z/2)^3 + \dots]$$

$$\boxed{\phi_1(z) = \frac{1}{z-2} = -\frac{1}{2} - \frac{z}{4} - \frac{z^2}{8} - \frac{z^3}{16} - \dots} \quad \text{و } 0$$

$$4 \quad \phi_2(z) = \frac{-1}{z-1} = -1(z-1)^{-1} = -z^{-1} (1 - 1/z)^{-1} = -\frac{1}{z} (1 - \frac{1}{z})^{-1}$$

$$\frac{-1}{z-1} = -\frac{1}{z} (1 - \frac{1}{z})^{-1} = -\frac{1}{z} [1 + \frac{1}{z} + \frac{1}{z^2 2!} + \frac{1}{z^3 3!} + \dots]$$

$$\boxed{\phi_2(z) = \frac{-1}{z-1} = -\frac{1}{z} - \frac{1}{z^2} - \frac{1}{z^3 3!} - \frac{1}{z^4 4!} - \dots} \quad \text{و } -1$$

$$f(z) = \phi_1(z) + \phi_2(z)$$

$$f(z) = -\frac{1}{2} - \frac{z}{4} - \frac{z^2}{8} - \frac{z^3}{16} - \dots - \frac{1}{z} - \frac{1}{z^2} - \frac{1}{z^3} \dots$$

+ve powers

-ve powers

Laurent's series

$$\text{co-efficient of } \frac{1}{z^2} = -1$$

Q.  $f(z) = \frac{e^{1/2}}{z-1}$ ; sum of all Residues.

$$f(z) = \frac{1}{z-1} \left[ 1 + \frac{1}{z} + \frac{1}{z^2 2!} + \frac{1}{z^3 3!} + \dots \right]$$

$$f(z) = \frac{1}{(z-1)} + \frac{1}{(z-1)z} + \frac{1}{(z-1)z^2 2!} + \frac{1}{(z-1)z^3 3!} + \dots$$

$z=1, 0$  : singularities

$z=1$  : simple pole  $\leadsto R_1 = \lim_{z \rightarrow 1} (z-1) f(z) = \lim_{z \rightarrow 1} \frac{\cancel{z-1} e^{1/2}}{\cancel{(z-1)}}$

$z=0$  : Essential Singularity

$\hookrightarrow R_2$  : coefficient of  $\frac{1}{z}$  in  $f(z)$

$R_1 = e \approx 2.7$

$$R_2 = \frac{1}{z-1}$$

X X X X

$$\begin{array}{c} |x| < 1 \\ \underbrace{\hspace{1cm}} \\ \hookrightarrow \underbrace{|z| < 1} \end{array}$$

$f(z)$  = Expanded ~~11~~

$$f(z) = \underbrace{(z-1)^{-1}} \cdot e^{1/z}$$

$z \rightarrow 0$ : Residue at  $z=0$

$$f(z) = -(1-z)^{-1} \cdot e^{1/z}$$

$$= -(1+z+z^2+z^3+z^4+\dots) \left( 1 + \frac{1}{z} + \frac{1}{z^2 2!} + \frac{1}{z^3 3!} + \frac{1}{z^4 4!} + \dots \right)$$

$$= - \left[ \frac{1}{z} + \frac{1}{2! z} + \frac{1}{3! z} + \frac{1}{4! z} + \dots \right]$$

$$= - \left[ 1 + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \dots \right] \frac{1}{z}$$

co. efficient of  $z^2$  in  $f(z)$

$$R_2 = - \left[ 1 + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \dots \right]$$

$$= - [e - 1]$$

$$R_2 = 1 - e$$

$$\begin{aligned} \Sigma R &= R_1 + R_2 \\ &= e + 1 - e \end{aligned}$$

$$\Sigma R = 1$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$e^x - 1 = x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$e^{-1} = 1 + \frac{1}{2!} + \frac{1}{3!} + \dots$$

Q. Let

$$f(z) = \frac{e^{\frac{1}{z-2}}}{z-1}.$$

Find the sum of the residues of all the singularities lying inside the circle

$$|z| = 3.$$

$$\hookrightarrow f(z) = \frac{e^{\frac{1}{z-2}}}{z-1}$$

$\hookrightarrow z=1$ : simple pole  $\hookrightarrow R_1 = \lim_{z \rightarrow 1} e^{\frac{1}{z-2}} = 1/e$   
 $z=2$ : Essential sing.  $\hookrightarrow R_2 = ?$

$z=2$

$$f(z) = \frac{1}{z-1} \left[ 1 + \frac{1}{z-2} + \frac{1}{(z-2)^2 2!} + \frac{1}{(z-2)^3 3!} + \dots \right]$$

$\downarrow$   
Expand about  $z=2$

$$\begin{aligned} z-2 &= 0 \\ |z-2| &= 0 \\ |z-2| &< 1 \end{aligned}$$



$$(z-1)^{-1} = (1 + \underline{z-2})^{-1} = 1 - (z-2) + (z-2)^2 - (z-2)^3 + \dots$$

$$f(z) = [1 - (z-2) + (z-2)^2 - (z-2)^3 + \dots] \left[ 1 + \frac{1}{z-2} + \frac{1}{(z-2)^2} 2! + \frac{1}{(z-2)^3} 3! + \frac{1}{(z-2)^4} 4! \dots \right]$$

$z-2=0$   
 $|z-2| < 1$

$$= \left[ 1 - \frac{1}{2!} + \frac{1}{3!} - \frac{1}{4!} \dots \right] \frac{1}{z-2}$$

$$= [1 - 1/e] \times \frac{1}{z-2}$$

$$R_2 = [1 - 1/e]$$



$$e^{-x} = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \frac{x^4}{4!} - \dots$$

$$e^{-1} = 1 - 1 + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \dots$$

$$\frac{1}{e} - 1 = -1 + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \dots$$

$$1 - \frac{1}{e} = 1 - \frac{1}{2!} + \frac{1}{3!} - \frac{1}{4!} + \dots$$

$$R_1 = 1/e$$

$$R_2 = 1 - \frac{1}{e}$$

$$\Sigma R = 1$$

Q. Let

$$f(z) = \frac{6z}{2z^4 - 3z^3 + 7z^2 - 3z + 5}$$

$$f(i) = \frac{6i}{0}$$

$\rightarrow \infty$   
Removable  $\times \times$

Find the residue of  $f(z)$  at the pole  $z = i$ .

↳  $Q(z) = 2z^4 - 3z^3 + 7z^2 - 3z + 5 = 0$

polynomial  
with real  
co-efficient

$$z = i, -i, \cancel{x}, -\cancel{x} \rightarrow \text{poles}$$

→ simple pole ✓  
→ Multiple pole  $\times \times$   
→ Removable pole  $\times$

$$\text{sum of roots} = \frac{3}{2} \neq 0$$

$$R = \lim_{z \rightarrow i} \frac{6z}{8z^3 - 9z^2 + 14z - 3} = \frac{6i}{-8i + 9 + 14i - 3} = \frac{i}{1+i} = \frac{1+i}{2}$$

$$f(z) = \frac{N(z)}{D(z)} \quad ; \quad D(z)=0 \Rightarrow z=z_0: \text{Simple pole}$$

↳ Residue

$$D(z_0) = 0$$

$$R = \lim_{z \rightarrow z_0} (z - z_0) \frac{N(z)}{D(z)}$$

$$= \frac{(z_0 - z_0) N(z_0)}{D(z_0)} = \frac{0}{0} \text{ } \hookrightarrow \text{L.H. Rule}$$

$$\Phi(z) = \frac{N(z)}{D'(z)}$$

$$R = \lim_{z \rightarrow z_0} \frac{(z - z_0) N'(z) + N(z) \cdot 1}{D'(z)} = \lim_{z \rightarrow z_0} \frac{N(z)}{D'(z)}$$

$$= \lim_{z \rightarrow z_0} \Phi(z) \quad \checkmark$$

Q.  $f(z) = \left(z + \frac{1}{z}\right)^2$  ; singularity and Residue?

↳  $f(z) = \frac{(z^2 + 1)^2}{z^2}$  ;  $z^2 = 0 \Rightarrow \underbrace{z=0, 0}_{\text{Repeated poles ; order 2}}$

$$R = \lim_{z \rightarrow 0} \frac{1}{1!} \frac{d}{dz} (z^2 + 1)^2 = \lim_{z \rightarrow 0} 2(z^2 + 1) 2z = 0$$

$$\boxed{R=0}$$

Q.

## Question

Let

$$f(z) = \frac{z^2}{(z^2 - 3z + 2)^2}.$$

Find the sum of all the residues of  $f(z)$ .

$$4 \quad f(z) = \frac{z^2}{[(z-2)(z-1)]^2} = \frac{z^2}{(z-2)^2(z-1)^2} = \frac{N(z)}{D(z)} \quad ; \quad D(z) = 0$$

$$[(z-2)(z-1)]^2 = 0$$

$$\boxed{z = 2, 2, 1, 1}$$

Repeated poles of order 2 =

Residue at  $z=2$

$$R_1 = \lim_{z \rightarrow 2} \frac{1}{1!} \frac{d}{dz} \left( \frac{z^2}{(z-1)^2} \right) = \lim_{z \rightarrow 2} \frac{-2z}{(z-1)^3} = -4$$

Residue at  $z=1$

$$R_2 = \lim_{z \rightarrow 1} \frac{1}{1!} \frac{d}{dz} \left( \frac{z^2}{(z-2)^2} \right) = \lim_{z \rightarrow 1} \frac{-4z}{(z-2)^3} = 4$$

$$\boxed{R_1 = -4}, \quad \boxed{R_2 = 4}$$

$$\boxed{\sum R = 0}$$



## Question

Let

$$f(z) = \frac{1}{z \sin z}.$$

Find the sum of all the residues of  $f(z)$  lying inside the circle

$$|z| = 7.$$

↳  $f(z) = \frac{1}{z \sin z}$

$\underbrace{z \sin z = 0}$

$z = 0$

$\sin z = 0$

$z = 0, \pm\pi, \pm 2\pi, \pm 3\pi, \dots$

$0, \pm 3.14, \pm 6.28, 9.42, \dots$

$z = 0, 0, +\pi, -\pi, -2\pi, 2\pi$

Multiple pole

simple poles



Residue at  $z=0$

$$R_1 = \lim_{z \rightarrow 0} \frac{1}{1!} \frac{d}{dz} z^2 f(z)$$

$$R_1 = \lim_{z \rightarrow 0} \frac{d}{dz} \frac{z^2 \times 1}{z \sin z}$$

$$= \lim_{z \rightarrow 0} \frac{d}{dz} \frac{z}{\sin z}$$

$$= \lim_{z \rightarrow 0} \frac{(\sin z) 1 - z \cos z}{(\sin z)^2} = \frac{0-0}{0} \rightarrow \text{h.h.}$$

$$= \lim_{z \rightarrow 0} \frac{\cos z - [-z \sin z + \cos z]}{2 \sin z \cos z}$$

simple pole  
multiple pole

Removable

$z = z_0$ : order  $n$

$$R = \lim_{z \rightarrow z_0} \frac{1}{(n-1)!} \frac{d^{n-1}}{dz^{n-1}} (z-z_0)^n f(z)$$

$$R_1 = 0$$

$$= \lim_{z \rightarrow 0} \frac{z \sin z}{2 \cancel{\sin z} z \cos z} = 0$$

$$f(z) = \frac{1}{z \sin z}$$

Residue at  $z = \pi$

$$R_2 = \lim_{z \rightarrow \pi} (z - \pi) f(z) = \lim_{z \rightarrow \pi} \frac{z - \pi}{z \sin z} \stackrel{\text{L.H.}}{=} \lim_{z \rightarrow \pi} \frac{1}{z \cos z + \sin z}$$

$$R_2 = -\frac{1}{\pi}$$

$$\boxed{\sum R = 0} \text{ Ans.}$$

M.2

$$f(z) = \frac{1/z}{\sin z}$$

$$R_2 = \lim_{z \rightarrow \pi} \frac{1/z}{\cos z} = -\frac{1}{\pi}$$

$$\boxed{R_3 = 1/\pi, R_4 = 1/2\pi, R_5 = -1/2\pi}$$

$$\text{at } z = \pi \Rightarrow \frac{1}{z} = \frac{1}{\pi}$$

$$\sin z = \sin \pi = 0$$

$$\begin{matrix} R_3 & R_4 & R_5 \\ z = -\pi, 2\pi, -2\pi \end{matrix}$$

$\nabla 1/z$ : no problem =

Q.  $f(z) = \frac{\sin z}{z^2}$  ; Residue = ?

$\hookrightarrow f(z) = \frac{\sin z}{z^2}$  ;  $\underbrace{z=0, 0}_{\text{multiple pole}}$

$$R = \lim_{z \rightarrow 0} \frac{1}{1!} \frac{d}{dz} z^2 f(z) = \lim_{z \rightarrow 0} \frac{d}{dz} \sin z = \lim_{z \rightarrow 0} \cos z = \boxed{1}$$

$M-2$

$z=0, 0 \hookrightarrow \text{Removable}$   
 $\downarrow$   
 simple  
 $\downarrow$  pole  
 $R_2 = ?$

$\Downarrow$   
 $\boxed{R_1 = 0}$

$$R_2 = \lim_{z \rightarrow 0} z f(z) = \lim_{z \rightarrow 0} z \times \frac{\sin z}{z^2}$$

$\boxed{R_2 = 1}$   $\boxed{\sum R = 1}$   $\hookrightarrow \lim_{z \rightarrow 0} \frac{\sin z}{z} = 1$

Q.  $f(z) = \frac{\sin z}{z^3}$  ;  $z=0$  ~~(a)~~ pole of order = 2  
~~(b)~~ pole of order = 3

↳  $z^3 = 0 \Rightarrow z = 0, 0, 0^{\rightarrow}$   
Multiple pole of order = 3 X X

↳  $f(z) = \frac{[z - \frac{z^3}{3!} + \frac{z^5}{5!} - \frac{z^7}{7!} \dots]}{z^3}$

$\lim_{z \rightarrow 0} \frac{\sin z}{z} = 1$   
 ↳  $z=0$   
 ↓  
 Removable

↳  $f(z) = \frac{1}{\boxed{z^2}} - \frac{1}{3!} + \frac{z^2}{5!} - \frac{z^4}{7!} \dots$

$z^2 = 0$   
 $z = 0, 0$  ∴ pole of order = 2

$$f(z) = \frac{\sin z}{z^3}; \text{ Residue} = ?$$

$$4 \text{ (i) } z=0, 0, 0 \rightarrow \text{Removable sing. } \boxed{R_1=0}$$

pole of order = 2

$$\boxed{\Sigma R = 0}$$

$$R_2 = \lim_{z \rightarrow 0} \frac{1}{1!} \frac{d}{dz} z^2 \cdot \frac{\sin z}{z^3} = \lim_{z \rightarrow 0} \frac{d}{dz} \frac{\sin z}{z} = \lim_{z \rightarrow 0} \frac{z \cos z - \sin z}{z^2} \quad \text{0/0}$$

$$(2) f(z) = \frac{\sin z}{z^3} \quad z=0, 0, 0 \text{ pole of order} = 3$$

$$= \lim_{z \rightarrow 0} \frac{-z \sin z + \cos z - \cos z}{2z}$$

$$R = \lim_{z \rightarrow 0} \frac{1}{2!} \frac{d^2}{dz^2} \sin z = \lim_{z \rightarrow 0} \frac{-\sin z}{2!} = 0$$

$$\boxed{R_2 = 0}$$

$$\boxed{R=0} \text{ ANS.}$$

## Complex Integral

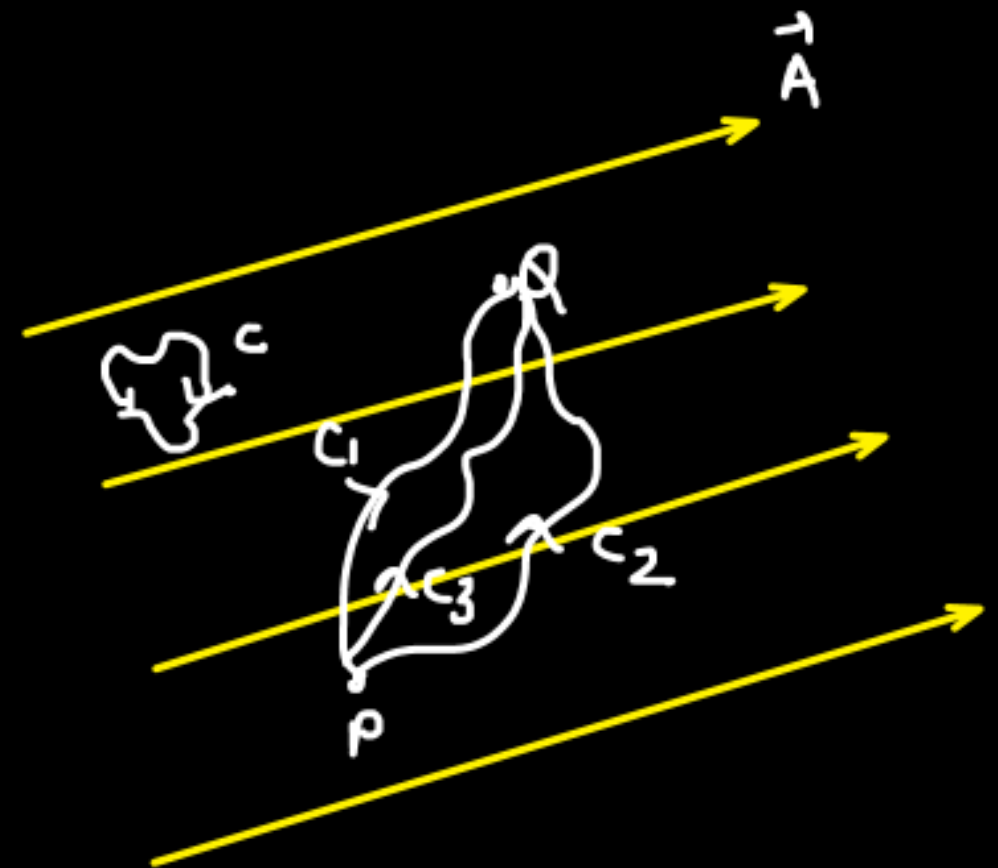
↳ Do you Remember:-

$$\oint_C \vec{A} \cdot d\vec{l} = 0$$

Any closed path

$$\int_{C_1} \vec{A} \cdot d\vec{l} = \int_{C_2} \vec{A} \cdot d\vec{l} = \int_{C_3} \vec{A} \cdot d\vec{l} = \alpha$$

$\vec{A}$ : conservative vector field





Q.  $f(z)$ : Analytic  $f^n$  in Domain  $D$ .

Find  $\oint_C f(z) \cdot dz = ?$

$$f(z) = u + iv$$

↳

$$I = \oint_C f(z) \cdot dz$$

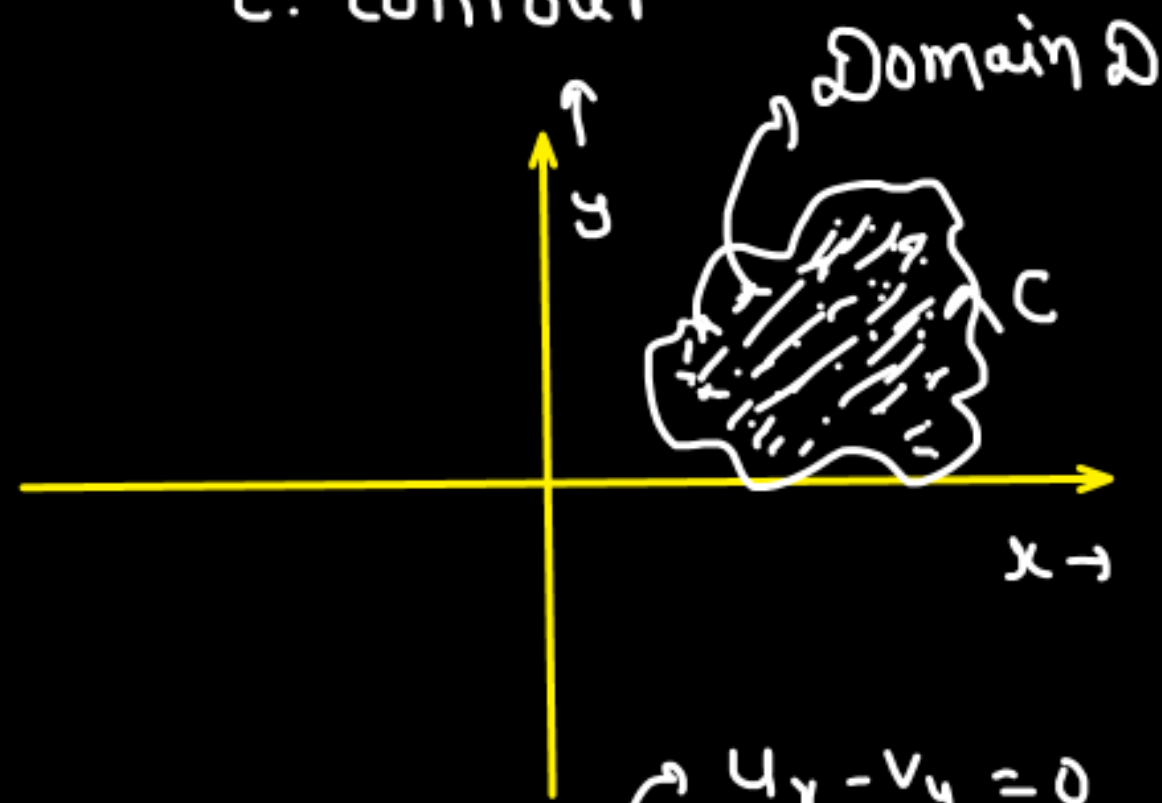
$$= \oint_C (u + iv)(dx + idy)$$

$$= \oint_C u dx + i \oint_C u dy + i \oint_C v dx - \oint_C v dy$$

$$I = \oint_C \underbrace{(u dx - v dy)}_{I_1} + i \oint_C \underbrace{(u dy + v dx)}_{I_2}$$

$$z = x + iy$$
$$dz = dx + i dy$$

$C$ : contour



$$\hookrightarrow u_x - v_y = 0$$

$$u_x = v_y$$

$$u_y = -v_x$$

$$u_y + v_x = 0$$



$$I_1 = \oint_C u dx - v dy = \oint_C (\underbrace{u \hat{i} - v \hat{j}}_{\vec{A}}) (\underbrace{dx \hat{i} + dy \hat{j}}_{\vec{dl}})$$

$$= \oint_C \vec{A} \cdot \vec{dl} = \iint_S (\nabla \times \vec{A}) \cdot \vec{ds}$$

$$I_1 = \iint_D \underbrace{-(v_x + u_y)}_0 dx dy = 0$$

$$\nabla \times \vec{A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ u & -v & 0 \end{vmatrix} \quad \vec{ds} = dx dy \hat{k} = \hat{k} (-v_x - u_y)$$

$$I_2 = \iint_D \underbrace{(u_x - v_y)}_0 dx dy = 0$$

$$I = I_1 + I_2 = 0$$

$$\frac{\partial v}{\partial z} = 0, \quad \frac{\partial u}{\partial z} = 0$$

$$v = v(x, y), \quad u = u(x, y)$$

$$\oint_C f(z) \cdot dz = 0; \text{ if } f(z) \text{ is analytic in}$$

Domain  $D$ : Region bounded by contour  $C$

$\checkmark f(z)$ : Analytic in domain  $D \Rightarrow \oint_C f(z) \cdot dz = 0$   $\uparrow y$

$f(z)$ : Analytic  $\equiv \vec{A}$  conservative



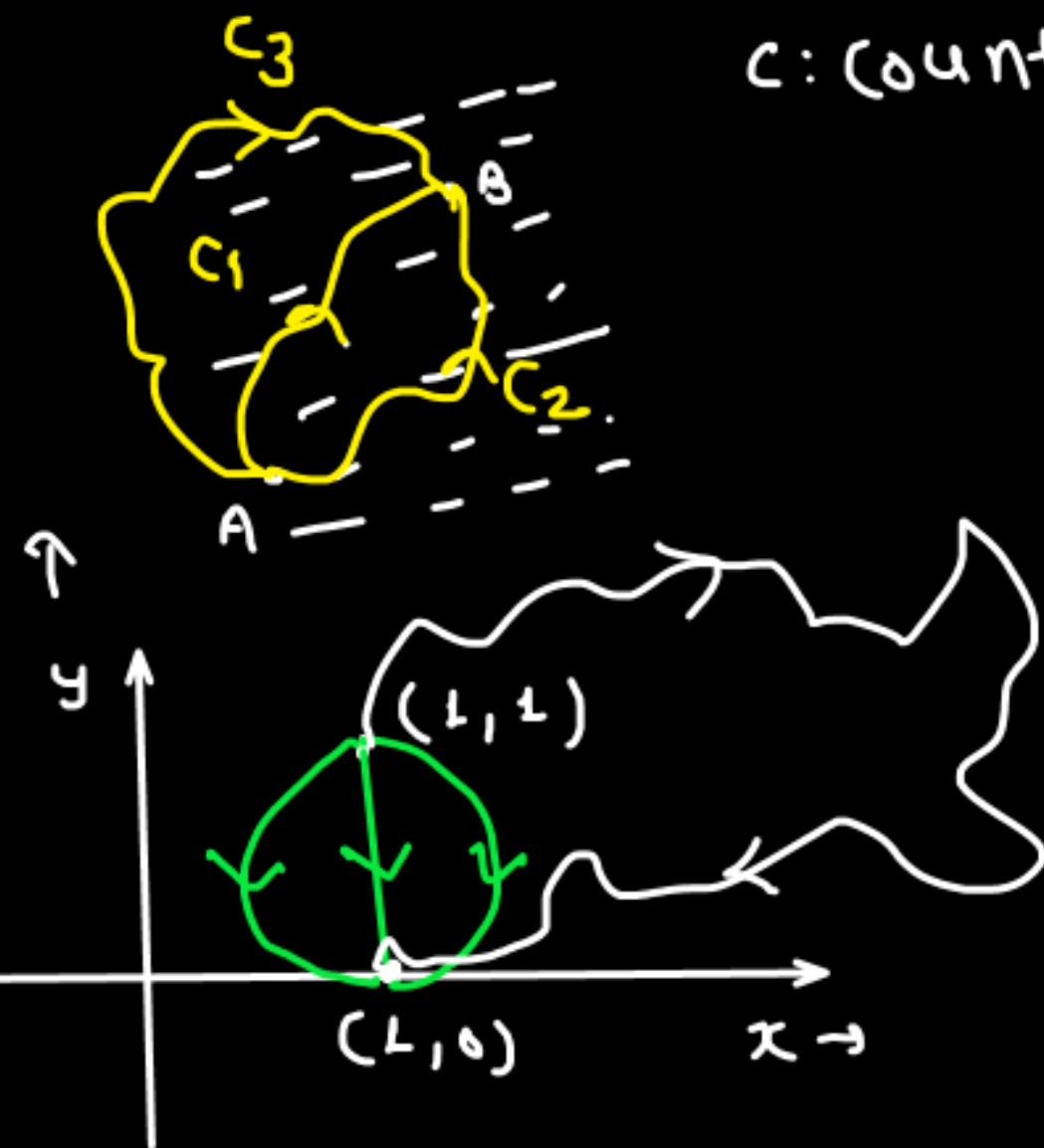
In domain  $D$ :-

$$\int_{C_1} f(z) \cdot dz = \int_{C_2} f(z) \cdot dz = \int_{C_3} f(z) \cdot dz$$

$C$ : contour

Q.  $I = \int_{1+i}^1 (z + e^z) dz$

Entire  $f^n$ : Analytic everywhere



$$I = \left[ \frac{z^2}{2} + e^z \right]_{1+i}^1 = \frac{1}{2} + e - \left[ \frac{(1+i)^2}{2} + e^{1+i} \right]$$

Ans.

Q.

Evaluate the contour integral

$$\int_C \underbrace{(z^2 + 2z + 3)}_{\text{Analytic}} dz,$$

where  $C$  is the upper semicircle

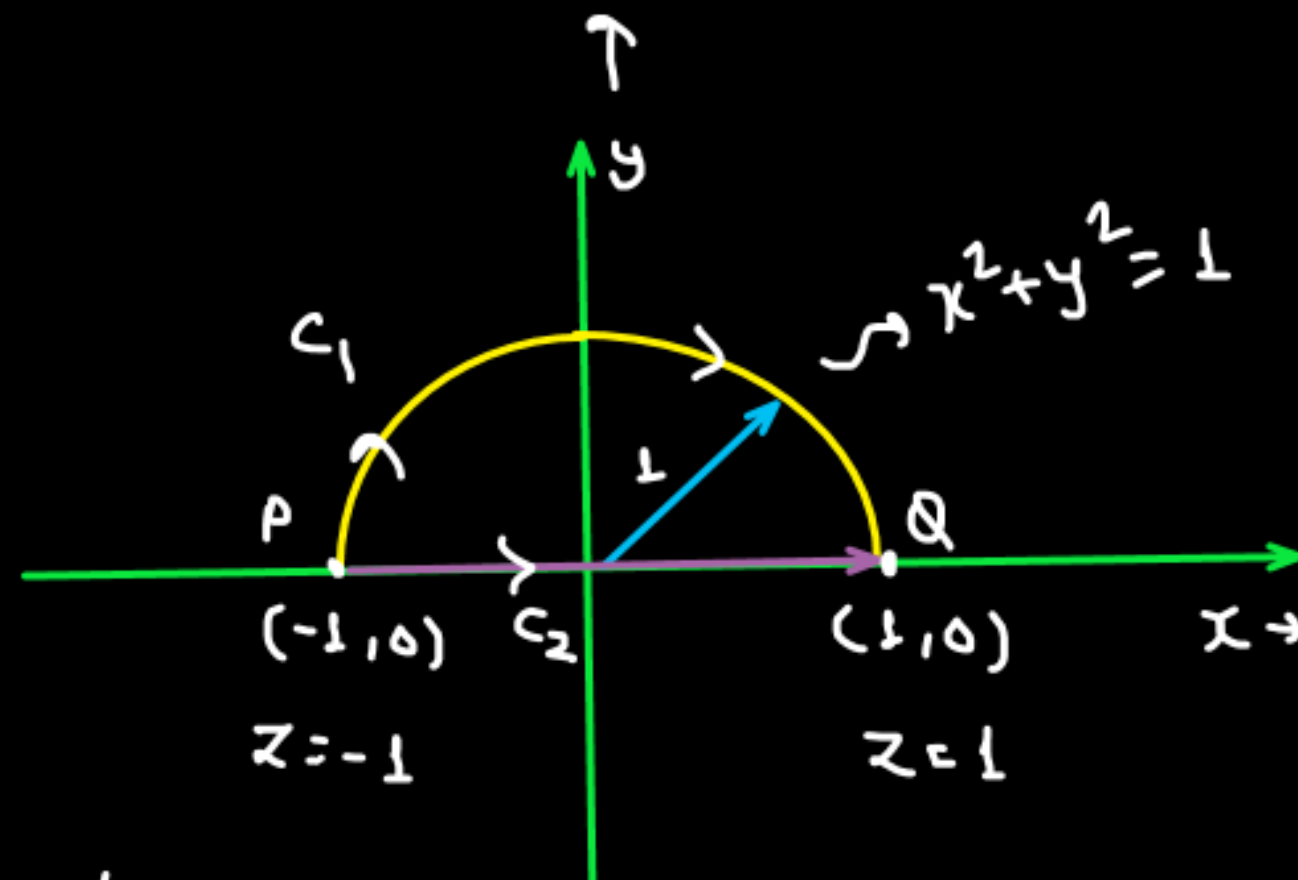
$$x^2 + y^2 = 1, \quad y \geq 0,$$

traversed from  $z = -1$  to  $z = 1$  (counterclockwise).

$$z = -1 = x + iy = -1 + i0$$

$$\stackrel{\text{M-R}}{=} \int_{C_1} f(z) \cdot dz = \int_{C_2} f(z) dz = \checkmark$$

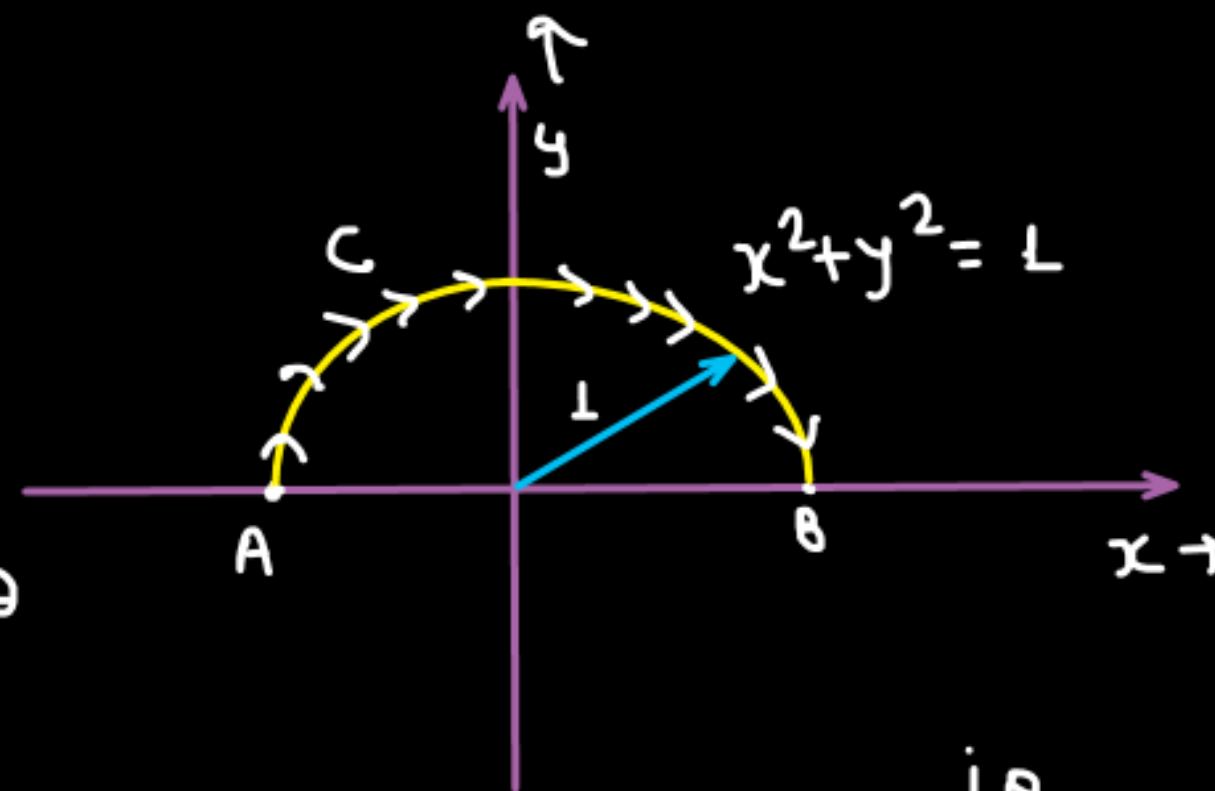
$$I = \int_{z=-1}^{z=1} z^2 + 2z + 3 \cdot dz = \left[ \frac{z^3}{3} + z^2 + 3z \right]_{-1}^1 = \left( \frac{1}{3} + 1 + 3 \right) - \left( -\frac{1}{3} + 1 - 3 \right) = \boxed{20/3} \text{ Ans.}$$



n=2 without the concept of analytic fn

$$I = \int_C (z^2 + 2z + 3) \cdot dz$$

$$z = e^{i\theta}$$
$$dz = ie^{i\theta} d\theta$$



$$I = \int_{\theta=\pi}^0 (e^{i2\theta} + 2e^{i\theta} + 3)(ie^{i\theta}) d\theta$$

$$z = re^{i\theta}$$

$$r = 1 ; \pi \leq \theta \leq 0$$

$$\Rightarrow z = e^{i\theta}$$

$$dz = e^{i\theta} i d\theta$$

$$\Rightarrow dz = ie^{i\theta} d\theta$$

$$= i \int_{\theta=\pi}^0 e^{i3\theta} + 2e^{i2\theta} + 3e^{i\theta} \cdot d\theta$$

$$= i \left[ \frac{e^{i3\theta}}{i3} + \frac{2e^{i2\theta}}{i2} + \frac{3e^{i\theta}}{i} \right]_{\pi}^0 =$$

$$I = \left[ \frac{e^{i3\theta}}{3} + e^{i2\theta} + 3e^{i\theta} \right]_0^\pi$$

$$= \left[ \frac{1}{3} + 1 + 3 - \left( -\frac{1}{3} + 1 - 3 \right) \right]$$

$$I = 20/3$$

$$e^{i2\pi} = 1$$

$$e^{i3\pi} = -1$$

$$e^{i\pi} = -1$$

Q.

Question

Evaluate the line integral:

$$z = x + iy$$

$$dz = dx + i dy$$

$$I = \int_C \underbrace{\left[ (x^2 - y^2 + 2x) + i(2xy + 2y) \right]}_{f(z)} dz$$

where  $C$  is the cubic curve  $y = x^3$  moving from the origin  $z_1 = 0$  to  $z_2 = 1 + i$ .

 $\hookrightarrow M-I$ 

$$I = \int_C (x^2 - y^2 + 2x) + i(2xy + 2y) \cdot (dx + i dy)$$

$$I = \int_0^1 (t^2 - t^6 + 2t^2) + i(2t^4 + 2t^3) \cdot (dt + i 3t^2 dt)$$

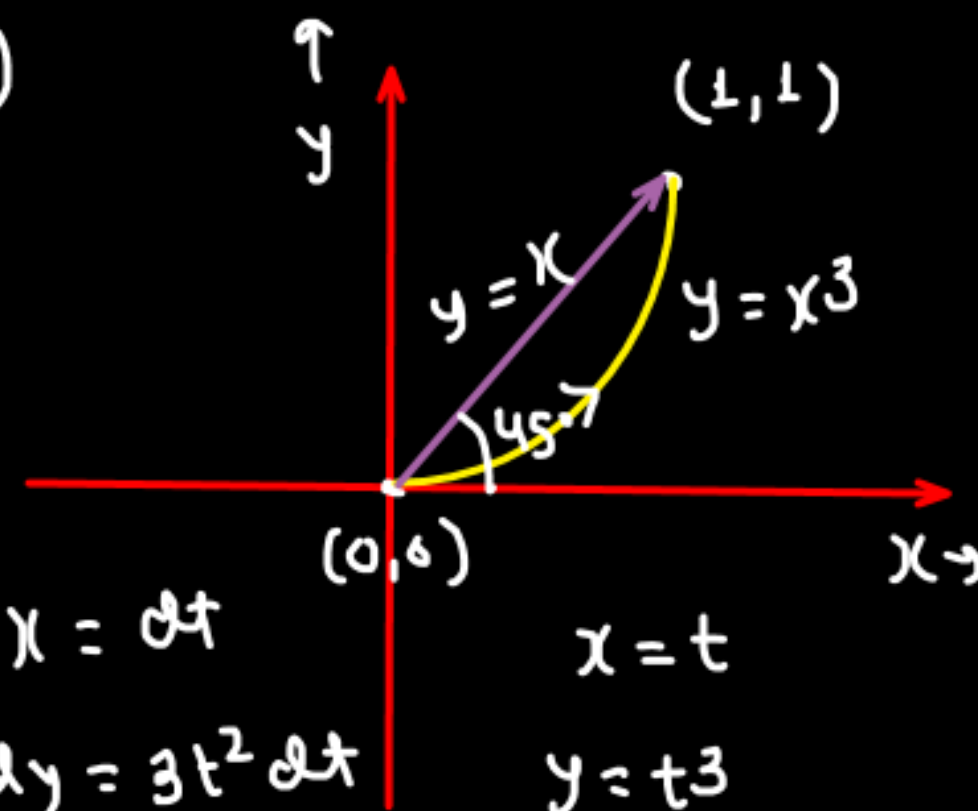
$$= \int_{t=0}^1 \left[ \underbrace{(t^2 - t^6 + 2t^2)}_{\text{lengthy}} + i \underbrace{(2t^4 + 2t^3)}_{\text{lengthy}} \right] \underbrace{(1 + i 3t^2)}_{\text{lengthy}} dt; \quad dx = dt$$

 $t=0$ 
lengthy =  $\checkmark$ 

$$dy = 3t^2 dt$$

$$x = t$$

$$y = t^3$$





$$\left\{ \begin{array}{l} f(z) = u + iv \\ u = x^2 - y^2 + 2x \quad ; \quad v = 2xy + 2y \\ u_x = 2x + 2 \quad \searrow \quad v_x = 2y \\ u_y = -2y \quad \nearrow \quad v_y = 2x + 2 \end{array} \right\} \text{Analytic}$$

$$I = \int_{t=0}^1 [(t^2 - t^2 + 2t) + i(2t^2 + 2t)] (dt + i dt) \quad \begin{array}{l} y = x = t \\ dx = dy = t \end{array}$$

$$= (1+i) \int_{t=0}^1 (2t + i2t^2 + i2t) \cdot dt = (1+i) \left[ t^2 + i \frac{2t^3}{3} + i t^2 \right]_0^1$$

$$= (1+i) \left[ 1 + i \frac{2}{3} + i \right]$$

$$= (1+i) \left[ 1 + i \frac{5}{3} \right]$$

$$= 1 + i \frac{5}{3} + i - \frac{5}{3} = \boxed{-\frac{2}{3} + i \frac{8}{3}} \text{Ans.}$$



M-3

$f(z)$ : Analytic ; MT said:-  $x \leftrightarrow z, y \leftrightarrow 0$

$$I = \int_{z=0}^{1+i} (z^2 + 2z) + i(0) \cdot dz$$

$$f(z) = z^2 + 2z$$

$$z=0 \rightarrow 1+i$$

$$= \left[ \frac{z^3}{3} + z^2 \right]_0^{1+i} = \frac{(1+i)^3}{3} + (1+i)^2$$

$$= \frac{2i-2}{3} + 2i$$

$$= \frac{2i}{3} - \frac{2}{3} + 2i$$

$$I = \frac{8i}{3} - \frac{2}{3}$$

Q.

Evaluate the line integral:

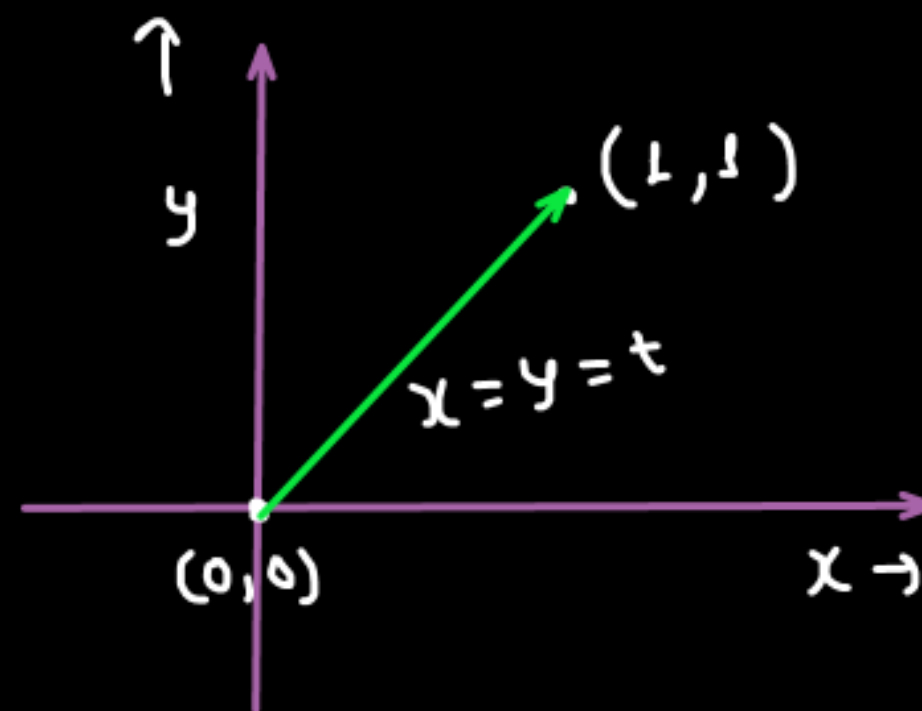
$$I = \int_C \underbrace{(x - iy)}_{\text{NOT analytic}} dz$$

where  $C$  is the straight line segment from  $z_1 = 0$  to  $z_2 = 1 + i$ .

$$\begin{aligned} z &= x + iy & dz &= dx + i dy \\ dz &= dt + i dt \\ dz &= (1 + i) \cdot dt \end{aligned}$$

$$\begin{aligned} I &= \int_{t=0}^1 (t - it)(1 + i) dt = \int_0^1 (1 - i)(1 + i)t \cdot dt \\ &= 2 \int_0^1 t \cdot dt = 2 \left[ \frac{t^2}{2} \right]_0^1 = \boxed{1} \text{ Ans.} \end{aligned}$$

$$f(z) = \bar{z}$$



Q.

### Question

Evaluate the line integral:

$$I = \int_C \underbrace{(x - iy)}_{\text{NOT analytic}} dz$$

where  $C$  is the upper half of the unit circle  $|z| = 1$  traversed in the counterclockwise direction from  $z_1 = 1$  to  $z_2 = -1$ .

u-1

$$f(z) = x - iy$$

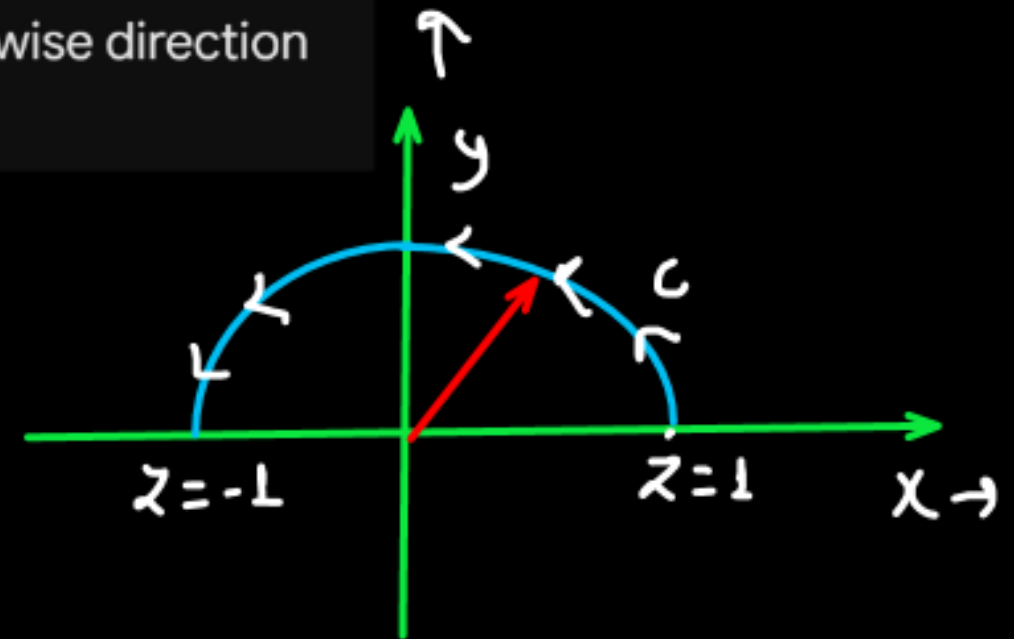
$$= \bar{z} = e^{-i\theta}$$

$$z = e^{i\theta}$$

$$dz = e^{i\theta} (i d\theta)$$

$$I = \int_{\theta=0}^{\pi} e^{-i\theta} \cdot i e^{i\theta} d\theta = i \int_0^{\pi} d\theta$$

$$= i(\pi) = \boxed{i\pi} \text{ Ans.}$$



$$z = r e^{i\theta}$$

$$r = 1 \Rightarrow z = e^{i\theta}$$

$$0 < \theta < \pi$$

$$\int (x - iy)(dx + i dy)$$

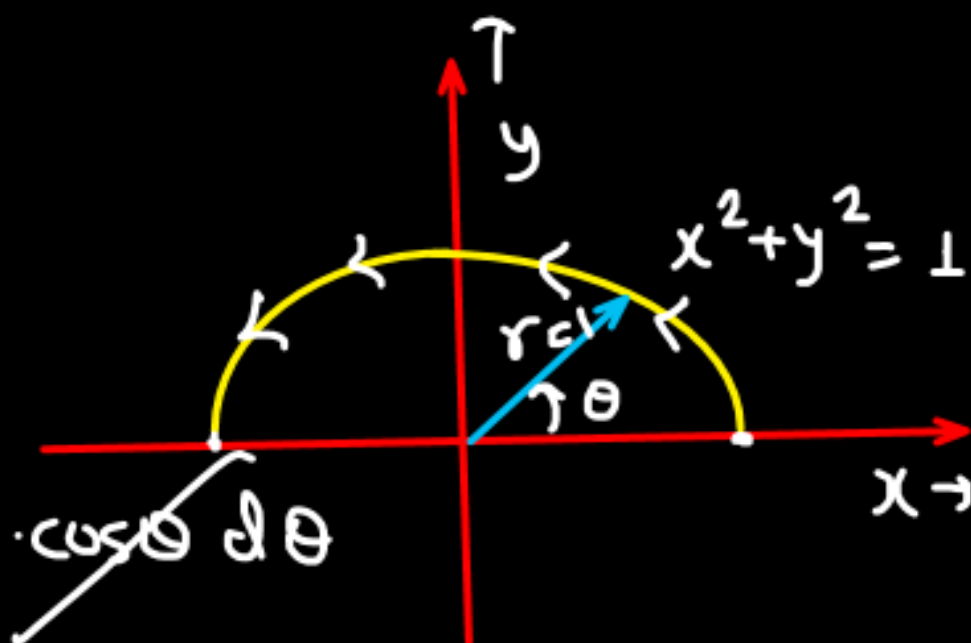
11-2

$$I = \int_C (x - iy)(dx + i dy)$$

$$= \int_C x dx + i x dy - i y dx + y dy$$

$$= \int_{\theta=0}^{\pi} -\cancel{\cos\theta \sin\theta d\theta} + i \cos^2\theta d\theta + i \sin^2\theta d\theta + \cancel{\sin\theta \cos\theta d\theta}$$

$$= i \int_{\theta=0}^{\pi} d\theta = \boxed{i\pi} \quad \text{Ans.}$$



$$x = r \cos\theta$$

$$y = r \sin\theta \quad r=1$$

$$x = \cos\theta \quad 0 \leq \theta \leq \pi$$

$$y = \sin\theta$$

$$dx = -\sin\theta d\theta$$

$$dy = \cos\theta d\theta$$

Q.

Question

Evaluate the line integral:

$$dz = dx + i dy$$

$$u = 2y, \quad v = x$$

$$u_x = 0$$

$$v_x = 1$$

$$u_y = 2$$

$$v_y = 0$$

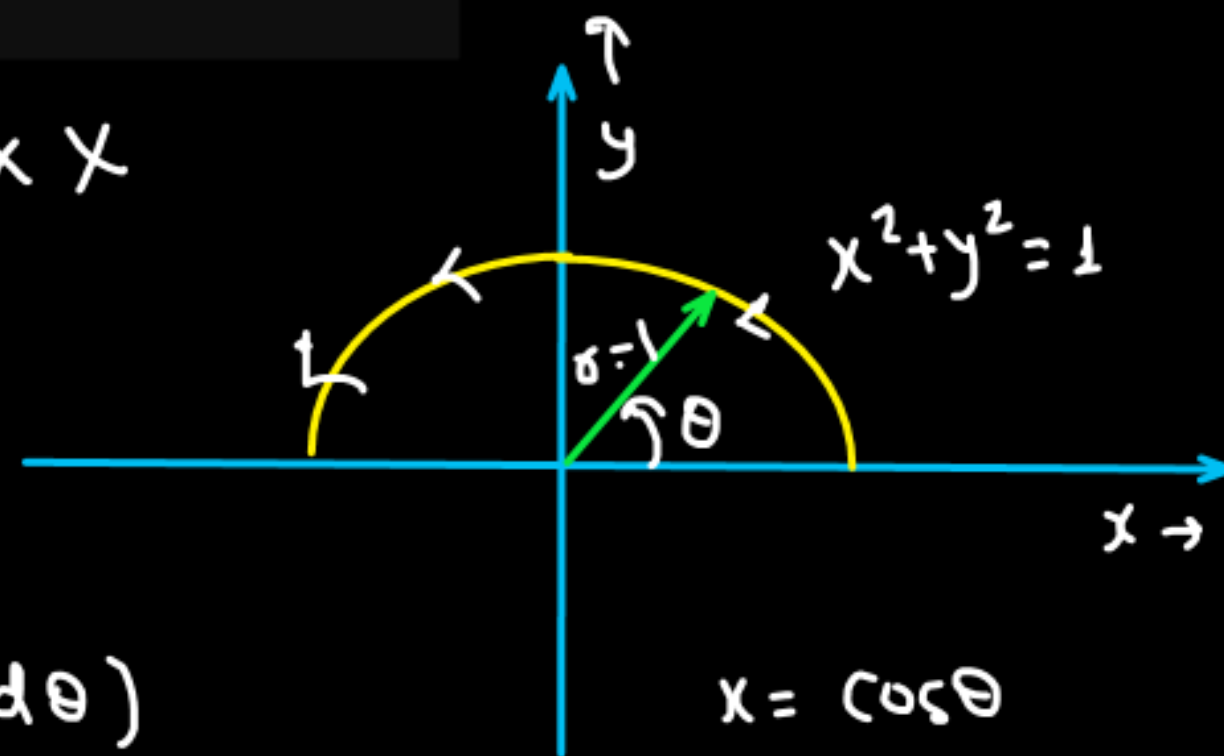
} NOT  
analytic

$$I = \int_C \underbrace{(2y + ix)}_{\text{NOT analytic}} dz$$

where  $C$  is the upper half of the unit circle  $|z| = 1$  traversed counterclockwise from  $z_1 = 1$  to  $z_2 = -1$ .

MT said: -  $x \leftrightarrow z, y \leftrightarrow 0 \quad f(z) = iz$  XXX

$\downarrow$   
 $f(z)$ : analytic



$$I = \int_0^\pi (2\sin\theta + i\cos\theta)(-\sin\theta d\theta + i\cos\theta d\theta)$$

$$= \int_0^\pi -2\sin^2\theta + i2\sin\theta\cos\theta - i\sin\theta\cos\theta - \cos^2\theta d\theta$$

$$x = \cos\theta$$

$$y = \sin\theta$$

$$dx = -\sin\theta d\theta$$

$$dy = \cos\theta d\theta$$

$$= \int_{\theta=0}^{\pi} -2\sin^2\theta \cdot d\theta + \int_{\theta=0}^{\pi} -\cos^2\theta \cdot d\theta + i \int_{\theta=0}^{\pi} \frac{\sin\theta}{2} \cdot \cos\theta d\theta$$

$$= \int_{\theta=0}^{\pi} -(1 - \cos 2\theta) \cdot d\theta - \int_{\theta=0}^{\pi} \frac{1 + \cos 2\theta}{2} d\theta + i \int_{\theta=0}^{\pi} \sin 2\theta d\theta$$

$$= -1(\pi) - 1/2(\pi)$$

$$\boxed{I = -3/2\pi}$$

$$\begin{aligned}\cos 2\theta &= 1 - 2\sin^2\theta \\ &= 2\cos^2\theta - 1\end{aligned}$$

$$\sin\theta \rightarrow 2\pi$$

$$\sin 2\theta \rightarrow \pi$$



Q.

# Question

Evaluate the line integral:

$$I = \int_C \left[ (x^3 - 3xy^2 + e^{2x} \cos 2y) + i (3x^2y - y^3 + e^{2x} \sin 2y) \right] dz$$

where  $C$  is the parabolic path  $y = x - x^2$  moving from  $z_1 = 0$  to  $z_2 = 1$ .

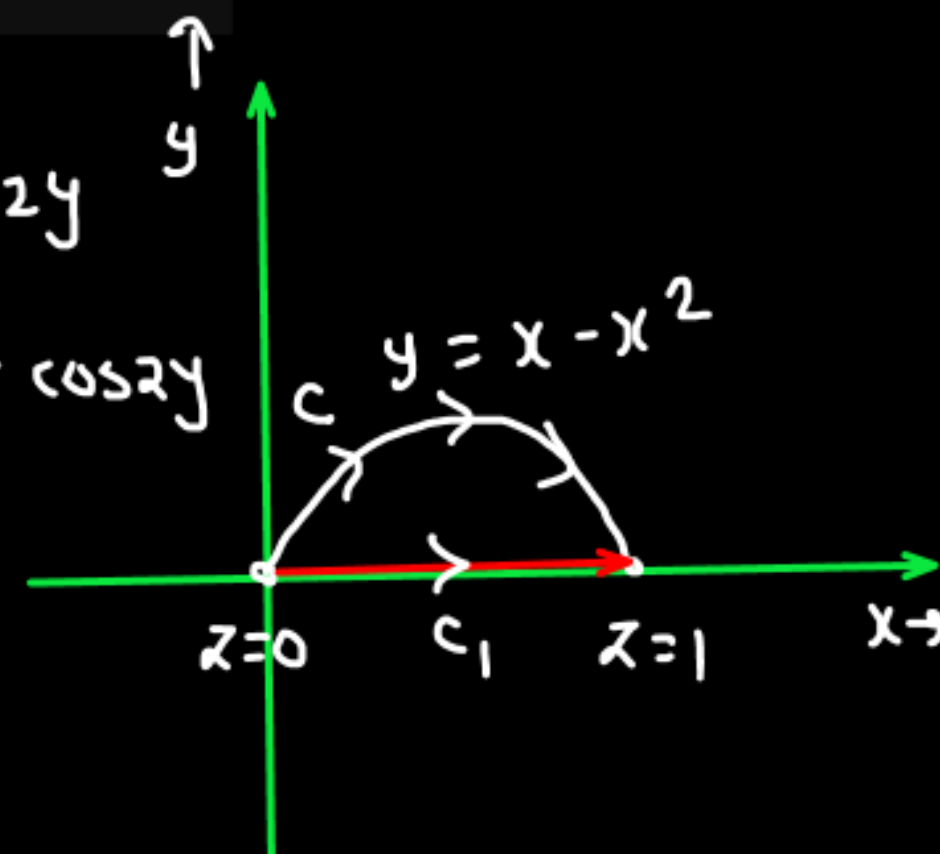
$$u_x = 3x^2 - 3y^2 + 2e^{2x} \cos 2y ; v_x = 6xy + 2e^{2x} \sin 2y$$

$$u_y = -6xy - 2e^{2x} \sin 2y \quad v_y = 3x^2 - 3y^2 + 2e^{2x} \cos 2y$$

analytic

MT :-  $x \leftrightarrow z, y \leftrightarrow 0$   
 $f(z) = z^3 + e^{2z}$

$$I = \int_{z=0}^1 (z^3 + e^{2z}) \cdot dz$$





$$I = \left[ \frac{z^4}{4} + \frac{e^{2z}}{2} \right]_0^1 = \left[ \frac{1}{4} + \frac{e^2}{2} - \left( \frac{1}{2} \right) \right]$$

$$= \underbrace{\frac{e^2}{2} - \frac{1}{4}}_{\text{Ans.}} = \boxed{3.44} \text{ Ans.}$$

M.2

$$I = \int_0^1 (x^3 + e^{2x}) \cdot dx$$

$$= \left[ \frac{x^4}{4} + \frac{e^{2x}}{2} \right]_0^1 = 3.44 \text{ Ans.}$$

$$dz = dx + i dy$$

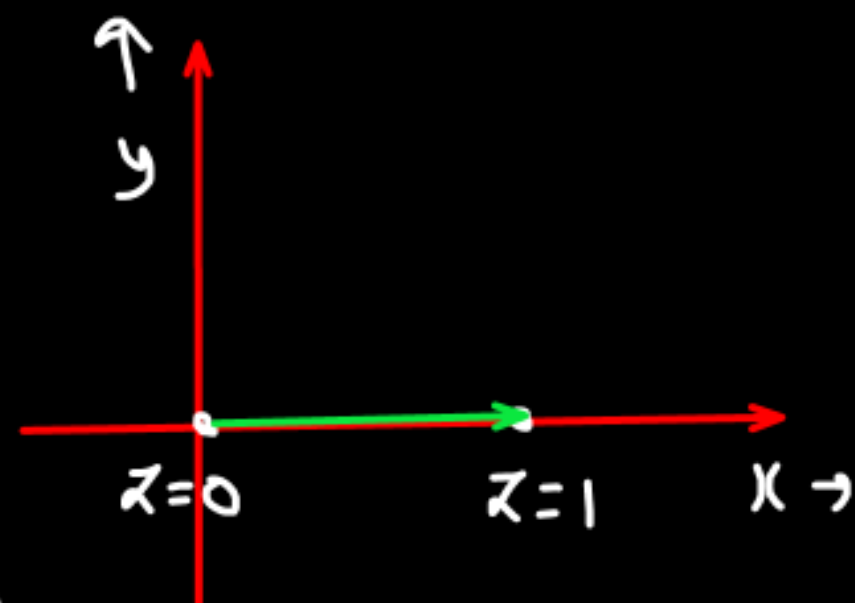
$$dz = dx$$

$$dx \neq 0$$

$$dy = 0$$

$$0 < x < 1$$

$$y = 0$$

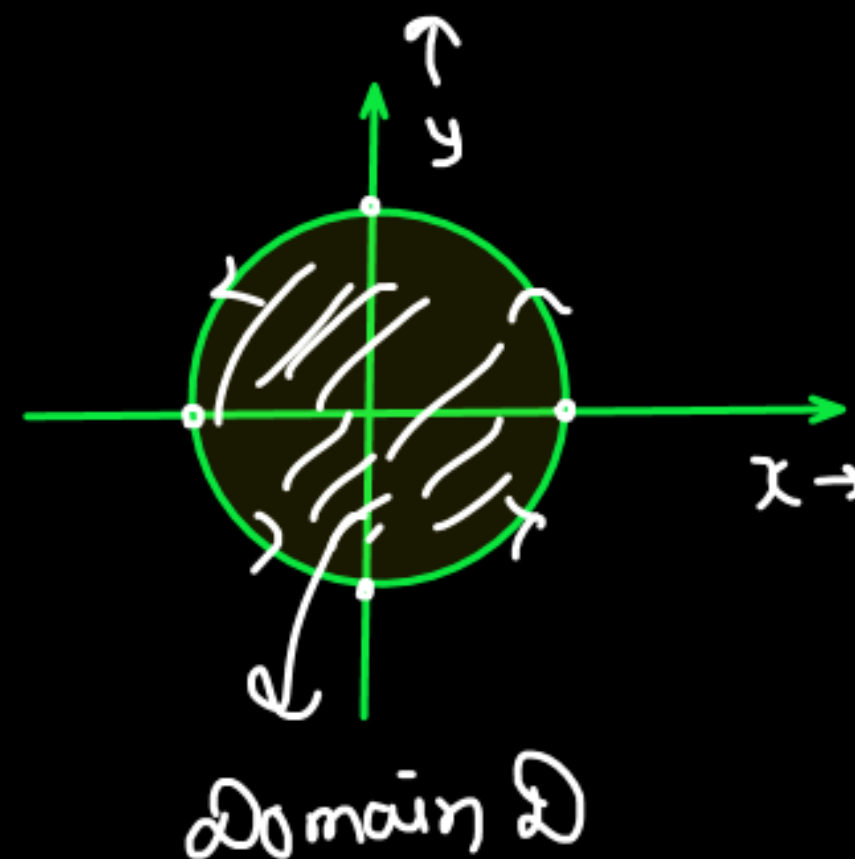


Q.  $\oint_C \underbrace{\cos z \cdot dz}_{\text{Entire}} = 0$

$\oint_C \underbrace{\sin z \cdot dz}_{\text{Entire}} = 0$

$\oint_C \underbrace{e^{z^2} \cdot \sin [\cosh(2z^2 + 3z)] \cdot dz}_{\text{Entire}} = 0$

$\underbrace{C: |z| = 1}_{\text{ACW}}$



Q.  $\oint_C \underbrace{\sec z}_{\text{NOT ENTIRE f'n}} \cdot dz = ?$

$C: |z| = 1$  ; ACW

$$f(z) = \sec z = \frac{1}{\cos z}$$

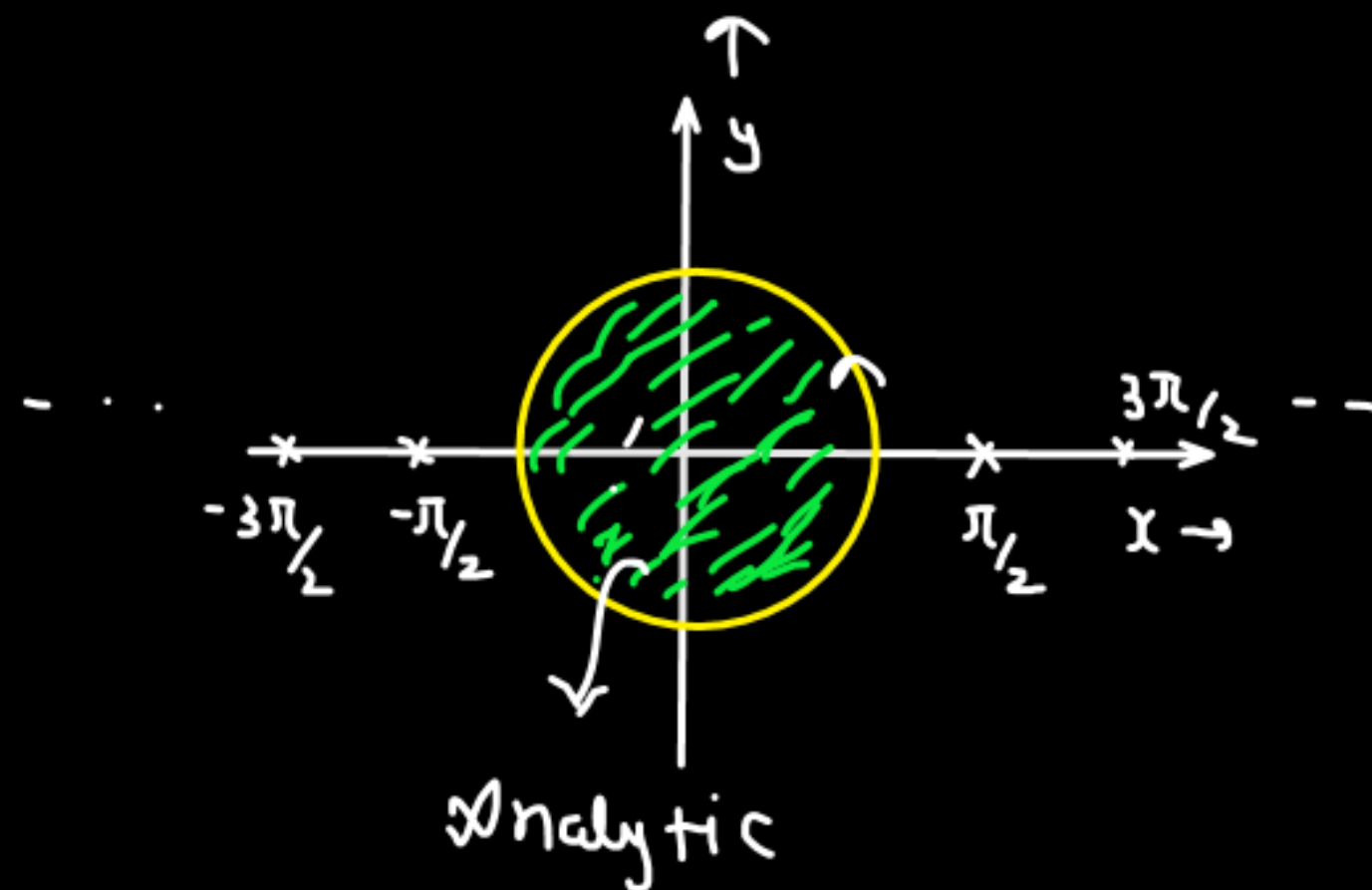
$$\cos z = 0 \Rightarrow z = \pm \pi/2, \pm 3\pi/2, \dots$$

NOT analytic

$$\pm 1.57, \dots$$

✓

$$\oint \sec z \cdot dz = 0$$



Q.  $I_1 = \oint_c \frac{z}{(z-1)(z+2)(z-4)} \cdot dz$ ;  $c: |z|=3$ , encirclement ACW

$I_1 = \oint_c \frac{z}{(z-1)(z+2)(z-4)} \cdot dz$ ;  $c: |z|=3$ , encirclement CW

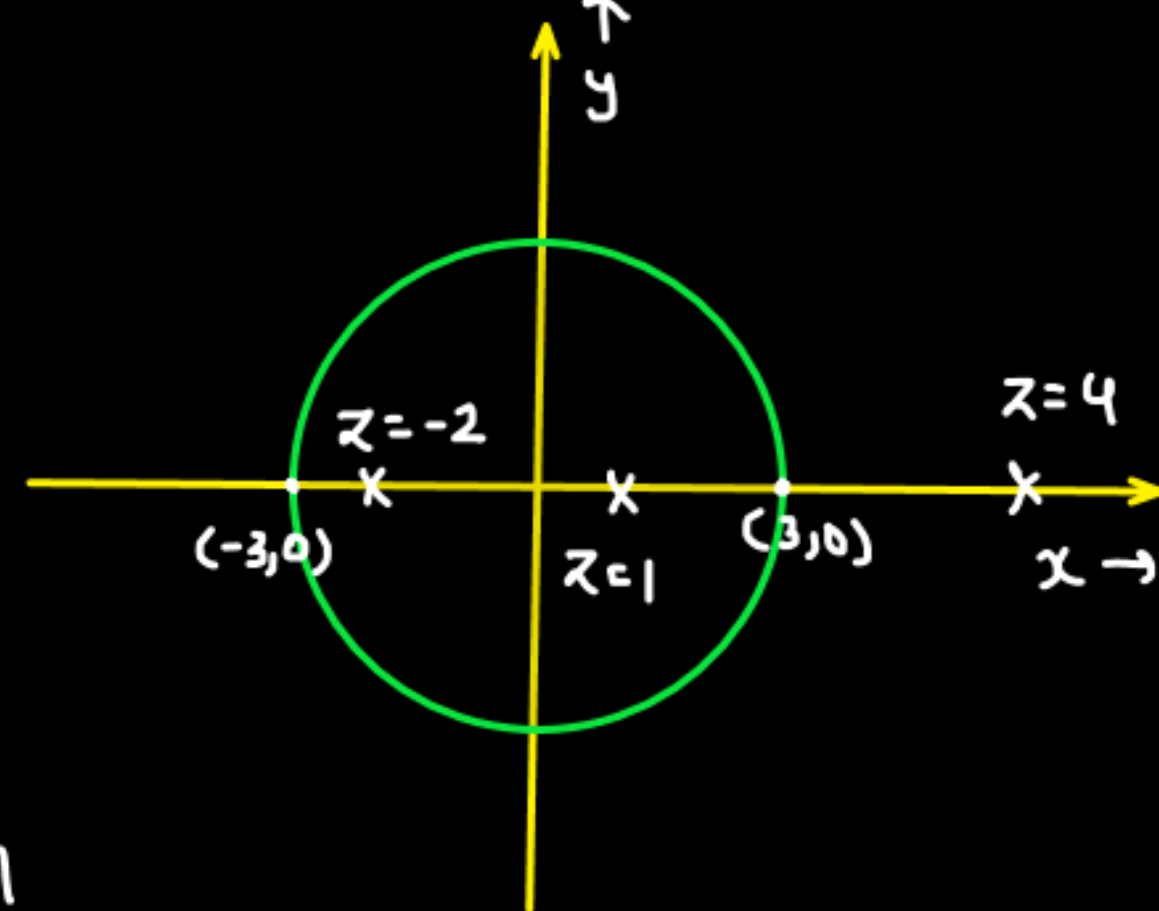
4  $f(z) = \frac{z}{(z-1)(z+2)(z-4)}$

$z = 1, -2, 4$

NOT analytic

$z = 1, -2$ : lying in domain  $D$ : Problem

$z = 4$ : NOT in domain  $D$ : no problem



$$4 \text{ Residue } z=1 \Rightarrow R_1 = \lim_{z \rightarrow 1} \frac{z}{(z+2)(z-4)} = -\frac{1}{9}$$

$$4 \text{ Residue } z=-2 \Rightarrow R_2 = \lim_{z \rightarrow -2} \frac{z}{(z-1)(z-4)} = -\frac{1}{9}$$

$$\Sigma R = R_1 + R_2 = -2/9$$

$$1. \quad c: \text{Anticlockwise}; \quad I = \oint_c f(z) \cdot dz = 2\pi i (\Sigma R) = 2\pi i \left(-\frac{2}{9}\right) = -\frac{4\pi i}{9}$$

$$2. \quad c: \text{clockwise}; \quad I = \oint_c f(z) \cdot dz = -2\pi i (\Sigma R) = -2\pi i \left(-\frac{2}{9}\right) = \frac{4\pi i}{9}$$

$$\hookrightarrow \oint_C f(z) \cdot dz = 0 \quad ; \quad f(z) \text{ is analytic in the region} \\ \text{bounded by closed contour } C. \\ \text{[Domain } \mathcal{D}]$$

$$\hookrightarrow \oint_C f(z) \cdot dz = 2\pi i (\sum R) \quad ; \quad \text{encirclement: ccw} \\ f(z) \text{ is not analytic in Domain } \mathcal{D} \\ \sum R = \text{sum of all Residue of } f(z) \\ \text{at all the singularities in the} \\ \text{domain } \mathcal{D}.$$

$$\hookrightarrow \oint f(z) \cdot dz = -2\pi i (\sum R) \quad ; \quad \text{encirclement: cw}$$



## ✓✓ Residue Theorem (Cauchy's Residue Theorem)

**Theorem:** Let  $f(z)$  be analytic on and inside a simple closed contour  $C$ , except at a finite number of isolated singularities  $z_1, z_2, \dots, z_n$  lying inside  $C$ . Then,

$$\oint_C f(z) dz = \begin{cases} +2\pi i \sum_{k=1}^n \text{Res}(f, z_k), & \text{if the encirclement is anticlockwise,} \\ -2\pi i \sum_{k=1}^n \text{Res}(f, z_k), & \text{if the encirclement is clockwise.} \end{cases}$$

Here,  $\text{Res}(f, z_k)$  denotes the residue of  $f(z)$  at the isolated singularity  $z_k$ .

$\text{Res}(f, z_1)$  = Residue of  $f$  at  $z=z_1$

$\sum_{k=1}^n \text{Res}(f, z_k)$  = Sum of all Residue of  $f$   
at all singularities in domain  $D$





Q.

Q5.24.1

MCQ

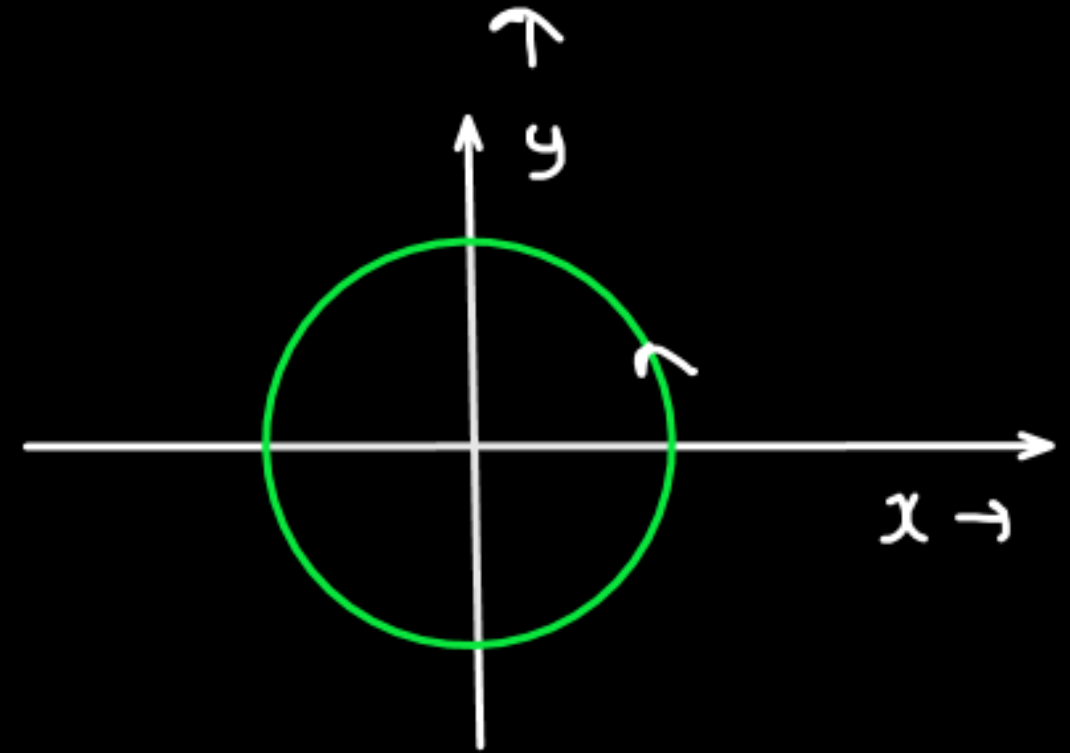
2024

2M



Let  $z$  be a complex variable. If  $f(z) = \frac{\sin(\pi z)}{z^2(z-2)}$  and  $C$  is the circle in the complex plane with  $|z| = 3$  then  $\oint_C f(z) dz$  is \_\_\_\_\_.

- A.  $\pi^2 j$     B.  $j\pi \left(\frac{1}{2} - \pi\right)$     C.  $j\pi \left(\frac{1}{2} + \pi\right)$     **D.  $-\pi^2 j$**



$$f(z) = \frac{\sin \pi z}{z^2(z-2)}$$

$$z^2(z-2) = 0 ; \quad z = 0, 0, 2$$

$z = 0, 0, 2$  : Inside  
Multiple pole      single pole

$$R_1 = [\text{Residue at } z=2] = \lim_{z \rightarrow 2}$$

$$R_1 = 0$$

$$|z| = 3$$

$|z| > 3$  : outside

$|z| < 3$  : Inside

$$\frac{\sin(\pi z)}{z^2} = \frac{\sin(2\pi)}{4} = 0$$

$$R_2 = \lim_{z \rightarrow 0} \frac{1}{1!} \frac{d}{dz} \left( \frac{\sin \pi z}{z-2} \right)$$

$$= \lim_{z \rightarrow 0} \frac{\pi(z-2) \cos \pi z - \sin \pi z}{(z-2)^2}$$

$$= \frac{\pi(-2)(1) - 0}{4} = -\pi/2$$

$$\begin{aligned} I &= \oint f(z) \cdot dz = 2\pi i (-\pi/2) \\ &= \underbrace{-\pi^2 i} \end{aligned}$$

$$\sum R = -\pi/2 + 0 = -\pi/2$$

Q.

Q5.23.2

MCQ

2023

1M



The value of the contour integral,  $\oint_C \left( \frac{z+2}{z^2+2z+2} \right) dz$ , where the contour  $C$  is  $\{z : |z + 1 - \frac{3}{2}j| = 1\}$ , taken in the counter clockwise direction, is

A.  $-\pi(1 + j)$

✓ B.  $\pi(1 + j)$

C.  $\pi(1 - j)$

D.  $-\pi(1 - j)$

$$|z + 1 - 3/2j| > 1 \rightarrow \text{outside}$$

$$|z + 1 - 3/2j| < 1 \rightarrow \text{Inside}$$

$$|-1+j+1-3/2j| = |-1/2j| = 1/2 < 1$$

↳

$$z^2 + 2z + 2 = 0$$

$$z = \frac{-2 \pm \sqrt{4-8}}{2} = \frac{-2 \pm 2j}{2} = \underbrace{-1+j, -1-j}_{\text{Inside}} \rightarrow \text{outside}$$

$$C: |z + 1 - 3/2j| = 1$$

$$\downarrow \text{centre } (-1, 3/2); \text{ radius} = 1$$

$$|-1-j+1-3/2j| = |-5/2j| = 5/2 > 1$$

Residue at  $z = -1+j$  : simple pole

$$R = \lim_{z \rightarrow -1+j} \frac{z+2}{2z+2} = \frac{-1+j+2}{2(-1+j)+2} = \frac{j+1}{2j}$$

$$= \frac{1}{2} - \frac{1}{2}j = \frac{1-j}{2}$$

$$I = 2\pi i \left( \frac{1-j}{2} \right) = \pi i (1-j) = \pi (i - i^2) \\ = \pi (1+i)$$

Q.

Q5.22.1

NAT

2022

1M



A simple closed path  $C$  in the complex plane is shown in the figure. If

$$\oint_C \frac{2^z}{z^2 - 1} dz = -i\pi A,$$

where  $i = \sqrt{-1}$ , then the value of  $A$  is \_\_\_\_\_ (rounded off to two decimal places).

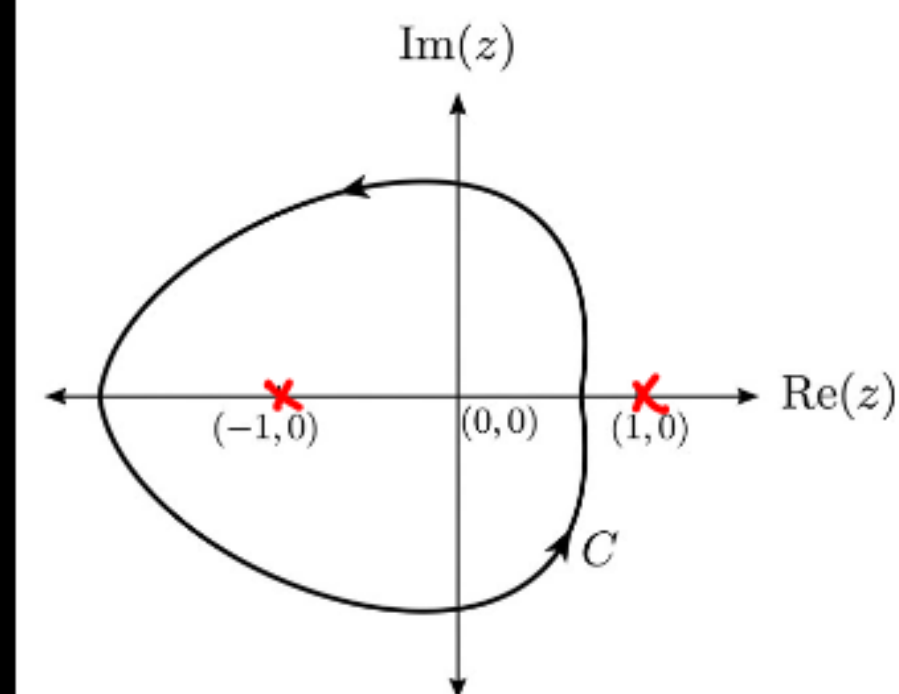


Fig. Q5.22.1

 $\mathcal{A} \subset \omega$ 
 $\hookrightarrow$ 

$$z^2 - 1 = 0 \Rightarrow z = 1, -1$$

x      ✓

$$\text{Residue at } z = -1: R = \lim_{z \rightarrow -1} \frac{2^z}{2z} = \frac{2^{-1}}{2(-1)} = -\frac{1}{4}$$

$$I = 2\pi i \left(-\frac{1}{4}\right) = -\frac{\pi i}{2}$$

$$\boxed{\mathcal{A} = \frac{1}{2}}$$

GATE  
Q  
pyQ  
=

Q5.21.1

MCQ

2021

2M



Consider the integral

$$\oint_C \frac{\sin(x)}{x^2(x^2 + 4)} dx$$

→ NO OPTION

CORRECT

where  $C$  is a counter-clockwise oriented circle defined as  $|x - i| = 2$ . The value of the integral is

A.  $-\frac{\pi}{8} \sin(2i)$

B.  $\frac{\pi}{8} \sin(2i)$

C.  $-\frac{\pi}{4} \sin(2i)$

D.  $\frac{\pi}{4} \sin(2i)$

$$|z - i| = 2$$

$$\hookrightarrow z = 0; |0 - i| = |-i| = 1 < 2 \quad \checkmark$$

$$\hookrightarrow z = 2i; |2i - i| = |i| = 1 < 2 \quad \checkmark$$

$$\hookrightarrow z = -2i; |-2i - i| = |-3i| = 3 > 2 \quad \times$$

$$I = \oint \frac{\sin z}{z^2(z^2 + 4)} dz$$

$$z^2(z^2 + 4) = 0$$

$$z = 0, 0, 2i, -2i$$

~~~~~  
Inside

Residue at $z = 0$:-

$$R_1 = \lim_{z \rightarrow 0} \frac{1}{1!} \frac{d}{dz} \left(\frac{\sin z}{z^2 + 4} \right)$$

$$= \lim_{z \rightarrow 0} \frac{(z^2 + 4)(\cos z - \sin z(2z))}{(z^2 + 4)^2} = \frac{1}{4}$$

Residue at $z = 2i$: simple pole

$$A_2 = \lim_{z \rightarrow 2i} \frac{\sin z}{4z^3 + 8z} = \frac{\sin(2i)}{4(2i)^3 + 8(2i)} = \frac{\sin(2i)}{-16i}$$

$$\Sigma R = \frac{1}{4} - \frac{\sin(2i)}{16i}$$

$$\oint f(z) \cdot dz = 2\pi i \left[\frac{1}{4} - \frac{\sin(2i)}{16i} \right]$$

=

~~Q5.18.1~~

NAT

2018

2M



The contour C given below is on the complex plane $z = x + jy$, where $j = \sqrt{-1}$. The value of the integral $\frac{1}{\pi j} \oint_C \frac{dz}{z^2 - 1}$ is _____.

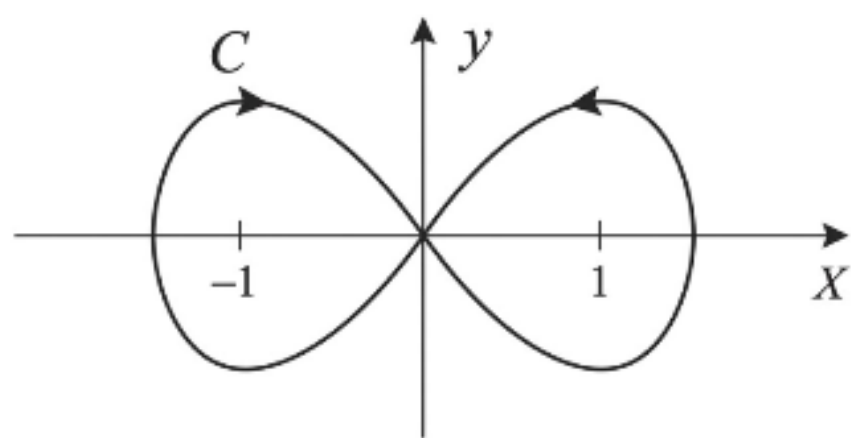
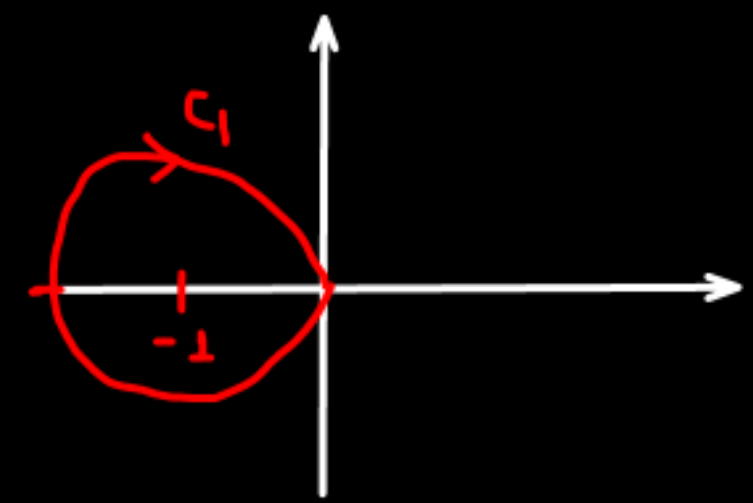


Fig. Q5.18.1

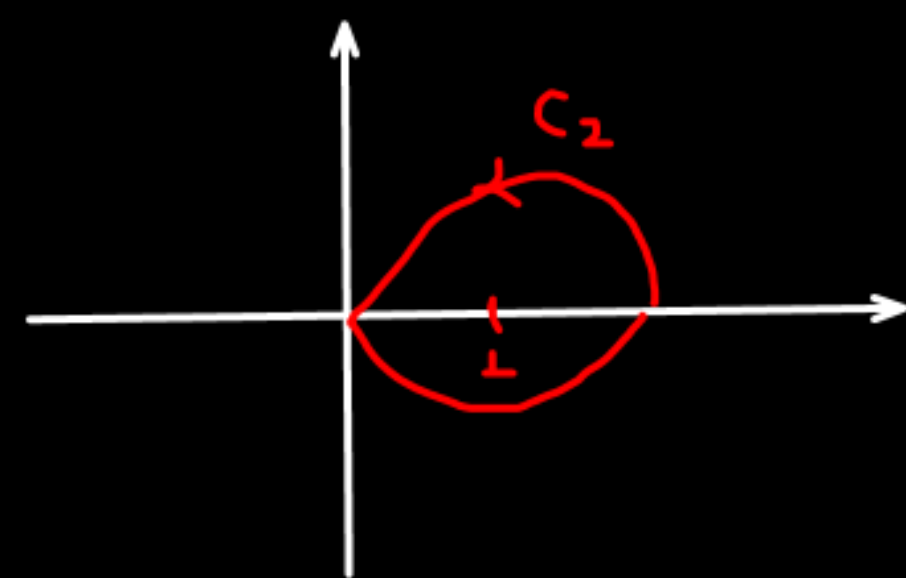
$$I = I_1 + I_2$$

$$z^2 - 1 = 0$$
$$z = \underline{1, -1}$$

$$I = 2$$
 Ans.



$$I_1 = \frac{1}{\pi j} \oint_{C_1} \frac{dz}{z^2 - 1}$$



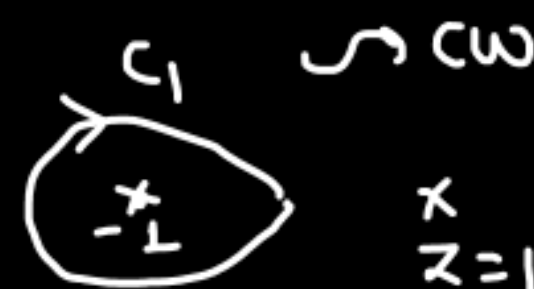
$$I_2 = \frac{1}{\pi j} \oint_{C_2} \frac{dz}{z^2 - 1}$$

$$I_1' = -2\pi i (\sum R)$$

Residue at $z = -1 \Rightarrow \lim_{z \rightarrow -1} \frac{1}{2z} = -\frac{1}{2}$

$$I_1' = -2\pi i \left(-\frac{1}{2}\right) = \pi i$$

$$I_1 = \frac{1}{\pi i} I_1' \Rightarrow \boxed{I_1 = 1}$$

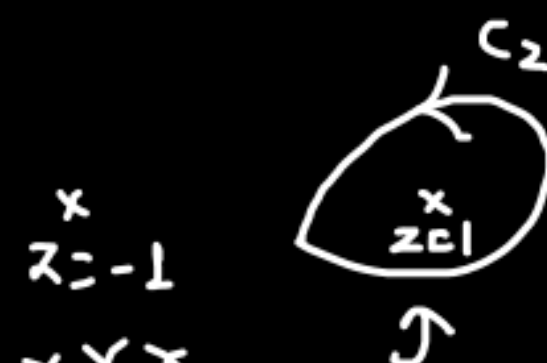


$$z = 1, -1$$

X ✓

$$I_2' = 2\pi i (\sum R)$$

$$R = \lim_{z \rightarrow 1} \frac{1}{2z} = \frac{1}{2} \leftarrow \text{Residue at } z = 1$$



$$z = -1$$

X X X

$$I_2' = 2\pi i \left(\frac{1}{2}\right) = \pi i$$

$$I_2 = \frac{1}{\pi i} I_2' = 1$$

$$\boxed{I_2 = 1}$$

Q.

Q5.17.2

MCQ

2017

1M



An integral I over a counterclockwise circle C is given by

$$I = \oint_C \frac{z^2 - 1}{z^2 + 1} e^z dz.$$

If C is defined as $|z| = 3$, then the value of I is

A. $-\pi i \sin(1)$ B. $-2\pi i \sin(1)$ C. $-3\pi i \sin(1)$ D. $-4\pi i \sin(1)$

$$\sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$$

$$z^2 + 1 = 0$$

$$z = i, -i$$

$$|z| = 3$$

$$|i| = 1 < 3$$

$$|-i| = 1 < 3$$

$$\text{Residue at } z=i \Rightarrow R_1 = \lim_{z \rightarrow i} \frac{(z^2 - 1)e^z}{2z} = \frac{(i^2 - 1)e^i}{2i} = ie^i$$

$$\text{Residue at } z=-i \Rightarrow R_2 = \lim_{z \rightarrow -i} \frac{(z^2 - 1)e^z}{2z} = \frac{(i^2 - 1)e^{-i}}{-2i} = -ie^{-i}$$

$$I = 2\pi i [ie^i - ie^{-i}] = -2\pi [\frac{e^i - e^{-i}}{2i}] \times 2i = \boxed{-4\pi i \sin 1} \text{ ans.}$$

$$\Sigma R = ie^i - ie^{-i}$$

Q.

Q5.25.1

MSQ

2025

2M



Which of the following statements involving contour integrals (evaluated counter-clockwise) on the unit circle C in the complex plane is/are TRUE?

☒ A. $\oint_C e^z dz = 0$

☒ B. $\oint_C z^n dz = 0$, where n is an even integer

☒ C. $\oint_C \cos z dz \neq 0 \rightarrow$

☒ D. $\oint_C \sec z dz \neq 0 \rightarrow$

$$\oint \cos z \cdot dz = 0$$

$$\oint \sec z \cdot dz = 0 ; |z| = 1$$

$\downarrow$ $\downarrow$
 $z = \pm \pi/2, \pm 3\pi/2 \dots$ counter
lying outside

1. $I = \oint \underbrace{z^2}_{\text{polynomial in } z} \cdot dz = 0$

$$I = \oint z^\eta \cdot dz = 0 ; \eta = \text{any positive Integer (odd or even)}$$

★ Negative Integers:-

$$\boxed{n = -1}$$

$$I = \oint_C \frac{1}{z} dz = 2\pi i(1) = 2\pi i \neq 0$$

↓
 $z=0$: single pole

$$R = \lim_{z \rightarrow 0} z f(z) = \lim_{z \rightarrow 0} z \times \frac{1}{z} = 1$$

$$\boxed{n = -2}$$

$$I = \oint \frac{1}{z^2} dz = 2\pi i(R) = 0$$

↓
 $z = \underbrace{0, 0}$: Multiple pole

$$R = \lim_{z \rightarrow 0} \frac{1}{1!} \frac{d}{dz} z^2 f(z) = \lim_{z \rightarrow 0} \frac{1}{1!} \frac{d}{dz} (1) = 0$$

$$\underline{\underline{\eta = -3}}$$

$$I = \oint \frac{1}{z^3} \cdot dz = 2\pi i(R) = 0$$

$z=0, 0, 0$: Multiple pole

$$\boxed{R=0}$$

$$R = \lim_{z \rightarrow 0} \frac{1}{2!} \frac{d^2}{dz^2} \left(\frac{1}{z} \right) = 0$$

$$\underline{\underline{\eta = -4}}$$

$$I = \oint \frac{1}{z^4} \cdot dz = 2\pi i(R) = 0$$

$z=0, 0, 0, 0$: Multiple

$$R = \lim_{z \rightarrow 0} \frac{1}{3!} \frac{d^3}{dz^3} \left(\frac{1}{z} \right) = 0$$

$$\times \quad I = \oint \bar{z}^\eta \cdot dz = \begin{cases} 0; & \eta \in \mathbb{I} - \{1\} \\ \pm 2\pi i; & \eta = 1 \end{cases}$$

Q5.15.4

NAT

2015

2M



If C denotes the counterclockwise unit circle, the value of the contour integral

$$\frac{1}{2\pi j} \oint_C \underbrace{\text{Re}\{z\}}_{\pi j} dz$$

is _____.

$$\begin{aligned} I &= \frac{1}{2\pi j} \int_{\theta=0}^{2\pi} \cos \theta \cdot i e^{i\theta} \cdot d\theta \\ &= \frac{1}{2\pi} \int_0^{2\pi} \left(\frac{e^{i\theta} + e^{-i\theta}}{2} \right) e^{i\theta} \cdot d\theta \\ &= \frac{1}{2\pi} \int_0^{2\pi} \frac{e^{2i\theta}}{2} + \frac{1}{2} \cdot d\theta = \frac{1}{2\pi} \times \frac{1}{2} [\theta]_0^{2\pi} = \boxed{\frac{1}{2}} \end{aligned}$$

$$z = x + iy$$

$$\text{Re}(z) = x$$

$$z = r e^{i\theta}$$

$$z = r \cos \theta + i r \sin \theta$$

$$|z| = 1$$

$$\text{Re}(z) = r \cos \theta$$

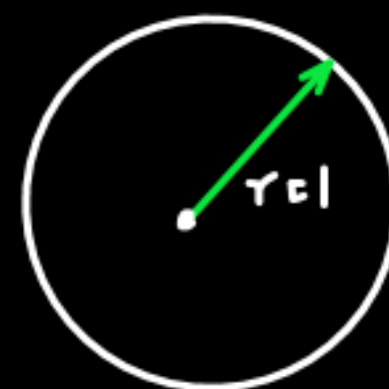
$$r = 1$$

$$0 \leq \theta \leq 2\pi$$

$$z = e^{i\theta}$$

$$; r = 1 ; 0 \leq \theta \leq 2\pi$$

$$dz = i e^{i\theta} d\theta$$



Q. **Q5.25.1**

NAT

2025

2M



Let C be a clockwise oriented closed curve in the complex plane defined by $|z| = 1$. Further, let $f(z) = jz$ be a complex function, where $j = \sqrt{-1}$. Then, $\oint_C f(z) dz = \underline{\hspace{2cm}}$ (round off to the nearest integer).

$$\oint_C f(z) \cdot dz$$

$$= \oint_C \underbrace{jz}_{p(z) \text{ with Img. coeff.}} \cdot dz = \boxed{0}$$

$p(z)$ with Img. coeff.

Q.

Q5.21.2

MCQ

2021

2M



Let $(-1-j)$, $(3-j)$, $(3+j)$ and $(-1+j)$ be the vertices of a rectangle C in the complex plane. Assuming that C is traversed in counter-clockwise direction, the value of the contour integral $\oint_C \frac{dz}{z^2(z-4)}$ is

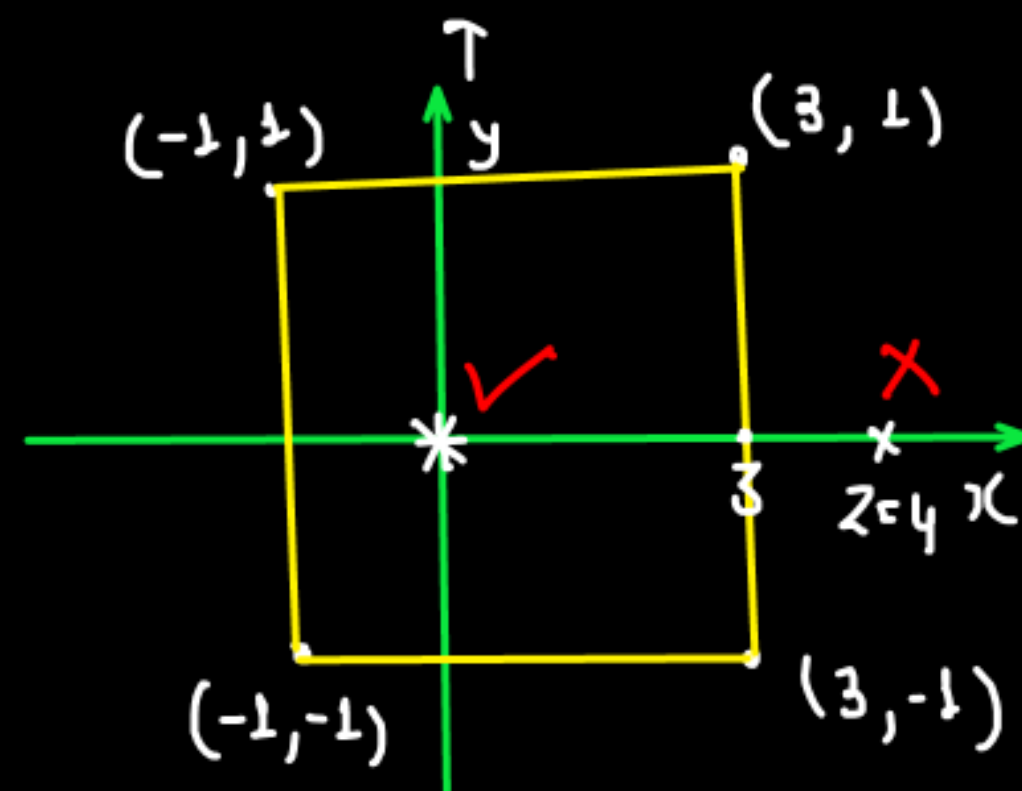
A. $j\pi/2$

B. 0

☒ C. $-j\pi/8$

D. $j\pi/16$

$$z^2(z-4)=0; \quad \underbrace{z=0, 0, 4}_x$$



Residue at $z=0$:-

$$R = \lim_{z \rightarrow 0} \frac{1}{1!} \frac{d}{dz} \frac{1}{z-4} \Rightarrow R = \lim_{z \rightarrow 0} -\frac{1}{(z-4)^2} = -\frac{1}{16}$$

$$I = 2\pi i \left(-\frac{1}{16}\right) = -\frac{\pi i}{8} \quad \underline{\text{Ans.}}$$

Q.

Q5.11.1

MCQ

2011

2M



The value of $\oint_C \frac{z^2}{z^4 - 1} dz$ using Cauchy's integral around the circle $|z + 1| = 1$, where $z = x + iy$, is

- A. $2\pi i$ **B. $-\frac{\pi i}{2}$** C. $-\frac{3\pi i}{2}$ D. $\pi^2 i$

$$|z+1| = 1$$

$$z^4 - 1 = 0 \quad \checkmark \quad z = re^{i\theta}$$

$$r^4 e^{i4\theta} = e^{i2n\pi}$$

$$r = 1, \quad \theta = n\pi/2$$

$$z = 1, -1, i, -i$$

$z = -1$: lying inside

$$|z+1| = 1$$

$$z = 1; \quad |2| > 1 \quad \times$$

$$z = -1; \quad |0| < 1 \quad \checkmark$$

$$z = i; \quad |i+1| = \sqrt{2} > 1 \quad \times$$

$$z = -i; \quad |-i+1| = \sqrt{2} > 1 \quad \times$$

Residue at $z = -1$

$$R = \lim_{z \rightarrow -1} \frac{z^2}{4z^3} = -\frac{1}{4}$$

$$I = 2\pi i \left(-\frac{1}{4} \right) = -\frac{\pi i}{2}$$

$$\text{Magnitude} = \left| -\frac{\pi i}{2} \right| = \pi/2$$

Q.

Q5.21.2

MCQ

2021

2M



$f(z) = (z-1)^{-1} - 1 + (z-1) - (z-1)^2 + \dots$ is the series expansion of

A. $\frac{-1}{z(z-1)}$ for $|z-1| < 1$

B. $\frac{1}{z(z-1)}$ for $|z-1| < 1$

C. $\frac{1}{(z-1)^2}$ for $|z-1| < 1$

D. $\frac{-1}{(z-1)}$ for $|z-1| < 1$

$$a + ar + ar^2 + ar^3 + \dots$$

$$= \frac{a}{1-r} = \frac{1/z-1}{1+(z-1)}$$

$$= \frac{1}{z(z-1)}$$

$$f(z) = \frac{1}{z-1} - 1 + (z-1) - (z-1)^2 + (z-1)^3 - \dots$$

$r = -(z-1)$

$\Rightarrow$ G.P.

$$f(z) = \frac{1/z-1}{1+(z-1)} = \frac{1}{z(z-1)}$$

✓

$$\hookrightarrow f(z) = \frac{-1}{z(z-1)} ; \text{ expand about } z=1$$

$$\hookrightarrow f(z) = \underbrace{\frac{1}{z}} - \underbrace{\frac{1}{z-1}}$$

$$\frac{1}{(z-1)+1} = [1 + (z-1)]^{-1} = \underbrace{1 - (z-1) + (z-1)^2 - (z-1)^3 + (z-1)^4 - \dots}$$

$$\frac{1}{z-1} = (z-1)^{-1}$$

$$\hookrightarrow f(z) = \underbrace{-(z-1)^{-1} + 1 - (z-1) + (z-1)^2 - (z-1)^3 + (z-1)^4 - \dots}$$